

Conductor Transport, Attained Profiles and the Covariance Return

Complete additive proof supplement

Split-Zero research programme

18 September 2026

This volume records the full continuation proofs integrated into the three cumulative source editions of 18 September 2026. It accompanies the accepted complete proof corpus and preserves its human-source and programme provenance. It has no newly assigned deposited DOI.

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Continuation of 18 September 2026

The following sections give the complete new weighted-conductor, transport, equilibrium, attained-fibre, covariance and joint-return calculations. Original coordinates, source orders, masses, conductor coefficients and endpoint signs are retained throughout. Older equation names identify their complete proofs in the accompanying cumulative sources. The programme origin and deposited attribution remain those documented in [H19, H20]; this continuation was derived and checked with AI assistance within that programme. Mathematical human-source citations remain attached to their actual uses.

1 A polynomial forward bound for the full weighted conductor

The original Split-Zero programme and the deposited source notes retain their existing attribution [H19, H20]. The new calculation in this section acts on the literal SGT10 matrix. It improves its forward bound at ranks of order a^2 . Its exact triangular determinant then controls every inverse exterior rank and supplies a two-sided four-endpoint comparison for the entire conductor block.

1.1 Original spaces, mass, phases and matrix

Fix the original hypothetical simple quartet, its original period and logarithm branch. Retain $0 < \delta < 1/2$, $\gamma > 2$, and all original coefficients a_{uw} , including repeated contributions. Put

$$\begin{aligned} a &= 4l + 1 \geq 9, \quad q = (a + 1)^2, \quad q' = (a - 7)^2, \quad \beta_{n,u,w} = n/2 + (2u - n)\delta + i(2w - n)\gamma, \\ E_A(z) &= \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \quad \mu_n = E_A^{(n)}(0), \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \quad A_{\text{abs}} = \sum_{u,w=0}^8 |a_{uw}|, \\ s &\in \{1, a\}, \quad b = s/2, \quad c_0 = a/2, \quad c_1 = a/2 - 4, \quad N = D + v, \quad D \geq 0. \end{aligned} \tag{WCF1}$$

The symbol and its coefficients are fixed as a varies. The Hermitian inner product is conjugate-linear in its first variable. The original polynomial norm, orthonormal frame, and leading coefficient are

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\ \|P\|_{s,c}^2 &= \int_{\mathbb{R}} |P(c + iy)|^2 dm_s(y), \quad p_{0,s} = 1, \quad p_{1,s} = y, \\ p_{n+1,s} &= yp_{n,s} - n(n-1 + s/2)p_{n-1,s}, \quad h_{n,s} = M_s n! (s/2)_n, \\ \phi_{n,s,c}(S) &= \frac{p_{n,s}((S-c)/i)}{\sqrt{h_{n,s}}}, \quad [S^n]\phi_{n,s,c} = \frac{i^{-n}}{\sqrt{h_{n,s}}}, \quad \rho_n = \sqrt{\frac{n!}{(s/2)_n}}. \end{aligned} \tag{WCF2}$$

The classical dictionary is $p_{n,s}(y) = n! P_n^{(s/4)}(y/2; \pi/2)$, with $dy = 2d(y/2)$. The weight, norm, recurrence and generating function are respectively DLMF (18.19.8)–(18.19.9), (18.22.8), and (18.23.7) [H21]. In the generating function take $\lambda = s/4 > 0$, $x = y/2$, $\varphi = \pi/2$, and $z = t$, $|t| < 1$. The two powers take their branches with value one at $t = 0$; their product is

$$\sum_{n \geq 0} \frac{p_{n,s}(y)}{n!} t^n = (1 + t^2)^{-s/4} e^{y \arctan t}. \tag{WCF3}$$

This also follows by multiplying WCF2's recurrence by $t^n/n!$ and solving $(1 + t^2)\partial_t F = (y - (s/2)t)F$, $F(y, 0) = 1$. Thus the stated generating identity holds at the complex polynomial arguments below, without a change of phase or of the measure.

For $\mathcal{P}_d = \mathbb{C}[S]_{\leq d}$, $\mathcal{P}_{-1} = 0$, the unscaled conductor map is

$$\mathcal{T}_A P(S') = \sum_{u,w=0}^8 a_{uw} P(S' + \beta_{8,u,w}), \quad A_{D,s} : (\mathcal{P}_{D+v}/\mathcal{P}_{v-1}, \|\cdot\|_{s,c_0,\text{quot}}) \longrightarrow (\mathbb{C}[S']_{\leq D}, \|\cdot\|_{s,c_1}). \quad (\text{WCF4})$$

Indeed $\mathcal{T}_A S^n = \sum_{j=v}^n \binom{n}{j} \mu_j S^{n-j}$. For $n \geq v$ the leading coefficient is $\binom{n}{v} \mu_v \neq 0$; for $n < v$ the image is zero. Successive elimination in degree proves that WCF4 is an isomorphism with precisely the displayed kernel. Its quotient frame is represented by $\phi_{v,s,c_0}, \dots, \phi_{D+v,s,c_0}$, the orthogonal complement of that full kernel. Set

$$w(t) = -i \arctan t = \frac{1}{2} \log \frac{1-it}{1+it}, \quad g(t) = t^{-v} e^{-4w(t)} E_A(w(t)), \quad (\text{WCF5})$$

$$g(0) = (-i)^v \mu_v / v!, \quad g(t) = \sum_{j \geq 0} g_j t^j.$$

The singularity at zero is removable. The displayed logarithm is the analytic branch equal to zero there. Apply the finite sum WCF4 to WCF3. Since $c_0 - c_1 = 4$, the result is $t^v g(t)$ times the generating function centred at c_1 . Comparing coefficients, including $n!/\sqrt{h_{n,s}} = \rho_n/\sqrt{M_s}$, proves

$$A_{D,s} = \text{diag}(\rho_0, \dots, \rho_D)^{-1} T_D(g) \text{diag}(\rho_v, \dots, \rho_{D+v}), \quad (A_{D,s})_{ij} = \begin{cases} g_{j-i} \rho_{j+v} / \rho_i, & i \leq j, \\ 0, & i > j. \end{cases} \quad (\text{WCF6})$$

Here $T_D(g)_{ij} = g_{j-i}$ above the diagonal. The two copies of $M_s^{-1/2}$ cancel in this matrix because the source order is the same on both sides. All norms and all endpoint Grams retain their original mass. No scalar $v!/\mu_v$ is inserted in WCF4 or WCF6.

1.2 Forward coefficient and weighted-shift estimates

For $|t| \leq r < 1$ the Cayley ratio $z = (1-it)/(1+it)$ satisfies

$$\Re z = \frac{1-|t|^2}{|1+it|^2} > 0, \quad \frac{1-r}{1+r} \leq |z| \leq \frac{1+r}{1-r}, \quad |\arg z| < \pi/2. \quad (\text{WCF7})$$

The first identity follows by multiplying by $1-i\bar{t}$; the modulus bounds follow from the triangle inequality and its reverse. The branch in WCF5 is therefore exactly the logarithm on the right half-plane. For $\beta_{8,u,w} - 4 = x + iy$, we have $|x| \leq 8\delta$, $|y| \leq 8\gamma$. Consequently

$$\left| e^{(\beta_{8,u,w}-4)w(t)} \right| = \exp\left(\frac{x}{2} \log |z| - \frac{y}{2} \arg z\right) \leq \left(\frac{1+r}{1-r}\right)^{4\delta} e^{2\pi\gamma}. \quad (\text{WCF8})$$

This estimate retains the imaginary shifts through $e^{2\pi\gamma}$; it does not replace them by a real-root symbol.

For $D \geq 1$, choose $r_D = D/(D+1)$ and define

$$C_D = e 2^v A_{\text{abs}} e^{2\pi\gamma} (2D+1)^{4\delta}. \quad (\text{WCF9})$$

On $|t| = r_D$, WCF5 and WCF8 bound $|g(t)|$ by $r_D^{-v} A_{\text{abs}} e^{2\pi\gamma} (2D+1)^{4\delta}$. Cauchy's coefficient integral gives $|g_l| \leq r_D^{-l} \max |g|$. This is the classical Cauchy coefficient formula on the positively oriented circle, with expansion centre zero [H25, eqs. (1.9.30)–(1.9.31)]. For $0 \leq l \leq D$, $r_D^{-l} \leq (1+1/D)^D \leq e$ and $r_D^{-v} \leq 2^v$. Hence

$$|g_l| \leq C_D \quad (0 \leq l \leq D). \quad (\text{WCF10})$$

The l th upper diagonal in WCF6 is a weighted shift whose norm is the maximum of ρ_{i+l+v}/ρ_i . Its nonzero columns have orthogonal images, which proves this norm assertion directly. If $s = a \geq$

9, then $b > 1$ and $\rho_{n+1}/\rho_n = \sqrt{(n+1)/(n+b)} \leq 1$, so this norm is at most one. If $s = 1$, the factors in

$$\frac{\rho_{i+d}^2}{\rho_i^2} = \prod_{t=0}^{d-1} \left(1 + \frac{1/2}{i+t+1/2} \right) \leq \frac{d!}{(1/2)_d} \leq 2d+1 \quad (d \geq 0) \quad (\text{WCF11})$$

decrease with i . For the last bound use induction: it is equality at $d = 0$, and multiplication by $(d+1)/(d+1/2)$ takes $2d+1$ to $2d+2 \leq 2d+3$. The following integral calculation avoids paying for the number of upper diagonals.

For $s = a$, so $b > 1$, give a polynomial $P(z) = \sum p_n z^n$ the norm

$$\|P\|_\rho^2 = \sum_n |p_n|^2 \rho_n^2 = (b-1) \int_0^1 \int_0^{2\pi} |P(\sqrt{x}e^{i\theta})|^2 \frac{d\theta}{2\pi} (1-x)^{b-2} dx. \quad (\text{WCF11a})$$

Angular integration kills unequal monomials. The radial identity $(b-1) \int_0^1 x^n (1-x)^{b-2} dx = n!/(b)_n$ follows either by integration by parts and its value at $n = 0$, or directly from Euler's beta integral, DLMF (5.12.1), at the positive parameters $n+1, b-1$ [H22]. Thus WCF11a is an isometric coefficient realization of these exact positive weights. It is an auxiliary proof space for WCF6; the original Gamma measure WCF2 is unchanged.

The disk-bound step below uses the classical maximum-modulus principle [H25, §1.10(v)]; the multiplier and compression calculations are performed in the displayed auxiliary norm. Multiplication by $g(rz)$, $0 < r < 1$, has norm at most $\max_{|t|=r} |g(t)|$ in WCF11a, since that pointwise modulus bound holds on the whole unit disk. Its power series converges uniformly on the closed unit disk. Orthogonal projection onto degrees at most D contracts the norm. In the orthonormal basis z^n/ρ_n , the transpose of this compressed multiplication matrix is

$$U_r = \text{diag}(\rho_0, \dots, \rho_D)^{-1} T_D(g(r \cdot)) \text{diag}(\rho_0, \dots, \rho_D). \quad (\text{WCF11b})$$

Transposition preserves operator norm. For $R_r = \text{diag}(1, r, \dots, r^D)$, the unscaled matrix is $U = R_r U_r R_r^{-1}$. Hence $\|U\| \leq r^{-D} \max_{|t|=r} |g(t)|$. Finally $A_{D,a} = U \text{diag}(\rho_{i+v}/\rho_i)$, whose last diagonal is a contraction. Taking $r = r_D$ and WCF8 proves $\|A_{D,a}\| \leq C_D$.

For $s = 1$ apply the same angular integration with the unweighted circle norm $\sum |p_n|^2$: multiplication and orthogonal compression give $\|T_D(g)\| \leq r^{-D} \max_{|t|=r} |g(t)|$ by the same radial similarity. In this case ρ_n increases, so the two diagonal operator norms in WCF6 have product ρ_{D+v} , bounded by $\sqrt{2(D+v)+1}$ in WCF11. We have therefore proved the sharper finite estimate

$$\|A_{D,s}\| \leq B_{D,s}^+ := \begin{cases} C_D \sqrt{2(D+v)+1}, & D \geq 1, s = 1, \\ C_D, & D \geq 1, s = a, \\ |\mu_v| \rho_v / v!, & D = 0. \end{cases} \quad (\text{WCF12})$$

The $D = 0$ case is the exact scalar modulus from WCF6; $\rho_0 = 1$. The estimate therefore includes $v = 0$ and the smallest cutoff. In particular, uniformly for $s \in \{1, a\}$ and $D \leq Ca^2$,

$$\log^+ B_{D,s}^+ = O_{\text{actual},C}(\log(a+1)). \quad (\text{WCF13})$$

For $D \geq 1$ this follows by taking the logarithm of WCF12 and using fixed $v, \delta, \gamma, A_{\text{abs}}$. At $D = 0$, $\rho_v \leq 1$ for $s = a$ and is fixed for $s = 1$. This proves the asserted uniformity.

1.3 A finite inverse estimate with every principal part

The following records the inverse terms needed at the negative endpoints. It also prevents an upper-triangular estimate from being applied to a lower-triangular correction. Retain the original

SGT zero-free constants

$$\begin{aligned}
B_0 &= \max\{1, \max_{u,w} |\beta_{8,u,w}|\}, \quad C_v = A_{\text{abs}} B_0^v / |\mu_v|, \\
R_0 &= B_0^{-1} \min\{1, (v+1)/(2eC_v)\}, \quad a_0 = \min\{1/2, \tanh(R_0/2)\}, \\
d_0 &= 1 - \frac{a_0^2}{3(1-a_0^2)}, \quad M_0 = \frac{2v!}{|\mu_v|} d_0^{-v} e^{2R_0}, \quad S_D(r) = \sum_{l=0}^D r^{-l}.
\end{aligned} \tag{WCF14}$$

Here $v!E_A(z)/(\mu_v z^v)$ differs from one by at most $C_v B_0 |z| e^{B_0 |z|} / (v+1)$, by its exponential Taylor series. This is at most $1/2$ for $|z| \leq R_0$. On $|t| \leq a_0$, $|w(t)| \leq R_0/2$ and $|w(t)/t| \geq d_0$, by the arctangent series. Factoring g as $(\mu_v/v!)(w(t))^v e^{-4w} [v!E_A(w)/(\mu_v w^v)]$ proves $|1/g(t)| \leq M_0$ on that entire disk, including zero.

Fix any $0 < \varrho < 1$ with no zero of g on $|t| = \varrho$. Let its interior zeros be ζ , of orders m_ζ , and put

$$\begin{aligned}
H_\varrho &= \sum_{\zeta} m_\zeta, \quad A_{\zeta,h} = \frac{1}{(m_\zeta - h)!} \frac{d^{m_\zeta - h}}{dt^{m_\zeta - h}} \frac{(t - \zeta)^{m_\zeta}}{g(t)} \Big|_{t=\zeta}, \\
f_\varrho(t) &= \frac{1}{g(t)} - \sum_{\zeta} \sum_{h=1}^{m_\zeta} \frac{A_{\zeta,h}}{(t - \zeta)^h}, \quad W_\varrho = \operatorname{arctanh} \varrho, \\
m_{E,\varrho} &= \min_{|t|=\varrho} |E_A(w(t))| > 0, \quad M_\varrho = \frac{\varrho^v e^{4W_\varrho}}{m_{E,\varrho}} + \sum_{\zeta,h} \frac{|A_{\zeta,h}|}{(\varrho - |\zeta|)^h}.
\end{aligned} \tag{WCF15}$$

All these pole sums are finite. Zero is not a pole, and f_ϱ is holomorphic on a neighbourhood of the closed circle and its interior. Its modulus on the circle is at most M_ϱ . The analytic framework here is the classical Laurent expansion and removal of its finite negative part at each pole [H25, §1.10(iii), eqs. (1.10.6)–(1.10.7)]. The weighted matrix decomposition below is calculated separately.

Truncated power-series multiplication gives $T_D(g)^{-1} = T_D(1/g)$. The inverse weighted upper diagonals have ratios ρ_{i+l}/ρ_{i+v} . WCF11 gives an upper bound $\sqrt{2l+1}$ at $s = 1$. At $s = a$, this ratio is at most $\rho_i/\rho_{i+v} \leq \sqrt{(b)_v/v!}$; the last inequality follows by decreasing each factor $(i+t+b)/(i+t+1)$ to its maximum at $i = 0$. Thus set

$$W_{D,s}^- = \begin{cases} \sqrt{2D+1}, & s = 1, \\ \sqrt{(s/2)_v/v!}, & s = a, \end{cases} \quad B_{D,s}^0 = W_{D,s}^- M_0 S_D(a_0). \tag{WCF16}$$

Cauchy's bound on $1/g$ at radius a_0 , followed by the sum of these upper-diagonal norms, proves $\|A_{D,s}^{-1}\| \leq B_{D,s}^0$. The coefficient estimate is again Cauchy's formula [H25, eqs. (1.9.30)–(1.9.31)], now applied to the reciprocal on the displayed zero-free disk, with the original weights retained.

For completeness, the Taylor coefficient of $A/(t - \zeta)^h$ is $c_n = A(-1)^h \zeta^{-n-h} \binom{n+h-1}{h-1}$. For nonnegative indices this is the classical binomial series [H25, eq. (1.10.4)]; the following extension to integer indices and its finite matrix corrections are computed explicitly here. Extend the binomial polynomial to every integer n . The full matrix $(c_{j-i})_{0 \leq i,j \leq D}$ has rank at most h , by expanding its polynomial in i . For all orders at one pole the combined column space is contained in $\operatorname{span}\{(\zeta^i i^d)_i : 0 \leq d < m_\zeta\}$, hence its rank is at most m_ζ . For a lower-diagonal gap $l = i - j \geq 1$, the omitted entry equals zero if $l < h$ and equals $-A\zeta^{l-h} \binom{l-1}{h-1}$ if $l \geq h$. Its cancellation therefore requires a lower-triangular correction with the opposite, positive displayed coefficient.

After multiplying by the actual inverse diagonal factors, that lower-diagonal shift has weight ρ_j/ρ_{j+l+v} . Its norm is at most one for $s = 1$, and at most $\sqrt{(b)_{l+v}/(l+v)!}$ for $s = a$, by the same

decreasing-factor argument. Define the finite, original-pole quantity

$$\begin{aligned}\Theta_{l,s} &= \begin{cases} 1, & s = 1, \\ \sqrt{(s/2)_{l+v}/(l+v)!}, & s = a, \end{cases} \\ Z_{D,s,\varrho} &= \sum_{\zeta,h} |A_{\zeta,h}| \sum_{l=h}^D |\zeta|^{l-h} \binom{l-1}{h-1} \Theta_{l,s}, \\ B_{D,s,\varrho}^{\text{rem}} &= W_{D,s}^- M_\varrho S_D(\varrho) + Z_{D,s,\varrho}, \quad B_{D,s,\varrho}^{\text{cap}} = \min\{B_{D,s}^0, B_{D,s,\varrho}^{\text{rem}}\}.\end{aligned}\tag{WCF17}$$

An empty inner sum is zero. The preceding exact principal-part calculation and the coefficient bound on f_ϱ prove a decomposition of the entire inverse into a matrix of rank at most H_ϱ and a remainder of norm at most $B_{D,s,\varrho}^{\text{rem}}$. Diagonal multiplication leaves this rank bound unchanged. No pole, lower-triangular correction, or original coefficient has been removed.

For c orthogonal copies, $c \in \{1, 4\}$, and a rank r plane or attained subquotient, put $h_c = \min\{r, cH_\varrho\}$ and

$$\mathcal{L}_{D,s,r,c}^{\text{wt}}(\varrho) = 2h_c \log^+ B_{D,s}^0 + 2(r - h_c) \log^+ B_{D,s,\varrho}^{\text{cap}}, \quad \mathcal{U}_{D,s,r}^{\text{wt}} = 2r \log^+ B_{D,s}^+.\tag{WCF18}$$

To prove the inverse exterior bound, take the kernel of the exceptional matrix. It has codimension at most cH_ϱ , and on it the full inverse has norm at most the remainder bound. Diagonalizing its positive squared norm shows that every singular value after the first cH_ϱ is at most that bound; all singular values are at most $B_{D,s}^0$. Multiplying the largest r singular values proves the exterior bound in WCF18, with $\log^+$ giving a nonnegative allowance.

The older inverse estimate remains available without alteration. In its literal NCT16–19 notation define

$$\begin{aligned}L_{s,D,v} &= \frac{1}{2} \max\{1, s/2 - 1\} \sum_{n=1}^{D+v} \frac{1}{n}, \quad G_{0,D} = M_0 S_D(a_0), \\ P_\varrho &= \sum_{\zeta,h} \frac{|A_{\zeta,h}|}{(1 - |\zeta|)^h}, \quad V_{\varrho,D} = M_\varrho S_D(\varrho) + P_\varrho, \\ \mathcal{L}_{D,s,r,c}^{\text{old}}(\varrho) &= 4r L_{s,D,v} + 2h_c \log^+ G_{0,D} + 2(r - h_c) \log^+ \min\{G_{0,D}, V_{\varrho,D}\}, \\ \mathcal{L}_{D,s,r,c}^{\text{pole}} &= \inf_{\varrho \text{ admissible}} \min\{\mathcal{L}_{D,s,r,c}^{\text{wt}}(\varrho), \mathcal{L}_{D,s,r,c}^{\text{old}}(\varrho)\}.\end{aligned}\tag{WCF19}$$

The old bound follows from the same pole decomposition before diagonal weighting: summing the lower shifts gives P_ϱ by the binomial series, and each of the two diagonal factors costs at most $e^{L_{s,D,v}}$. The latter follows by writing ρ_n as its product and using $\log(1+x) \leq x$. Applying the exterior argument then gives exactly WCF19. Thus taking the displayed minimum and infimum keeps two independently proved finite bounds. Admissible radii exist arbitrarily close to one because zeros are isolated. The infimum is finite and nonnegative and need not be attained.

1.4 The full determinant controls every inverse exterior rank

There is a stronger estimate that uses the entire original triangular matrix and needs no separation of its poles. Put $d = D + 1$ and $M_{D,s} = \max\{1, B_{D,s}^+\}$. Triangularity of WCF6 proves the exact complex determinant identity

$$\det A_{D,s} = g(0)^d \prod_{i=0}^D \frac{\rho_{i+v}}{\rho_i}.\tag{WCF19a}$$

It retains the complete phase $g(0) = (-i)^v \mu_v / v!$. For $s = 1$ the product of positive ratios is at least one. For $s = a$ its reciprocal satisfies

$$\begin{aligned} \log \prod_{i=0}^D \frac{\rho_i}{\rho_{i+v}} &= \frac{1}{2} \sum_{i=0}^D \sum_{t=0}^{v-1} \log \left(1 + \frac{s/2 - 1}{i + t + 1} \right) \\ &\leq \frac{1}{2} (s/2 - 1) v \sum_{j=1}^d \frac{1}{j}. \end{aligned} \quad (\text{WCF19b})$$

For $v = 0$ both sums are empty. Define the nonnegative finite quantity

$$J_{D,s} = d \log^+ \frac{1}{|g(0)|} + \frac{1}{2} (s/2 - 1) v \sum_{j=1}^d \frac{1}{j}. \quad (\text{WCF19c})$$

Equations WCF19a–b prove $|\det A_{D,s}|^{-1} \leq e^{J_{D,s}}$.

Let $B = (A_{D,s}^\top)^{\oplus c}$, of dimension cd , with singular values $\sigma_1 \geq \dots \geq \sigma_{cd} > 0$. Its inverse exterior norm is

$$\left\| \bigwedge^r B^{-1} \right\| = \frac{\sigma_1 \cdots \sigma_{cd-r}}{\sigma_1 \cdots \sigma_{cd}} = \frac{\|\bigwedge^{cd-r} B\|}{|\det B|} \leq M_{D,s}^{cd-r} e^{cJ_{D,s}} \quad (0 \leq r \leq cd). \quad (\text{WCF19d})$$

The first equality multiplies the r largest inverse singular values; the denominator is exactly the modulus of the full determinant. Thus this estimate includes every direction of the original inverse, including all directions caused by every symbol zero. It has imposed no zero-free unit disk or bound on its pole count.

For $r > 0$ set

$$\mathcal{L}_{D,s,r,c}^{\det} = 2((cd - r) \log M_{D,s} + cJ_{D,s}), \quad \mathcal{L}_{D,s,r,c}^* = \min\{\mathcal{L}_{D,s,r,c}^{\text{pole}}, \mathcal{L}_{D,s,r,c}^{\det}\}. \quad (\text{WCF19e})$$

At $r = 0$ define both allowances to be zero, as the actual empty exterior norm and determinant are one. This sharpens rather than replaces the preserved finite principal-part bounds. For fixed $C_{\text{cut}} > 0$, if $D \leq C_{\text{cut}} a^2$, $c \leq 4$, $s \in \{1, a\}$, then WCF13 and WCF19c give

$$\mathcal{L}_{D,s,r,c}^* \leq \mathcal{L}_{D,s,r,c}^{\det} = O_{\text{actual}, C_{\text{cut}}}(a^2 \log(a+1)) = o(aq) \quad (0 \leq r \leq c(D+1)). \quad (\text{WCF19f})$$

Indeed $cd - r \leq 4(D+1)$, the first term is $O(a^2 \log(a+1))$, and $J_{D,s} = O_{\text{actual}, C_{\text{cut}}}(a^2 + a \log(a+1))$. The bounds hold uniformly over every retained rank. No limiting choice of radius is needed for WCF19f.

1.5 The original lower evaluation space and the conductor Gram

Now take the original cutoffs $N \geq q - 1$ and $D = N - v$. Define the full observation matrices, with the indicated rows and fixed root order,

$$(\mathbf{E}_-)_{n,(u,w)} = \phi_{n,s,c_1}(\beta_{a-8,u,w}), \quad 0 \leq n \leq D, \quad (\mathbf{E}_+)_{n,(u,w)} = \phi_{n,s,c_0}(\beta_{a,u,w}), \quad 0 \leq n \leq N. \quad (\text{WCF20})$$

The lower dimension is q' . Since $q - q' = 16a - 48 \geq 96 > v$, $D \geq q' - 1$, so the lower matrix has full column rank by its nonzero leading coefficients and the lower Vandermonde. The upper matrix likewise has rank q . Let the full original conductor matrix C_a be defined by collecting every original shift:

$$\sum_{u,w} (C_a^\top \eta)_{uw} e^{\beta_{a,u,w} z} = E_A(z) \sum_{u,w=0}^{a-8} \eta_{uw} e^{\beta_{a-8,u,w} z}. \quad (\text{WCF21})$$

The identity $\beta_{a-8,u,w} + \beta_{8,r,t} = \beta_{a,u+r,w+t}$ proves the formula, retaining all terms with the same resulting root. The map $C_a^\top$ is injective: a vanishing product makes the lower exponential sum zero on an open set, hence everywhere, and its distinct-root Vandermonde forces $\eta = 0$.

Complex-linear functionals transform by ordinary transpose. In fact evaluating $\mathcal{T}_A \phi_{v+j,s,c_0}$ at the lower roots and using WCF6 gives exactly

$$\boxed{\mathbf{E}_+ C_a^\top = \begin{bmatrix} 0_{v \times q'} \\ A_{D,s}^\top \mathbf{E}_- \end{bmatrix}, \quad K_{C,N,s} = \overline{C}_a(\mathbf{E}_+^* \mathbf{E}_+) C_a^\top = \mathbf{E}_-^* \overline{A_{D,s}} A_{D,s}^\top \mathbf{E}_-.} \quad (\text{WCF22})$$

The first v rows vanish because those polynomials lie in the full kernel of $\mathcal{T}_A$. This is the exact map from the original lower evaluation space to the conductor-multiple subspace. Its lower Gram is $K_{-,D,s} = \mathbf{E}_-^* \mathbf{E}_- > 0$ and its upper Gram is the literal $K_{C,N,s} > 0$ of TAC5. No new lower source order is introduced: the two orders remain 1 and a , even though the lower root packet has degree $a - 8$. The shift of its centre is exactly four.

For a full-column-rank matrix E of rank r and an invertible B , the Gram determinant is the squared norm of the wedge of its columns. Applying $\bigwedge^r B$ to this wedge and then applying $\bigwedge^r B^{-1}$ to its image proves the two directed determinant bounds. The transpose has the same singular values as its matrix: if $A = U\Sigma V^*$ then $A^\top = \bar{V}\Sigma U^\top$. Using WCF18–19 in WCF22 therefore proves

$$\boxed{-\mathcal{L}_{D,s,q',1}^* \leq e_{N,s}^C := \log \det K_{C,N,s} - \log \det K_{-,N-v,s} \leq \mathcal{U}_{D,s,q'}^{\text{wt}}} \quad (\text{WCF23})$$

The same proof applies to restrictions and attained quotients. To see the precise map, write $K \subset Y$ in the original space, identify Y/K with $Y \cap K^\perp$, and identify BY/BK with $BY \cap (BK)^\perp$. The induced map and its inverse are the restrictions of B and B^{-1} followed by orthogonal projections. Each projection contracts every exterior norm, which follows in an orthonormal basis from its zero or unit singular values. Thus the same bounds use the actual quotient rank, with the full original denominator retained. The determinant/wedge identity is classical; compare Lyons, §4, (4.1) [H3]. The exact maps, weights and their bounds have been proved here. For the exterior power of an orthogonal projection, compare also Lyons's Lemma 4.1 [H3, §4, Lemma 4.1, preprint p. 12]. The actual quotient maps and complementary inverse comparison remain the explicit calculations above, not assertions imported from that lemma.

For any rank $r \leq C_r a^2$, $D \leq C_{\text{cut}} a^2$, both original orders, and $c = 1, 4$, WCF13 gives the new uniform forward conclusion

$$\mathcal{U}_{D,s,r}^{\text{wt}} = O_{\text{actual}, C_{\text{cut}}, C_r}(a^2 \log(a+1)) = o(aq). \quad (\text{WCF24})$$

It applies in particular to the complete q' -dimensional conductor restriction, as well as to all earlier rank- $O(a)$ native subquotients.

1.6 The four signs and the remaining calculation

Retain the four actual endpoints and the TAC dual orientation:

$$N_i = (q-1, q, 2q-1, 2q)_i, \quad D_i = N_i - v, \quad \mathcal{R}f = -f(N_0) - f(N_1) + f(N_2) + f(N_3). \quad (\text{WCF25})$$

Set $U_i^C = \mathcal{U}_{D_i,s,q'}^{\text{wt}}$, $L_i^C = \mathcal{L}_{D_i,s,q',1}^*$, and $\mathcal{C}_{a,s}^- = \mathcal{R} \log \det K_{-,N-v,s}$. Subtracting the two low endpoint inequalities WCF23 and adding the two high endpoint inequalities gives the exact finite receiver

$$\boxed{\mathcal{C}_{a,s}^- - U_0^C - U_1^C - L_2^C - L_3^C \leq \mathcal{C}_{a,s} \leq \mathcal{C}_{a,s}^- + L_0^C + L_1^C + U_2^C + U_3^C.} \quad (\text{WCF26})$$

Here $\mathcal{C}_{a,s} = \mathcal{R} \log \det K_{C,N,s}$ is the exact conductor term in TAC10 and TAC15. Every U_i^C and L_i^C is $O_{\text{actual}}(a^2 \log(a+1)) = o(aq)$, by WCF19f and WCF24. Consequently the full conductor return transfers to the literal lower-root covariance return with a proved two-sided error on that scale. The receiver respects all four signs and every actual correlated endpoint.

The finite endpoint dimensions give a sharper upper-directed statement. At the two low endpoints $d_i = D_i + 1$ satisfies $d_0 - q' = m$ and $d_1 - q' = m + 1$, where $m = 16a - 48 - v$. Thus

$$\begin{aligned} \mathcal{C}_{a,s} - \mathcal{C}_{a,s}^- &\leq 2m \log M_{D_0,s} + 2(m+1) \log M_{D_1,s} + 2J_{D_0,s} + 2J_{D_1,s} \\ &\quad + 2q' \log M_{D_2,s} + 2q' \log M_{D_3,s}. \end{aligned} \quad (\text{WCF26a})$$

This follows from WCF26 by using the determinant bound only on the two negatively signed low endpoints. Its first line is $O_{\text{actual}}(q + a \log(a+1))$. Its second line is at most $32\delta q \log a + O_{\text{actual}}(q)$ for $s = a$, and at most $(32\delta + 4)q \log a + O_{\text{actual}}(q)$ for $s = 1$. To verify the coefficients, use $D_2, D_3 = 2(a+1)^2 - v + O(1)$ in WCF9 and WCF12: $\log M_{D_i,a} \leq 8\delta \log a + O_{\text{actual}}(1)$ and $\log M_{D_i,1} \leq (8\delta + 1) \log a + O_{\text{actual}}(1)$. There are two upper terms, each multiplied by $2q' \leq 2q$. These are upper bounds on an actual signed correction, not assigned values of independent endpoint determinants.

For the analytic quotient of rank $m = q - q' - v = 16a - 48 - v$, replace q' by m in WCF18–19. Its new forward allowance is $O(a \log(a+1))$. The retained old inverse estimates at rank m have the already proved NCT21 radius-infimum limit $o(aq)$: at any fixed admissible radius their limsup divided by a^3 is bounded by a constant times $\log(1/\varrho)$, while the exceptional rank is fixed; only afterwards let ϱ approach one. Thus the TAC10a–11 analytic correction may use the smaller forward constants without changing its existing two-sided conclusion. The complete conductor has rank q' and is addressed instead by WCF26.

Finally let $F_{L_a}^{(s)}$, $\delta_{a,s}^{\text{ar}}$, $\Xi_{a,s}$, $\mathcal{T}_{a,s}^{\leq R}$, $\mathcal{J}_{a,s,R}$, and $W_a, P_{a,\pm}$ be the unchanged terms defined by TAC9–15. Substitution into TAC15 gives the exact identity

$$\begin{aligned} J_a &= \mathcal{C}_{a,s}^- + \mathcal{R}e_{N,s}^C + F_{L_a}^{(s)} + \delta_{a,s}^{\text{ar}} + \Xi_{a,s} + \mathcal{T}_{a,s}^{\leq R} + \mathcal{J}_{a,s,R} \\ &\quad + W_a - P_{a,-} - P_{a,+}. \end{aligned} \quad (\text{WCF27})$$

In particular the upper complete-action estimate receives $L_0^C + L_1^C + U_2^C + U_3^C$ with coefficient one, and WCF26a retains its more precise endpoint dimensions. The bound $o(aq)$ does not make this correction negligible relative to TAC20's actual benchmark $4q \log(\delta q(a+1)/2) = 12q \log a + O_{\text{actual}}(q)$. The explicit $q \log a$ coefficients and the coupled actual terms must therefore be retained. The next quantitative calculation is the transferred lower-root covariance return together with the surviving native correlated tail and arithmetic correction in WCF27; the generic inverse transport estimate is now proved. WCF22 specifies its literal lower space, source order, centre and shifted cutoffs. WCF15–17 retain every pole and lower-triangular correction when finer signed control is needed. The original independent degree- k proper-source matrices and all CPQ complements continue to obey their existing exact receiving maps; this degree- a calculation has not substituted the target packet for that proper source.

2 Uniform all-rank transport for the original conductor

Fix the original simple quartet, period, branch and conductor coefficients. The programme origin retains its original attribution [H19, H20]. Put

$$\begin{aligned} a &= 4\ell + 1 \geq 9, \quad 0 < \delta < \tfrac{1}{2}, \quad \gamma > 2, \quad q = (a+1)^2, \quad Q = (a-7)^2, \\ \beta_{n,u,w} &= \tfrac{n}{2} + (2u-n)\delta + i(2w-n)\gamma, \quad E_A(z) = \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \\ \mu_j &= E_A^{(j)}(0), \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \quad A_{\text{abs}} = \sum_{u,w=0}^8 |a_{uw}|, \\ s &\in \{1, a\}, \quad b = s/2, \quad c_0 = a/2, \quad c_1 = a/2 - 4, \quad D \in \mathbb{Z}_{\geq 0}, \quad N = D + v, \quad d = D + 1. \end{aligned} \quad (\text{LT1})$$

The actual coefficients, v, δ, γ are fixed as a varies. The inner product is conjugate-linear in its first argument. Retain the complete original measure and all its factors:

$$\begin{aligned}
dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\
\|P\|_{s,c}^2 &= \int_{\mathbb{R}} |P(c + iy)|^2 dm_s(y), \quad p_{0,s} = 1, \quad p_{1,s} = y, \\
p_{n+1,s} &= yp_{n,s} - n(n-1 + s/2)p_{n-1,s}, \quad h_{n,s} = M_s n! (s/2)_n, \\
\phi_{n,s,c}(S) &= p_{n,s}((S-c)/i)/\sqrt{h_{n,s}}, \quad [S^n]\phi_{n,s,c} = i^{-n}/\sqrt{h_{n,s}}, \quad \rho_n = \sqrt{n!/(s/2)_n}.
\end{aligned} \tag{LT2}$$

These are the original orthonormal frames. To check the classical dictionary with its mass, take $p_{n,s}(y) = n!P_n^{(s/4)}(y/2; \pi/2)$ in the Meixner–Pollaczek orthogonality formulas, DLMF (18.19.8)–(18.19.9) [H21]. Their hypotheses hold because $s/4 > 0$. With $y = 2x$, the squared norm is exactly

$$\frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} 2(n!)^2 \frac{2\pi\Gamma(n + s/2)}{2^{s/2}n!} = M_s n! (s/2)_n.$$

The generating identity at these same parameters is DLMF (18.23.7) [H21]:

$$\sum_{n \geq 0} \frac{p_{n,s}(y)}{n!} t^n = (1 + t^2)^{-s/4} e^{y \arctan t}, \quad |t| < 1. \tag{LT3}$$

It also follows directly by multiplying the recurrence in LT2 by $t^n/n!$: the series satisfies $(1 + t^2)\partial_t F = (y - (s/2)t)F$, $F(y, 0) = 1$. The analytic factors and arctan take their branches fixed at zero.

For $\mathcal{P}_j = \mathbb{C}[S]_{\leq j}$ and $\mathcal{P}_{-1} = 0$, the literal conductor is

$$(\mathcal{T}_A P)(S') = \sum_{u,w=0}^8 a_{uw} P(S' + \beta_{8,u,w}), \quad A_{D,s} : (\mathcal{P}_N/\mathcal{P}_{v-1}, \|\cdot\|_{s,c_0,\text{quot}}) \longrightarrow (\mathbb{C}[S']_{\leq D}, \|\cdot\|_{s,c_1}). \tag{LT4}$$

Indeed $\mathcal{T}_A S^n = \sum_{j=v}^n \binom{n}{j} \mu_j S'^{n-j}$. The leading coefficient for $n \geq v$ is $\binom{n}{v} \mu_v \neq 0$, and all polynomials of degree below v vanish. Elimination by degree proves surjectivity and that the full kernel is $\mathcal{P}_{v-1}$. The quotient is represented isometrically by $\phi_{v,s,c_0}, \dots, \phi_{D+v,s,c_0}$. Set

$$w(t) = -i \arctan t = \frac{1}{2} \log \frac{1 - it}{1 + it}, \quad g(t) = t^{-v} e^{-4w(t)} E_A(w(t)), \quad g(0) = (-i)^v \mu_v / v!. \tag{LT5}$$

The zero-order definition of v removes the singularity at zero. Apply $\mathcal{T}_A$ to LT3 at $y = (S - c_0)/i$. Since $c_0 - c_1 = 4$, its image is $t^v g(t)$ times the same generating function at $y = (S' - c_1)/i$. Comparing coefficients, including $n!/\sqrt{h_{n,s}} = \rho_n/\sqrt{M_s}$, gives the exact matrix

$$A_{D,s} = [\text{diag}(\rho_i)_{i=0}^D]^{-1} T_D(g) \text{diag}(\rho_{i+v})_{i=0}^D, \quad T_D(g)_{ij} = \begin{cases} g_{j-i}, & i \leq j, \\ 0, & i > j. \end{cases} \tag{LT6}$$

Both copies of M_s remain in the original norms; they cancel only in this matrix connecting two frames with the same original order. No scalar $v!/\mu_v$ is inserted.

2.1 The finite analytic constants in LRC6

Define, exactly as in the incoming calculation,

$$\begin{aligned}
g_b(t) &= (1 + t^2)^4 g(t), \quad F_0 = 4 + 4\sqrt{\delta^2 + \gamma^2}, \\
B_0^b &= 2^{v+8} A_{\text{abs}} e^{2\pi\gamma}, \quad B_2^b = B_0^b \{F_0(F_0 + 1) + 4vF_0 + 4v(v+1)\}.
\end{aligned} \tag{LT7}$$

The superscript on B_0^b, B_2^b only prevents collision with WCF14's different zero-free-disk constant B_0 ; their numerical values are the incoming LRC6 constants, without alteration. For $\lambda_{uw} = (\beta_{8,u,w} - 4)/2$, put $\alpha_{uw} = 4 + \lambda_{uw}$ and $\beta_{uw}^b = 4 - \lambda_{uw}$. Then

$$t^v g_b(t) = H_b(t) := \sum_{u,w} a_{uw} (1-it)^{\alpha_{uw}} (1+it)^{\beta_{uw}^b}, \quad (\text{LT8})$$

$$|\Re \lambda_{uw}| \leq 4\delta < 2, \quad |\Im \lambda_{uw}| \leq 4\gamma, \quad |\alpha_{uw}|, |\beta_{uw}^b| \leq F_0.$$

For $|t| < 1$, $z = (1-it)/(1+it)$ satisfies $\Re z = (1-|t|^2)/|1+it|^2 > 0$. Each $1 \pm it$ has positive real part, and the branches fixed at zero obey $\log z = \log(1-it) - \log(1+it)$ throughout the disk. For nonnegative integers j, k with $j+k \leq 2$, the real exponents in $(1-it)^{\alpha_{uw}-j} (1+it)^{\beta_{uw}^b-k}$ are positive: the smallest possible one is $2-4\delta > 0$. Their sum is $8-j-k$, while the imaginary contribution to its modulus is $\exp[-\Im \lambda_{uw} \arg z] \leq e^{2\pi\gamma}$. Consequently

$$|(1-it)^{\alpha_{uw}-j} (1+it)^{\beta_{uw}^b-k}| \leq 2^{8-j-k} e^{2\pi\gamma}. \quad (\text{LT9})$$

For $f = (1-it)^\alpha (1+it)^\beta$ its first derivative has coefficients $-i\alpha, i\beta$, and its second derivative has coefficients $-\alpha(\alpha-1), 2\alpha\beta, -\beta(\beta-1)$. LT9 therefore proves

$$\begin{aligned} |H_b| &\leq 2^8 A_{\text{abs}} e^{2\pi\gamma}, \\ |H'_b| &\leq 2^7 (2F_0) A_{\text{abs}} e^{2\pi\gamma}, \\ |H''_b| &\leq 2^6 \{2F_0(F_0+1) + 2F_0^2\} A_{\text{abs}} e^{2\pi\gamma} \leq 2^8 F_0(F_0+1) A_{\text{abs}} e^{2\pi\gamma}. \end{aligned} \quad (\text{LT10})$$

On $1/2 \leq |t| < 1$, use $g_b = t^{-v} H_b$ and

$$g_b'' = t^{-v} H_b'' - 2vt^{-v-1} H'_b + v(v+1)t^{-v-2} H_b.$$

It follows that $|g_b| \leq B_0^b$ and $|g_b''| \leq B_2^b$ on this annulus. Both functions are holomorphic on the disk, since H_b has a zero of order v at zero. Applying the maximum-modulus principle on each circle of radius $r \in [1/2, 1)$ proves

$$\boxed{|g_b(t)| \leq B_0^b, \quad |g_b''(t)| \leq B_2^b \quad (|t| < 1).} \quad (\text{LRC6})$$

If $g_b(t) = \sum_{n \geq 0} b_n t^n$, Cauchy's integral formula on $|t| = r < 1$, followed by $r \uparrow 1$, gives

$$|b_0|, |b_1| \leq B_0^b, \quad |b_n| \leq B_2^b / (n(n-1)) \quad (n \geq 2). \quad (\text{LT11})$$

This argument has retained every interior zero of g .

2.2 Exact matrix factorization and every exterior rank

Let S_D be the upper shift, with entries 1 in positions $(i, i+1)$, and set

$$R = \text{diag}(\rho_i)_{i=0}^D, \quad W = R^{-1} S_D R, \quad V_v = \text{diag}(\rho_{i+v}/\rho_i)_{i=0}^D, \quad U_b = R^{-1} T_D(g_b) R. \quad (\text{LT12})$$

For any two series analytic at zero, $T_D(fh) = T_D(f)T_D(h)$: the (i, j) entry is exactly the finite convolution for coefficient $j-i$, and all entries below the diagonal vanish. Since $S_D^{D+1} = 0$, all power-series evaluations at S_D and at W are finite. The identity $g = g_b(1+t^2)^{-4}$ and LT6 thus give

$$\boxed{A_{D,s} = U_b (I + W^2)^{-4} V_v.} \quad (\text{LRC7})$$

This fixes the position of V_v on the right. It is never commuted through the other two matrices.

The nonzero images of different coordinate vectors under W^n are orthogonal, so its norm is the maximum of ρ_{i+n}/ρ_i . At $s = a$, each factor $(i+j+1)/(i+j+s/2)$ is at most one, hence $\|W^n\| \leq 1$ and $\|V_v\| \leq 1$. At $s = 1$,

$$\frac{\rho_{i+n}^2}{\rho_i^2} = \prod_{j=0}^{n-1} \frac{i+j+1}{i+j+1/2} \leq \frac{n!}{(1/2)_n} \leq 2n+1. \quad (\text{LT13})$$

The first inequality follows because every factor decreases with i . For the second, induction starts at $n = 0$; multiplying $2n+1$ by $(n+1)/(n+1/2)$ gives $2n+2 \leq 2n+3$. In particular $\|V_v\| \leq \sqrt{2v+1}$. Now $U_b = \sum_{n=0}^D b_n W^n$. For $n \geq 2$,

$$\frac{\sqrt{2n+1}}{n(n-1)} \leq \sqrt{10} n^{-3/2}, \quad \sum_{n=2}^{\infty} n^{-3/2} \leq \int_1^{\infty} x^{-3/2} dx = 2.$$

LT11–13 prove the original finite constant

$$\|U_b\| \leq H := B_0^b(1 + \sqrt{3}) + 2\sqrt{10}B_2^b. \quad (\text{LT14})$$

Put

$$H_s = \begin{cases} \max\{1, H\sqrt{2v+1}\}, & s = 1, \\ \max\{1, H\}, & s = a, \end{cases} \quad \kappa_s = \begin{cases} 1 + \sqrt{8/3}, & s = 1, \\ 2, & s = a. \end{cases} \quad (\text{LT15})$$

Writing $Q_D = I + W^2$, triangularity gives $\det Q_D = 1$. For $s = 1$, the decreasing-factor argument gives the sharper $\|W^2\| \leq \rho_2/\rho_0 = \sqrt{8/3}$, including $D < 2$ when $W^2 = 0$. For $s = a$ it gives $\|W^2\| \leq 1$. Thus $\|Q_D\| \leq \kappa_s$ in both cases.

For any invertible complex $n \times n$ matrix T with singular values $\sigma_1 \geq \dots \geq \sigma_n > 0$, diagonalizing its positive squared norm, or taking its singular-value decomposition, gives

$$\|\bigwedge^r T\| = \prod_{j=1}^r \sigma_j, \quad \|\bigwedge^r T^{-1}\| = \frac{\|\bigwedge^{n-r} T\|}{|\det T|} \quad (0 \leq r \leq n). \quad (\text{LT16})$$

The second equality multiplies the largest inverse singular values; its denominator is the full determinant, not a truncated product. Apply it to $Q_D^{\oplus c}$, where $c \in \{1, 4\}$, whose dimension is cd and determinant one. Exterior functoriality and submultiplicativity give

$$\|\bigwedge^r ((Q_D^{-4})^{\oplus c})\| \leq \|\bigwedge^r ((Q_D^{-1})^{\oplus c})\|^4 = \|\bigwedge^{cd-r} (Q_D^{\oplus c})\|^4 \leq \kappa_s^{4(cd-r)}. \quad (\text{LT17})$$

This is valid at $r = cd$, where the right side equals one. For a complex matrix $A = U\Sigma V^*$, $A^T = \bar{V}\Sigma U^T$ is another singular-value decomposition; ordinary transposition therefore preserves every exterior norm. Apply LRC7 to c orthogonal copies, use LT14 for the first factor and LT13 for the last, and then LT17. For $1 \leq r \leq cd$ this proves exactly

$$\log \|\bigwedge^r ((A_{D,s}^T)^{\oplus c})\| \leq F_{D,s,r,c} := r \log H_s + 4(cd-r) \log \kappa_s. \quad (\text{LRC8})$$

Set $F_{D,s,0,c} = 0$, the exact logarithm of the empty exterior norm.

LT6 also proves the full complex determinant identity

$$\det A_{D,s} = ((-i)^v \mu_v / v!)^d \prod_{i=0}^D \frac{\rho_{i+v}}{\rho_i}. \quad (\text{LT18})$$

The phase is retained in this equality. At $s = 1$ the product of positive ratios is at least one. At $s = a$,

$$\log \prod_{i=0}^D \frac{\rho_i}{\rho_{i+v}} = \frac{1}{2} \sum_{i=0}^D \sum_{j=0}^{v-1} \log \left(1 + \frac{s/2-1}{i+j+1} \right) \leq \frac{1}{2}(s/2-1)v \sum_{j=1}^d \frac{1}{j}. \quad (\text{LT19})$$

Here the elementary inequality $\log(1+x) \leq x$ applies to each nonnegative x ; the sums are empty if $v = 0$. Define $\log^+ x = \max\{0, \log x\}$, $H_d^{\text{harm}} = \sum_{j=1}^d 1/j$, and retain WCF's exact finite allowance

$$J_{D,s} = d \log^+(1/|g(0)|) + \frac{1}{2}(s/2 - 1)_+ v H_d^{\text{harm}}. \quad (\text{LT20})$$

LT18–19 give $|\det A_{D,s}|^{-1} \leq e^{J_{D,s}}$. Apply LT16 to the whole matrix $B = (A_{D,s}^\top)^{\oplus c}$ and LRC8 to its complementary rank. For $1 \leq r \leq cd$,

$$\log \left\| \bigwedge^r ((A_{D,s}^\top)^{\oplus c}) \right\| \leq I_{D,s,r,c} := cJ_{D,s} + F_{D,s,cd-r,c}. \quad (\text{LRC9})$$

Set $I_{D,s,0,c} = 0$. This does not assert that the displayed positive-rank formula extends to rank zero. At full rank it gives $I_{D,s,cd,c} = cJ_{D,s}$; at $D = 0$ the original matrix is the scalar $g(0)\rho_v$. Thus the smallest cutoff, $v = 0$, empty exterior powers and full exterior powers are all included. No assumption on absence or number of interior zeros of g has entered these estimates.

There is an explicit constant for the claimed uniform order. Suppose $D \leq C_{\text{cut}} a^2$ with fixed $C_{\text{cut}} \geq 0$, and put

$$\begin{aligned} K &= C_{\text{cut}} + 1, & G &= \log^+(1/|g(0)|), & H_* &= \max\{1, H\sqrt{2v+1}\}, & \kappa_* &= 1 + \sqrt{8/3}, \\ L_* &= \max\{\log H_*, 4 \log \kappa_*\}, & J_* &= KG + \frac{v}{4}(3 + \log K). \end{aligned} \quad (\text{LT21})$$

Then $d \leq Ka^2 \leq Kq$ and $H_d^{\text{harm}} \leq 1 + \log d \leq 1 + \log K + 2 \log a$. Since $\log a \leq a$, $a \leq q$ and $a^2 \leq q$, LT20 implies $J_{D,s} \leq J_* q$. Moreover the expression in LRC8 is bounded by cdL_* . Therefore for every retained rank, including zero,

$$0 \leq F_{D,s,r,c} \leq cKL_*q, \quad 0 \leq I_{D,s,r,c} \leq c(J_* + KL_*)q. \quad (\text{LT22})$$

All constants depend only on the actual conductor and the displayed cutoff constant. These inequalities prove uniform $O(q)$ bounds at all ranks and both original source orders, with no rank restriction beyond $0 \leq r \leq c(D+1)$.

2.3 The literal lower covariance and every attained quotient

Now impose the original endpoint domain $N \geq q - 1$, $D = N - v$. Let $\mathbf{E}_-$ and $\mathbf{E}_+$ be the full original observation matrices in the unchanged root order:

$$(\mathbf{E}_-)_{n,(u,w)} = \phi_{n,s,c_1}(\beta_{a-8,u,w}), \quad 0 \leq n \leq D, \quad (\mathbf{E}_+)_{n,(u,w)} = \phi_{n,s,c_0}(\beta_{a,u,w}), \quad 0 \leq n \leq N. \quad (\text{LT23})$$

The roots are distinct because their real and imaginary parts recover the two indices. Since $q - Q = 16a - 48 \geq 96 > v$, we have $D \geq Q - 1$. The leading coefficients in LT2 and the Vandermonde determinant prove that $\mathbf{E}_-$ has rank Q and $\mathbf{E}_+$ has rank q . Define the original conductor matrix C_a by collecting every term in the identity

$$\sum_{u,w} (C_a^\top \eta)_{uw} e^{\beta_{a,u,w} z} = E_A(z) \sum_{u,w=0}^{a-8} \eta_{uw} e^{\beta_{a-8,u,w} z}. \quad (\text{LT24})$$

This uses exactly $\beta_{a-8,u,w} + \beta_{8,r,t} = \beta_{a,u+r,w+t}$. All contributions producing the same root are retained. If the left side is zero, the lower exponential sum vanishes on an open set where $E_A \neq 0$, and hence everywhere. Its derivative Vandermonde forces $\eta = 0$; thus $C_a^\top$ is injective.

For a lower functional Λ , complex linearity gives $\Lambda(\mathcal{T}_A \phi_{v+j,s,c_0}) = \sum_i (A_{D,s})_{ij} \Lambda(\phi_{i,s,c_1})$. This proves the ordinary-transpose square, including the vanishing first v rows,

$$\mathbf{E}_+ C_a^\top = \begin{bmatrix} 0_{v \times Q} \\ A_{D,s}^\top \mathbf{E}_- \end{bmatrix}, \quad K_{-,D,s} = \mathbf{E}_-^* \mathbf{E}_- > 0, \quad K_{C,N,s} = \mathbf{E}_-^* \overline{A_{D,s}} A_{D,s}^\top \mathbf{E}_- > 0. \quad (\text{LT25})$$

This is the original WCF22 map with the unchanged orders $s = 1, a$, the lower centre $a/2 - 4$, and the shifted cutoff $N - v$.

For independent columns $e_1, \dots, e_r$, the determinant of their Gram matrix is $\|e_1 \wedge \dots \wedge e_r\|^2$. This follows by expanding both sides as the determinant of their pairwise inner products; it is the classical exterior Gram identity, for example Lyons, §4, (4.1) [H3]. Apply $\bigwedge^r B$ to this wedge and $\bigwedge^r B^{-1}$ to the image. LRC8–9 give the two directed determinant bounds. Taking $E = E_-$, $B = A_{D,s}^\top$ and $r = Q$ proves

$$\boxed{-2I_{D,s,Q,1} \leq e_{N,s}^C := \log \det K_{C,N,s} - \log \det K_{-,N-v,s} \leq 2F_{D,s,Q,1}.} \quad (\text{LRC10})$$

For completeness, the same argument controls every compatible attained quotient, rather than only a restriction. Let $K_0 \subset Y \subset H$ be subspaces of a finite-dimensional Hilbert space and $B : H \rightarrow H'$ an invertible linear map. Put $Y_0 = Y \cap K_0^\perp$ and $Y_1 = BY \cap (BK_0)^\perp$. Every affine fibre modulo K_0 has a unique minimum-norm element in Y_0 , by its orthogonal decomposition. The induced map and its inverse in these minimum representatives are exactly

$$\bar{B} = P_{Y_1} B|_{Y_0}, \quad \bar{B}^{-1} = P_{Y_0} B^{-1}|_{Y_1}. \quad (\text{LT26})$$

Indeed subtraction in the first formula lies in BK_0 and subtraction in the second lies in K_0 , proving that the compositions are identities on quotient classes and hence on their unique minimum representatives. Projections and isometric inclusions contract every exterior norm, as is immediate from their zero and unit singular values. It follows that each exterior norm of $\bar{B}$ and its inverse is bounded by the corresponding exterior norm of B and B^{-1} . For any common quotient coordinate frame of rank r , the quotient Gram determinant ratio therefore lies in

$$-2I_{D,s,r,c} \leq \log \frac{\det G_1}{\det G_0} \leq 2F_{D,s,r,c} \quad (\text{LT27})$$

when $B = (A_{D,s}^\top)^{\oplus c}$. A succession of restrictions and quotients reduces to the full preimage pair $K_0 \subset Y$ in the original ambient space: the union of the affine fibres minimized at successive steps is exactly the final affine fibre. Its minimum is attained by the same orthogonal projection. Thus the complete denominators remain, and no extra factor is paid for the dimension of a denominator or the number of steps.

Retain $N_i = (q - 1, q, 2q - 1, 2q)_i$, $D_i = N_i - v$ and $\mathcal{R}f = -f(N_0) - f(N_1) + f(N_2) + f(N_3)$. With $f_i^C = F_{D_i,s,Q,1}$ and $i_i^C = I_{D_i,s,Q,1}$, LRC10 gives

$$-2(f_0^C + f_1^C + i_2^C + i_3^C) \leq \mathcal{R}e^C \leq 2(i_0^C + i_1^C + f_2^C + f_3^C). \quad (\text{LT28})$$

The actual cutoffs have $D_i \leq 2(a + 1)^2 \leq 3a^2$ for $a \geq 9$; LT22 proves that both finite allowances are $O_{\text{actual}}(q)$. The precise native Ξ application of LT27, including its full annihilators, coordinate section and original lower Gram denominators, is given next. All earlier independently proved upper and lower allowances may be retained by taking the minimum separately on each directed side. The symbol, its zeros and their pole directions have never been removed from the original map.

2.4 The exact analytic comparison and its four signed endpoints

Retain the actual simple quartet, original period and logarithm branch, all coefficients of the original nonzero conductor, and the original root order. Write

$$\begin{aligned} a &= 4l + 1 \geq 9, \quad q = (a + 1)^2, \quad Q = (a - 7)^2 = q', \\ m &= q - Q - v, \quad N \geq q - 1, \quad D = N - v, \quad d = D + 1, \quad t = N - q + 1, \\ s &\in \{1, a\}, \quad c_0 = a/2, \quad c_1 = a/2 - 4, \quad \beta_{n,u,w} = n/2 + (2u - n)\delta + i(2w - n)\gamma, \\ E_A(z) &= \sum_{r,h=0}^8 a_{rh} e^{\beta_{8,r,h} z}, \quad \mu_j = E_A^{(j)}(0), \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \quad 0 < \delta < 1/2, \quad \gamma > 2. \end{aligned} \quad (\text{XIM1})$$

Thus $t \geq 0$, $q - Q = 16a - 48 \geq 96$, and $m > 0$. All polynomial spaces have complex coefficients. The original Hermitian norm and frame are

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, & M_s &= (2\pi)^{s/2}, \\ p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+s/2)p_{n-1,s}, \\ h_{n,s} &= M_s n! (s/2)_n, \quad \phi_{n,s,c}(S) = p_{n,s}((S-c)/i)/\sqrt{h_{n,s}}, \quad \|P\|_{s,c}^2 = \int_{\mathbb{R}} |P(c+iy)|^2 dm_s(y). \end{aligned} \tag{XIM2}$$

The frame is the original orthonormal frame, with the unchanged Meixner–Pollaczek dictionary and norm [H21, eqs. (18.19.8)–(18.19.9), (18.22.8)]; its leading coefficient is $i^{-n}/\sqrt{h_{n,s}}$. Neither the mass nor this phase is changed. The duals below are complex-linear duals, with the dual Hilbert norms.

Put $\mathcal{P}_j = \mathbb{C}[S]_{\leq j}$, with $\mathcal{P}_{-1} = 0$, and use S' for the target variable. The unscaled conductor map and its matrix in the displayed orthonormal frames are

$$\begin{aligned} \mathcal{T}_A P(S') &= \sum_{r,h=0}^8 a_{rh} P(S' + \beta_{8,r,h}), \\ A &= A_{D,s} : \mathcal{P}_N / \mathcal{P}_{v-1} \longrightarrow \mathbb{C}[S']_{\leq D}, \\ \mathcal{T}_A \phi_{v+j,s,c_0} &= \sum_{i=0}^D A_{ij} \phi_{i,s,c_1}, \quad 0 \leq j \leq D, \\ \rho_n &= \sqrt{n!/(s/2)_n}, \quad w(z) = -i \arctan z, \quad g(z) = z^{-v} e^{-4w(z)} E_A(w(z)) = \sum_{j \geq 0} g_j z^j, \end{aligned} \tag{XIM3}$$

$$g_0 = (-i)^v \mu_v / v!, \quad A_{ij} = \begin{cases} g_{j-i} \rho_{j+v} / \rho_i, & i \leq j, \\ 0, & i > j. \end{cases}$$

These are the literal NCT3–5 and WCF4–6 matrices. Indeed $\mathcal{T}_A S^n = \sum_{j=v}^n \binom{n}{j} \mu_j S'^{n-j}$, with nonzero leading coefficient $\binom{n}{v} \mu_v$ when $n \geq v$. It follows by elimination in degree that the kernel is exactly $\mathcal{P}_{v-1}$ and that A is an isomorphism. The quotient norm uses its original orthogonal representatives $\phi_{v,s,c_0}, \dots, \phi_{N,s,c_0}$. For the matrix entries, multiplying the recurrence in XIM2 by $z^n/n!$ gives the differential equation $(1+z^2)\partial_z G = (y - (s/2)z)G$, $G(y,0) = 1$, whose solution is $G(y,z) = (1+z^2)^{-s/4} \exp(y \arctan z)$. Applying $\mathcal{T}_A$ to this generating identity centred at c_0 multiplies the same identity centred at c_1 by $z^v g(z)$, since $c_0 - c_1 = 4$. Coefficient comparison and $n!/\sqrt{h_{n,s}} = \rho_n/\sqrt{M_s}$ give exactly XIM3.

Here are the full relation and annihilator spaces. Define

$$\chi_a(S) = \prod_{u,w=0}^a (S - \beta_{a,u,w}), \quad \chi_{-}(S') = \prod_{u,w=0}^{a-8} (S' - \beta_{a-8,u,w}). \tag{XIM4}$$

Use brackets $[P]_-$ for the coefficient column of P in $\phi_{0,s,c_1}, \dots, \phi_{D,s,c_1}$, and $[P]_+$ for the coefficient column of its class modulo $\mathcal{P}_{v-1}$ in $\phi_{v,s,c_0}, \dots, \phi_{N,s,c_0}$. Form the matrices

$$\begin{aligned} \mathbf{X}_N &= [[\chi_a S^j]_+]_{0 \leq j < t} \in \mathbb{C}^{d \times t}, \quad \mathbf{R}_N = [[\mathcal{T}_A(\chi_a S^j)]_-]_{0 \leq j < t} = \mathbf{A} \mathbf{X}_N \in \mathbb{C}^{d \times t}, \\ \mathbf{L}_N &= [[\chi_{-} S'^j]_-]_{0 \leq j < d-Q} \in \mathbb{C}^{d \times (d-Q)}. \end{aligned} \tag{XIM5}$$

At $t = 0$, the first two matrices have no columns. The upper relation columns are independent modulo $\mathcal{P}_{v-1}$ because their nonzero linear combinations have degree at least $q > v-1$. The native relation columns have distinct degrees $q+j-v$ and nonzero leading coefficients $\binom{q+j}{v} \mu_v$. The ideal columns are monic of distinct degrees $Q+j$. Thus their respective ranks are $t, t, d-Q$.

For every lower root, the identity $\beta_{a-8,u,w} + \beta_{8,r,h} = \beta_{a,u+r,w+h}$ shows that $\mathcal{T}_A(\chi_a S^j)$ vanishes at that root. Distinctness of the roots follows by comparing real and imaginary parts. Division by the monic χ_- therefore has zero remainder. Consequently $\text{im } \mathbf{R}_N \subset \text{im } \mathbf{L}_N$. For a functional with coordinate column $z = (\Lambda(\phi_{i,s,c_1}))_{i=0}^D$, evaluation on a polynomial with coefficient column p is $z^\top p$. Its norm is $\|z\|_2$. The two exact annihilator spaces are therefore

$$\begin{aligned}\mathcal{Y}_D &= \ker \mathbf{R}_N^\top = (\text{im } \overline{\mathbf{R}_N})^\perp, & \dim \mathcal{Y}_D &= d - t = q - v, \\ \mathcal{X}_D &= \ker \mathbf{L}_N^\top = (\text{im } \overline{\mathbf{L}_N})^\perp, & \dim \mathcal{X}_D &= Q, \\ \mathcal{X}_D &\subset \mathcal{Y}_D, & \dim(\mathcal{Y}_D/\mathcal{X}_D) &= m.\end{aligned}\tag{XIM6}$$

Complex conjugation in the orthogonal-complement expressions is necessary because the algebraic annihilator uses ordinary transpose.

We identify these spaces with the original analytic functionals and lower evaluations, including their frames. Let $\mathbb{V}[n, \beta] = \beta^n$, $0 \leq n < q$, be the upper Vandermonde in the original root order, and set

$$\begin{aligned}\mathcal{N}_v &= \{\nu \in \mathbb{C}^q : (\mathbb{V}\nu)_n = 0 \ (0 \leq n < v)\}, \\ C_{(u,w),(i,j)} &= \begin{cases} a_{i-u,j-w}, & 0 \leq i-u, j-w \leq 8, \\ 0, & \text{otherwise,} \end{cases} \quad C \in \mathbb{C}^{Q \times q}, \\ \mathcal{N}_\nu(z) &= \sum_\beta \nu_\beta e^{\beta z}, \quad \mathcal{N}_{C^\top \eta}(z) = E_A(z) \sum_{u,w=0}^{a-8} \eta_{uw} e^{\beta_{a-8,u,w} z}.\end{aligned}\tag{XIM7}$$

The last identity collects all original shift coefficients. Its zero left side forces the lower exponential sum to vanish wherever $E_A \neq 0$, hence everywhere by analyticity; the lower Vandermonde then forces $\eta = 0$. Thus $C^\top$ is injective and its image lies in $\mathcal{N}_v$.

Let P_v insert the last $q - v$ coordinates into $\mathbb{C}^q$ and put $J_v = P_v^\top \mathbb{V} C^\top$. Retain the original first nonzero Q -row minor $\mathcal{I}$ of J_v , its complementary row set $\mathcal{J}$, and the coordinate insertion $E_{\mathcal{J}}$. Then

$$Z = \mathbb{V}^{-1} P_v E_{\mathcal{J}} \in \mathbb{C}^{q \times m}, \quad [C^\top, Z] \text{ is a frame of } \mathcal{N}_v, \quad F_b = \mathcal{N}_{Z_{e_b}}/E_A.\tag{XIM8}$$

The frame assertion follows by applying $\mathbb{V}$, taking the $\mathcal{I}$ rows first, and then taking the complementary rows. Each F_b is analytic at zero because the first v numerator derivatives vanish.

These are also exactly the two packet annihilators in TAC1g-h. Put $E = \mathbb{C}[S]/(\chi_a)$, $E_- = \mathbb{C}[S]/(\chi_-)$, $L_v = \mathbb{C}[S]_{<v} \subset E$, and $V_A = \ker(\overline{\mathcal{T}}_A : E \rightarrow E_-)$. Root addition proves that this packet map is well defined, and in root-value coordinates it is C . Since $C^\top$ is injective, the map is onto and $\dim V_A = q - Q$. The degree calculation in XIM3 gives $L_v \subset V_A$. The exact complex-linear pairing is

$$\begin{aligned}\Phi : \mathbb{C}^q &\rightarrow E^\vee, & \Phi(\nu)([P]) &= \sum_\beta \nu_\beta P(\beta), \\ \Phi(\mathcal{N}_v) &= \text{ann}(L_v), & \Phi(\text{im } C^\top) &= \text{ann}(V_A), \\ \Psi : \mathcal{N}_v / \text{im } C^\top &\rightarrow (V_A/L_v)^\vee, & \Psi([\nu])([P] + L_v) &= \sum_\beta \nu_\beta P(\beta).\end{aligned}\tag{XIM8a}$$

The upper Vandermonde makes Φ invertible. Its first v moment equations are precisely vanishing on L_v . The conductor dual functionals vanish on V_A , and both that image and $\text{ann}(V_A)$ have dimension Q , proving their equality. Restriction therefore has exactly this kernel. Every functional on V_A/L_v extends linearly from V_A to E while vanishing on L_v , proving that Ψ is onto and hence an isomorphism. At $v = 0$, $L_v = 0$ and the corresponding moment conditions are empty.

For $\nu \in \mathcal{N}_v$, let $\Lambda_{\mathcal{N}_v/E_A}(S^r) = (\mathcal{N}_v/E_A)^{(r)}(0)$. Leibniz's finite identity proves

$$\Lambda_{\mathcal{N}_v/E_A}(\mathcal{T}_A P) = \Lambda_{\mathcal{N}_v}(P), \quad \deg P \leq N.\tag{XIM9}$$

It vanishes for $P = \chi_a S^j$, so the native functional belongs to $\mathcal{Y}_D$. If its first $D + 1$ derivatives vanish, XIM9 makes all numerator derivatives through N vanish. The first q such equations have invertible Vandermonde, so $\nu = 0$. The dimension $q - v$ now proves that these are all of $\mathcal{Y}_D$. Every lower evaluation annihilates the ideal generated by χ_- . Their first Q evaluation rows form an invertible triangular change of the lower Vandermonde, since $D \geq Q - 1$. They are independent, and dimension Q proves that they are all of $\mathcal{X}_D$.

For completeness, the finite analytic observation matrix can be constructed without an unevaluated division of germs. Put $n_{jb} = \sum_{\beta} Z_{\beta b} \beta^j$ and $f_{rb} = F_b^{(r)}(0)$. For $0 \leq r \leq D$, successively set

$$f_{rb} = \frac{n_{v+r,b} - \sum_{j=v+1}^{v+r} \binom{v+r}{j} \mu_j f_{v+r-j,b}}{\binom{v+r}{v} \mu_v}. \quad (\text{XIM10})$$

At $r = 0$ the sum is empty. All denominators are nonzero. This is exactly the derivative product identity, hence retains every original coefficient. Writing $\phi_{n,s,c_1}(S') = \sum_r \kappa_{nr} S'^r$, define

$$\begin{aligned} H &= \mathbf{E}_- \in \mathbb{C}^{d \times Q}, & H_{n,\beta'} &= \phi_{n,s,c_1}(\beta'), \\ J &\in \mathbb{C}^{d \times m}, & J_{nb} &= \sum_{r=0}^n \kappa_{nr} f_{rb}, & \mathcal{X}_D &= \text{im } H, & \mathcal{Y}_D &= \text{im}[H, J]. \end{aligned} \quad (\text{XIM11})$$

Here $[H, J]$ has full column rank $Q + m = q - v$.

Let $\mathbf{E}_+[n, \beta] = \phi_{n,s,c_0}(\beta)$, $0 \leq n \leq N$, and let $\mathbf{E}_+^{\text{red}}$ consist of rows $v, \dots, N$. The upper observation matrix has full column rank because its first q rows are an invertible triangular change of the upper Vandermonde. The first v rows vanish on $\mathcal{N}_v$. The ordinary dual matrix $B = A^\top$ therefore gives the exact equations

$$\begin{aligned} BH &= \mathbf{E}_+^{\text{red}} C^\top, & BJ &= \mathbf{E}_+^{\text{red}} Z, & \mathbf{E}_+ C^\top &= \begin{bmatrix} 0_{v \times Q} \\ BH \end{bmatrix}, & \mathbf{E}_+ Z &= \begin{bmatrix} 0_{v \times m} \\ BJ \end{bmatrix}, \\ B\mathcal{Y}_D &= \ker \mathbf{X}_N^\top, & B\mathcal{X}_D &= \ker(A^{-1} \mathbf{L}_N)^\top = \text{im}(\mathbf{E}_+^{\text{red}} C^\top). \end{aligned} \quad (\text{XIM12})$$

For the first line, apply each functional to the basis equation XIM3 and use XIM9. For the second line, use $\mathbf{R}_N = A \mathbf{X}_N$ and $B = A^\top$; invertibility proves both inclusions are equalities. This proves the full commutative square on actual finite jets. No adjoint and no scalar $\alpha_A = v!/\mu_v$ occurs.

The polynomial source of the analytic quotient is also exact. Restriction of functionals induces isometric isomorphisms

$$\frac{\text{ann}(\text{im } \mathbf{R}_N)}{\text{ann}(\text{im } \mathbf{L}_N)} \simeq \left(\frac{\text{im } \mathbf{L}_N}{\text{im } \mathbf{R}_N} \right)^\vee, \quad \frac{\text{ann}(\text{im } \mathbf{X}_N)}{\text{ann}(\text{im } A^{-1} \mathbf{L}_N)} \simeq \left(\frac{\text{im } A^{-1} \mathbf{L}_N}{\text{im } \mathbf{X}_N} \right)^\vee. \quad (\text{XIM13})$$

The primal metrics first restrict the original polynomial metric and then minimize over the full relation space. To prove the dual metric claim, extend a functional from the larger displayed polynomial subspace by composition with its orthogonal projection. This is the minimum extension, since every other extension differs on the orthogonal complement and the squared norms add. It still vanishes on the relation subspace. All such extensions differ by precisely the displayed denominator annihilator. This proves isometry as well as the algebraic quotient assertion. The primal map is A on these two polynomial subquotients, and XIM12 is its contravariant ordinary dual. Each primal subquotient has dimension $(d - Q) - t = m$. The original packet identification is

$$\begin{aligned} V_A/L_v &\longrightarrow \text{im}(A^{-1} \mathbf{L}_N)/\text{im } \mathbf{X}_N, & [P] + L_v &\longmapsto [[P]_+], \\ Y_N &= \mathbb{C}[S']_{\leq D-Q}/\{\mathcal{T}_A(\chi_a h)/\chi_- : h \in \mathcal{P}_{t-1}\}, \\ \theta_N : V_A/L_v &\longrightarrow Y_N, & [P] + L_v &\longmapsto [\mathcal{T}_A P/\chi_-]. \end{aligned} \quad (\text{XIM13a})$$

for the upper polynomial quotient and its divided native counterpart, respectively. A representative of degree at most N exists since $N \geq q - 1$. Changing it by $\chi_a h$ gives exactly an XIM5 relation,

and changing it by a polynomial of degree below v gives zero. Conversely, every polynomial in $\text{im}(A^{-1}\mathbf{L}_N)$ has a degree-at-most- N representative whose conductor vanishes on the lower roots, hence represents V_A . A zero quotient class differs from a multiple of χ_a by a polynomial of degree below v . This proves both surjectivity and injectivity. Dividing the native relation by χ_- is legitimate by XIM5; its relation map is the original $h \mapsto \mathcal{T}_A(\chi_a h)/\chi_-$. Thus XIM13a is exactly the TAC1d map, including its full relation space and scalar one.

We now give both attained coefficient minima. Put

$$\begin{aligned} K_0 &= H^*H > 0, & K_1 &= H^*B^*BH > 0, & P_0 &= HK_0^{-1}H^*, & P_1 &= BHK_1^{-1}H^*B^*, \\ W_0 &= (I - P_0)J, & W_1 &= (I - P_1)BJ, \\ D_{a,D}^{(s)} &= W_0^*W_0, & \widehat{D}_{a,D}^{(s)} &= W_1^*W_1. \end{aligned} \tag{XIM14}$$

The first metric is the original NCT10 native analytic metric. The second is precisely NCT8/TAC5: setting $K_+ = \mathbf{E}_+^* \mathbf{E}_+$, XIM12 gives

$$\widehat{D}_{a,D}^{(s)} = Z^* [K_+ - K_+ C^\top (\overline{C} K_+ C^\top)^{-1} \overline{C} K_+] Z. \tag{XIM15}$$

For every $x \in \mathbb{C}^m$, completing the two positive squares in the Schur-complement calculation [H23, Appendix A.5.5, p. 650; Appendix C.4.1, p. 673] proves

$$\begin{aligned} x^* D_{a,D}^{(s)} x &= \min_{\eta \in \mathbb{C}^Q} \|Jx + H\eta\|_2^2, & \eta_0(x) &= -K_0^{-1} H^* Jx, \\ x^* \widehat{D}_{a,D}^{(s)} x &= \min_{\eta \in \mathbb{C}^Q} \|B(Jx + H\eta)\|_2^2, & \eta_1(x) &= -K_1^{-1} H^* B^* B Jx. \end{aligned} \tag{XIM16}$$

Both minima retain every lower evaluation and all their cross terms. Each minimizer is unique because K_0, K_1 are positive definite. More explicitly, the difference between the first squared norm and its displayed minimum is $(\eta - \eta_0(x))^* K_0 (\eta - \eta_0(x))$, and the difference for the second is $(\eta - \eta_1(x))^* K_1 (\eta - \eta_1(x))$. These identities follow on expanding, using the two displayed formulas for η_0, η_1 ; each cross term cancels exactly. If either minimum were zero, invertibility of B and independence of $[H, J]$ would imply $x = 0$. Thus both analytic metrics are positive definite. The two minimizing coefficient columns need not coincide.

The induced map on the quotient has the following exact domain, codomain, and inverse:

$$\begin{aligned} Q_0 &= \mathcal{Y}_D \cap \mathcal{X}_D^\perp = \text{im } W_0, & Q_1 &= B\mathcal{Y}_D \cap (B\mathcal{X}_D)^\perp = \text{im } W_1, \\ \overline{B} : Q_0 &\longrightarrow Q_1, & \overline{B} &= P_{Q_1} B|_{Q_0}, & \overline{B}^{-1} &= P_{Q_0} B^{-1}|_{Q_1}, & \overline{B}(W_0 x) &= W_1 x. \end{aligned} \tag{XIM17}$$

The quotient identifications are isometric because each representative has a unique orthogonal minimum lift. Projection after B subtracts an element of $B\mathcal{X}_D$; applying B^{-1} turns that into an element of $\mathcal{X}_D$, which the second projection removes. This proves both inverse identities. It also proves the last equation using XIM14. In the orthonormal minimum-lift frames $W_0(D_{a,D}^{(s)})^{-1/2}$ and $W_1(\widehat{D}_{a,D}^{(s)})^{-1/2}$, the matrix is

$$[\overline{B}] = (\widehat{D}_{a,D}^{(s)})^{1/2} (D_{a,D}^{(s)})^{-1/2}, \quad [\overline{B}]^* [\overline{B}] = (D_{a,D}^{(s)})^{-1/2} \widehat{D}_{a,D}^{(s)} (D_{a,D}^{(s)})^{-1/2}. \tag{XIM18}$$

Indeed $W_1^* B W_0 = W_1^* W_1 = \widehat{D}_{a,D}^{(s)}$, because the discarded part lies in $B\mathcal{X}_D$. Thus the original TAC9 correction is exactly

$$\epsilon_{a,N,s} = \log \det \widehat{D}_{a,D}^{(s)} - \log \det D_{a,D}^{(s)} = 2 \log |\det [\overline{B}]|. \tag{XIM19}$$

Write the established all-rank logarithmic exterior estimates for the actual ambient matrix, including its original weights, as

$$\log \left\| \bigwedge^r ((A_{D,s}^\top)^{\oplus c}) \right\| \leq F_{D,s,r,c}, \quad \log \left\| \bigwedge^r ((A_{D,s}^\top)^{\oplus c})^{-1} \right\| \leq I_{D,s,r,c}, \quad 0 \leq r \leq cd. \tag{XIM20}$$

The F and I in this display are logarithms of exterior norms, without the Gram factor two. Orthogonal projections contract every exterior power: in orthonormal bases their singular values are zero or one. Restrictions and inclusions of Hilbert subspaces also contract exterior powers. Applying XIM17 in both directions and then XIM20 at $c = 1$, $r = m$, proves the finite bound

$$\boxed{-2I_{D,s,m,1} \leq \epsilon_{a,N,s} \leq 2F_{D,s,m,1}.} \quad (\text{XIM21})$$

There is no added Q -rank allowance: the complete denominator has already been retained in the exact quotient map and in both minima. For c copies, the denominator has dimension cQ and the full analytic quotient dimension cm . Any further restriction followed by an attained quotient has its full preimages $K' \subset Y'$ in the ambient c copies; necessarily K' contains the original cQ -dimensional denominator. Nested minimization over these fibres equals minimization over K' , because the union of the original affine fibres is the final affine fibre. XIM17 applies to $K' \subset Y'$ and proves the same determinant interval $[-2I_{D,s,r,c}, 2F_{D,s,r,c}]$ with its actual final rank r , including $r = 0$ with empty determinant one. No intermediate dimension is charged a second time.

Finally retain the actual TAC endpoints and dual sign orientation:

$$N_i = (q-1, q, 2q-1, 2q)_i, \quad D_i = N_i - v, \quad \Xi_{a,s} = -\epsilon_{a,N_0,s} - \epsilon_{a,N_1,s} + \epsilon_{a,N_2,s} + \epsilon_{a,N_3,s}. \quad (\text{XIM22})$$

For these four endpoints, $d_i = (q-v, q+1-v, 2q-v, 2q+1-v)_i$ and $t_i = (0, 1, q, q+1)_i$. The annihilator dimensions remain exactly $q-v$ and Q , and the analytic quotient rank remains m at every endpoint. Set $F_i = F_{D_i,s,m,1}$ and $I_i = I_{D_i,s,m,1}$. Negating the two low-endpoint inequalities in XIM21 and adding the two high-endpoint inequalities gives

$$\boxed{-2(F_0 + F_1 + I_2 + I_3) \leq \Xi_{a,s} \leq 2(I_0 + I_1 + F_2 + F_3).} \quad (\text{XIM23})$$

These are bounds on the same four correlated matrices appearing in TAC9. A separate radius infimum, minimum of established allowances, or improved ambient estimate can be inserted in each F_i or I_i provided it bounds XIM20; no attainment of an infimum is used. The source order remains 1 or a at the lower grid, its centre remains $a/2 - 4$, and its cutoff remains $N_i - v$. In particular no replacement by the order $a - 8$ or by an unshifted cutoff occurs. The explicit uniform bounds LT21–22 of the preceding transport proof apply because $D_i \leq 3a^2$ for $a \geq 9$. Inserting them in XIM23 proves $\Xi_{a,s} = O_{\text{actual}}(q)$ for both original source orders. The identical endpoint bound XIM21 holds at every cutoff $q-1 \leq N \leq 2q$, including each growing-prefix endpoint $N = q + R$ with $0 \leq R < q$, with the same fixed constant.

3 Quantile discrepancy and the original full-polynomial norms

3.1 A uniform midpoint-quantile logarithmic discrepancy

Let $A < B$ be real numbers, and let μ be the probability measure on $[A, B]$ with density

$$d\mu(y) = H(y) \sqrt{(B-y)(y-A)} dy, \quad H \in C^1([A, B]), \quad H(y) > 0.$$

The probability normalization is part of the hypothesis; no scalar rescaling of μ will be made. Put

$$c = \frac{B-A}{2} > 0, \quad m = \frac{A+B}{2}, \quad T(t) = m + ct \quad (-1 \leq t \leq 1).$$

The map $T : [-1, 1] \rightarrow [A, B]$ is an increasing bijection, with inverse $T^{-1}(y) = (y - m)/c$. Its inverse pushforward of μ has the density

$$\rho(t) = h(t) \sqrt{1-t^2}, \quad h(t) = c^2 H(T(t)).$$

Indeed, $dy = c dt$ and $\sqrt{(B - T(t))(T(t) - A)} = c\sqrt{1 - t^2}$, so $d\mu(T(t)) = \rho(t) dt$. In particular, $\int_{-1}^1 \rho(t) dt = 1$, $h > 0$, and $h \in C^1([-1, 1])$. Define

$$F(t) = \int_{-1}^t \rho(s) ds \quad (-1 \leq t \leq 1), \quad q = F^{-1} : [0, 1] \longrightarrow [-1, 1],$$

and the constants

$$Q_h = \frac{\max_{[-1, 1]} h}{\min_{[-1, 1]} h}, \quad M_h = \max_{[-1, 1]} \left| \frac{h'}{h} \right|, \quad K_h = Q_h(1 + M_h).$$

These constants are finite; $Q_h \geq 1$ and $K_h \geq 1$. The function F is continuous and strictly increasing on $[-1, 1]$, with endpoint values 0, 1, because $\rho > 0$ on $(-1, 1)$. Thus its inverse q is well defined and continuous, and $q \in C^2((0, 1))$.

For an integer $n \geq 1$, define the original physical midpoint probability quantiles by

$$u_j = \frac{j - \frac{1}{2}}{n}, \quad y_j = T(q(u_j)), \quad 1 \leq j \leq n.$$

We shall prove, for every real x ,

$$\sum_{j=1}^n \log |x - y_j| \leq n \int_A^B \log |x - y| d\mu(y) + 24K_h^2 + 30Q_h. \quad (1)$$

In fact the proof yields the stronger remainder $16K_h^2 + 24Q_h$. As usual $\log 0 = -\infty$; the right-hand integral is finite for every real x , since the density is bounded and $\log |x - y|$ is locally Lebesgue integrable. Hence a coincidence $x = y_j$ immediately proves (1) in the extended-real sense.

Exact invariance of the discrepancy under the affine map. Let $z = T^{-1}(x) = (x - m)/c$, with x allowed anywhere on $\mathbb{R}$. For each j and each t away from the respective possible zero,

$$\log |x - y_j| = \log c + \log |z - q(u_j)|, \quad \log |x - T(t)| = \log c + \log |z - t|.$$

Consequently the $n \log c$ contributions to the sum and to n times the probability integral agree exactly. The original discrepancy therefore equals

$$\sum_{j=1}^n \log |z - q(u_j)| - n \int_{-1}^1 \log |z - t| \rho(t) dt. \quad (2)$$

The correspondence also proves directly that y_j are precisely the midpoint quantiles of the original measure. It suffices to bound (2); the original coordinates and probability normalization are recovered by the displayed exact map.

The inverse-quantile derivative estimate. For $-1 < t \leq 0$, write $h_{\max} = \max h$ and $h_{\min} = \min h$. Then

$$F(t) \leq h_{\max} \sqrt{2} \int_{-1}^t \sqrt{1 + s} ds = \frac{2\sqrt{2}}{3} h_{\max} (1 + t)^{3/2}, \quad \rho(t) \geq h_{\min} \sqrt{1 + t}.$$

Since $2\sqrt{2}/3 < 1$, it follows that

$$\frac{F(t)}{\rho(t)} \leq Q_h(1 + t) \quad (-1 < t \leq 0). \quad (3)$$

For $0 \leq t < 1$, integrating from t to 1 instead gives

$$1 - F(t) \leq h_{\max} \sqrt{2} \int_t^1 \sqrt{1 - s} ds = \frac{2\sqrt{2}}{3} h_{\max} (1 - t)^{3/2}, \quad \rho(t) \geq h_{\min} \sqrt{1 - t},$$

and hence

$$\frac{1 - F(t)}{\rho(t)} \leq Q_h(1 - t) \quad (0 \leq t < 1). \quad (4)$$

On $(-1, 1)$,

$$\frac{\rho'(t)}{\rho(t)} = \frac{h'(t)}{h(t)} - \frac{t}{1 - t^2}.$$

For $t \leq 0$, using $\min(F(t), 1 - F(t)) \leq F(t)$ and (3), we obtain

$$\min(F(t), 1 - F(t)) \frac{|\rho'(t)|}{\rho(t)^2} \leq Q_h \left(M_h(1 + t) + \frac{-t}{1 - t} \right) \leq Q_h \left(M_h + \frac{1}{2} \right) \leq K_h.$$

For $t \geq 0$, using $\min(F(t), 1 - F(t)) \leq 1 - F(t)$ and (4), the corresponding calculation is

$$\min(F(t), 1 - F(t)) \frac{|\rho'(t)|}{\rho(t)^2} \leq Q_h \left(M_h(1 - t) + \frac{t}{1 + t} \right) \leq Q_h \left(M_h + \frac{1}{2} \right) \leq K_h.$$

In particular, no assumption that $F(0) = 1/2$ was made. Differentiating $F(q(u)) = u$ yields

$$q'(u) = \frac{1}{\rho(q(u))}, \quad \frac{q''(u)}{q'(u)} = -\frac{\rho'(q(u))}{\rho(q(u))^2}.$$

We have therefore proved

$$\left| \frac{d}{du} \log q'(u) \right| \leq \frac{K_h}{\min(u, 1 - u)} \quad (0 < u < 1). \quad (5)$$

A second-derivative bound away from the singular probability. Fix $z \in \mathbb{R}$, and put

$$v = \begin{cases} 0, & z \leq -1, \\ F(z), & -1 < z < 1, \\ 1, & z \geq 1. \end{cases}$$

Thus v is the actual probability coordinate when z is in the support, and the nearest probability endpoint when z is exterior. For $u \in (0, 1) \setminus \{v\}$, set

$$f(u) = \log |z - q(u)|, \quad d(u) = \min\{u, 1 - u, |u - v|\} > 0,$$

and define

$$\ell = \min \left\{ |u - v|, \frac{u}{2K_h}, \frac{1 - u}{2K_h} \right\} > 0.$$

Consider the segment of length ℓ starting at u and directed towards v . Every point w of this segment satisfies $w \geq u/2$ and $1 - w \geq (1 - u)/2$, because $K_h \geq 1$. Integrating (5) along any initial part of this segment therefore gives

$$|\log q'(w) - \log q'(u)| \leq \frac{2K_h |w - u|}{\min(u, 1 - u)} \leq 1.$$

It follows that $q'(w) \geq e^{-1} q'(u)$ on the segment. For $-1 \leq z \leq 1$, $q(v) = z$; for $z < -1$ or $z > 1$,

$$|z - q(u)| \geq |q(v) - q(u)|.$$

In all cases, monotonicity of q shows that the distance on the left is at least the change of q along the chosen segment. Consequently,

$$|z - q(u)| \geq e^{-1} q'(u) \ell, \quad \frac{q'(u)}{|z - q(u)|} \leq \frac{e}{\ell} \leq \frac{2eK_h}{d(u)}.$$

Since

$$f'(u) = -\frac{q'(u)}{z - q(u)}, \quad f''(u) = -\frac{q''(u)}{z - q(u)} - \frac{q'(u)^2}{(z - q(u))^2},$$

equation (5) gives

$$|f''(u)| \leq \frac{K_h}{d(u)} \frac{2eK_h}{d(u)} + \frac{4e^2 K_h^2}{d(u)^2} = \frac{(2e + 4e^2)K_h^2}{d(u)^2} \leq \frac{48K_h^2}{d(u)^2}.$$

The last numerical inequality follows already from $e < 3$, which gives $2e + 4e^2 < 42 < 48$. For completeness, $e = \sum_{k \geq 0} 1/k! < 1 + 1 + \sum_{k \geq 2} 2^{-(k-1)} = 3$.

Precise cell selection and the inverse-square sum. Let

$$I_j = \left[\frac{j-1}{n}, \frac{j}{n} \right], \quad 1 \leq j \leq n.$$

Mark a cell if $j = 1$, if $j = n$, or if $\text{dist}(v, I_j) \leq 1/n$. The endpoint equalities in this definition are included. With $w = nv$, the distance condition is equivalent to

$$j \in [w - 1, w + 2].$$

A closed interval of length 3 contains at most four integers. Together with the first and last cells, at most six distinct cells are marked. This also includes $n = 1$, when its only cell is marked.

Let U be the set of unmarked indices, and for $j \in U$ write

$$e_j = \text{dist}(nv, [j-1, j]), \quad m_j = n \inf_{u \in I_j} d(u) = \min\{j-1, n-j, e_j\}.$$

For these indices $2 \leq j \leq n-1$, $e_j > 1$, and $m_j > 0$. Therefore

$$\frac{1}{m_j^2} \leq \frac{1}{(j-1)^2} + \frac{1}{(n-j)^2} + \frac{1}{e_j^2}.$$

The first two sums over U are each at most 2, since their denominators run over distinct positive integers and

$$\sum_{k=1}^{\infty} \frac{1}{k^2} \leq 1 + \sum_{k=2}^{\infty} \frac{1}{k(k-1)} = 2.$$

Every unmarked cell lies strictly on one side of nv . On the left its distance is $nv - j$; on the right its distance is $j - 1 - nv$. On either side, arrange the distances in increasing order $a_1, \dots, a_N$. The first is greater than 1, and successive distances differ by positive integers; thus $a_k > k$. Each side contributes at most 2 to $\sum_{j \in U} e_j^{-2}$. It follows that

$$\sum_{j \in U} \frac{1}{m_j^2} \leq 2 + 2 + 2 + 2 = 8. \quad (6)$$

If there are no unmarked cells, this sum is zero and the assertion remains valid.

The discrepancy on the unmarked cells. For a twice continuously differentiable function on I_j with $\sup_{I_j} |f''| \leq M$, Taylor's formula about the midpoint u_j gives

$$|f(u) - f(u_j) - f'(u_j)(u - u_j)| \leq \frac{M}{2}(u - u_j)^2.$$

The linear term integrates to zero, and

$$n \int_{I_j} (u - u_j)^2 du = \frac{1}{12n^2}.$$

Hence

$$\left| f(u_j) - n \int_{I_j} f(u) du \right| \leq \frac{M}{24n^2}.$$

For every unmarked cell the preceding second-derivative bound applies on the closed cell and implies $M \leq 48K_h^2 n^2 / m_j^2$. Summing and using (6) proves

$$\sum_{j \in U} \left(f(u_j) - n \int_{I_j} f(u) du \right) \leq 2K_h^2 \sum_{j \in U} \frac{1}{m_j^2} \leq 16K_h^2. \quad (7)$$

Conditional physical densities on any cell. For any cell, marked or otherwise, put

$$a_j = q((j-1)/n), \quad b_j = q(j/n), \quad L_j = b_j - a_j > 0.$$

Its physical probability mass is exactly $1/n$, and the conditional probability density on $[a_j, b_j]$ is

$$p_j(t) = n\rho(t) = \frac{h(t)\sqrt{1-t^2}}{\int_{a_j}^{b_j} h(s)\sqrt{1-s^2} ds}.$$

The function $g(t) = \sqrt{1-t^2}$ is nonnegative and concave on $[-1, 1]$: in the interior its second derivative is $g''(t) = -(1-t^2)^{-3/2} < 0$, and it is continuous at the endpoints. Let $t_0 \in [a_j, b_j]$ be a point at which g is maximal. On each of $[a_j, t_0]$ and $[t_0, b_j]$, concavity places its graph above the chord joining the endpoint values; these chords in turn lie above the corresponding line segments with height zero at a_j, b_j and height $g(t_0)$ at t_0 . Integrating the two triangles (allowing a zero-length triangle when t_0 is an endpoint) gives

$$\int_{a_j}^{b_j} g(t) dt \geq \frac{L_j}{2} \max_{[a_j, b_j]} g.$$

Since this maximum is positive, comparison of h with its global maximum and minimum proves

$$0 \leq p_j(t) \leq \frac{2Q_h}{L_j} \quad (t \in [a_j, b_j]). \quad (8)$$

An interval-logarithm bound for every evaluation point. For any $a < b$, $L = b - a$, and $z \in \mathbb{R}$, set $d = \max\{|z - a|, |z - b|\} > 0$. We claim

$$\int_a^b \log \frac{d}{|z - t|} dt \leq 2L. \quad (9)$$

All these improper integrals converge, and their integrands are nonnegative. If $z \in [a, b]$, let $p = z - a$, $q_0 = b - z$, and $s = \min(p, q_0)$; then $p + q_0 = L$ and $d = \max(p, q_0)$. Using

$$\int_0^r \log(d/t) dt = r(1 + \log(d/r)) \quad (r > 0)$$

and its value 0 at $r = 0$, the integral equals

$$L + s \log(d/s),$$

with the second summand interpreted as zero when $s = 0$. If $s > 0$, the inequality $\log z_0 \leq z_0 - 1$ for $z_0 > 0$ gives

$$L + s \log(d/s) \leq L + d - s = 2d \leq 2L.$$

That elementary logarithm inequality follows, for example, by minimizing $z_0 - 1 - \log z_0$, whose derivative is $1 - z_0^{-1}$. If $s = 0$, the integral is L . For an exterior point, let $r = \text{dist}(z, [a, b]) > 0$, so $d = r + L$. A direct integral of the same primitive gives

$$\int_a^b \log \frac{d}{|z - t|} dt = L - r \log \frac{r + L}{r} \leq L.$$

This proves (9) for interior, endpoint, and exterior points, with no restriction on how close z is to the interval.

The discrepancy on the marked cells and conclusion. Continue with z not equal to a selected quantile, since all such coincidences were already resolved. On a cell I_j let $d_j = \max\{|z - a_j|, |z - b_j|\}$. Because $q(u_j) \in [a_j, b_j]$, $f(u_j) \leq \log d_j$. The exact change of variables $u = F(t)$ and the fact that p_j is a probability density give

$$\begin{aligned} f(u_j) - n \int_{I_j} f(u) du &\leq \log d_j - \int_{a_j}^{b_j} \log |z - t| p_j(t) dt \\ &= \int_{a_j}^{b_j} \log \frac{d_j}{|z - t|} p_j(t) dt \\ &\leq \frac{2Q_h}{L_j} \int_{a_j}^{b_j} \log \frac{d_j}{|z - t|} dt \leq 4Q_h. \end{aligned}$$

The penultimate inequality is legitimate because its logarithmic integrand is nonnegative. At most six cells are marked, so their total contribution is at most $24Q_h$. Combining this with (7) proves

$$\sum_{j=1}^n f(u_j) - n \int_0^1 f(u) du \leq 16K_h^2 + 24Q_h \leq 24K_h^2 + 30Q_h.$$

Finally, $\int_0^1 f(u) du = \int_{-1}^1 \log |z - t| \rho(t) dt$, and (2) returns this result to the original measure and original physical quantiles. This proves (1) for every $n \geq 1$ and every $x \in \mathbb{R}$, including support endpoints and exterior points.

3.2 The original EIQ density and the explicit LRC12 constant

For the original EIQ equilibrium for $V_{\alpha,\beta}(y) = \beta\sqrt{y} - \alpha \log y$, let its positive support endpoints be $A = u^2$ and $B = v_*^2$, with $0 < A < B$ and $\beta > 0$. The exact original density in EIQ10–EIQ11 is

$$\frac{d\mu}{dy} = H(y) \sqrt{(B-y)(y-A)}, \quad H(y) = \frac{\beta}{4\pi^2 y} \int_0^\infty \frac{\sqrt{s}}{(y+s)\sqrt{(A+s)(B+s)}} ds. \quad (10)$$

In the source notation the function called h in EIQ10 is $2\pi H$; the factor $1/(2\pi)$ in EIQ11 gives exactly (10). No factor of 2π , no mass, and no term of the original potential has been changed.

Set

$$I(y) = \int_0^\infty \frac{\sqrt{s}}{(y+s)\sqrt{(A+s)(B+s)}} ds.$$

This integral is finite and positive for $y > 0$. On a compact positive y interval, its integrand is bounded near zero by a constant times $\sqrt{s}$, and near infinity by a constant times $s^{-3/2}$. Its derivative in y is bounded on the same interval near zero by a constant times $\sqrt{s}$ and near infinity by a constant times $s^{-5/2}$. These bounds are integrable and uniform in that compact interval; differentiation under the integral is therefore justified and gives

$$I'(y) = - \int_0^\infty \frac{\sqrt{s}}{(y+s)^2 \sqrt{(A+s)(B+s)}} ds.$$

One may justify this differentiation directly by applying dominated convergence to the difference quotient and the mean-value theorem on a slightly larger compact positive y interval. The same majorants give continuity of both I and I' . In particular $H > 0$ and $H \in C^1([A, B])$.

Define, for this calculation only, the probability measure on $s \in (0, \infty)$

$$d\omega_y(s) = \frac{1}{I(y)} \frac{\sqrt{s} ds}{(y+s)\sqrt{(A+s)(B+s)}}.$$

Its integral equals one by definition; it is an exact expression for the logarithmic derivative of the original I , not a change in the original equilibrium measure. Since $0 < (y + s)^{-1} < y^{-1}$ for $s > 0$, we obtain

$$\frac{H'(y)}{H(y)} = -\frac{1}{y} + \frac{I'(y)}{I(y)} = -\frac{1}{y} - \int_0^\infty \frac{d\omega_y(s)}{y + s}, \quad -\frac{2}{y} \leq \frac{H'(y)}{H(y)} \leq -\frac{1}{y}.$$

Thus H is decreasing. Writing $R = B/A > 1$ and integrating the displayed derivative bound over $[A, B]$ gives

$$\log \frac{H(A)}{H(B)} = - \int_A^B \frac{H'(y)}{H(y)} dy \leq 2 \log \frac{B}{A}, \quad \frac{\max_{[A,B]} H}{\min_{[A,B]} H} \leq R^2.$$

For the exact affine pushforward already established, $h(t) = c^2 H(m + ct)$, and hence

$$Q_h = \frac{\max H}{\min H} \leq R^2, \quad \left| \frac{h'(t)}{h(t)} \right| = c \left| \frac{H'(m + ct)}{H(m + ct)} \right| \leq \frac{2c}{A} = R - 1.$$

It follows that

$$K_h = Q_h(1 + M_h) \leq R^2(1 + R - 1) = R^3.$$

Substitution in (1) proves the exact constant asserted in LRC12:

$$\sum_{j=1}^n \log |x - y_j| \leq n \int_A^B \log |x - y| d\mu(y) + C_{\text{disc}}, \quad C_{\text{disc}} = 24(B/A)^6 + 30(B/A)^2. \quad (11)$$

This holds for the original EIQ measure and its original midpoint probability quantiles, simultaneously for every positive integer n and every real x . All endpoint and exterior cases have already been covered by the preceding proof. The stronger constant $16(B/A)^6 + 24(B/A)^2$ is also proved, but is not needed to preserve the stated downstream formulas.

3.3 Original full-polynomial comparison and single-norm audit

The scalar estimates below retain the real coordinate, the source order, the original opposite roots, and the complete Gamma measure. Their providers in the cumulative source are EIQ1–33 and ECL1–14, restated with the exact measure conventions in GRF1–5 and GRE9–31. In particular,

$$d\sigma(y) = \sigma(y) dy = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \sigma(\mathbb{R}) = \sqrt{2\pi},$$

$$c_\Gamma e^{-\pi|y|/2}(1 + |y|)^{-1/2} \leq \sigma(y) \leq C_\Gamma e^{-\pi|y|/2}(1 + |y|)^{-1/2}, \quad c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}. \quad (\text{EQ1})$$

No probability renormalization of this measure is made. GRE13–16 prove its mass from the beta integral and its Fourier transform; GRE10–11 derive these exact pointwise constants from Binet's integral formula.

The exact coordinate map and the original roots. For a grid of degree b , put $Q_b = (b+1)^2$, $c_b = b/2$, and

$$\chi_b(S) = \prod_{a,d=0}^b (S - c_b - (2a - b)\delta - i(2d - b)\gamma).$$

Here the packet is the stipulated simple quartet, so that every displayed factor occurs once. The polynomial

$$\mathcal{Q}_b(y) = i^{-Q_b} \chi_b(c_b + iy) = \prod_{a,d=0}^b (y - (2d - b)\gamma + i(2a - b)\delta) \quad (\text{EQ2})$$

is monic in the same real coordinate y . Its roots occur in opposite pairs, with their full multiplicities, by the involution $(a, d) \mapsto (b-a, b-d)$, and have modulus at most $b\sqrt{\delta^2 + \gamma^2}$. For a monic degree- r polynomial $P(S)$ the map

$$\tilde{P}(y) = i^{-r} P(c_b + iy), \quad P(S) = i^r \tilde{P}((S - c_b)/i) \quad (\text{EQ3})$$

is a bijection of the two monic affine fibres. The leading coefficients are one in both directions; expansion in powers proves that the inverse formula holds on the whole fibre. The pointwise product gains the phase i^{Q_b+r} , of modulus one, so the norm is unchanged. This explains the real-coordinate notation in every norm below without deleting an affine phase.

Keep $s = 1$ or the original upper degree a , even when $b = a - 8$, and put

$$\ell_s = (s - 1)/4, \quad n = Q_b + \ell_s, \quad R = \max\{b\sqrt{\delta^2 + \gamma^2}, 2\ell_s + 1/2\}, \quad \epsilon_* = 2^{-16}.$$

The exact recurrence of the original Gamma density is

$$dm_s(y) = \beta_s \prod_{t=0}^{\ell_s-1} (y^2 + (2t + 1/2)^2) d\sigma(y), \quad \beta_s = \frac{(2\pi)^{s/2}}{\sqrt{2}\Gamma(s/2)}. \quad (\text{EQ4})$$

In the original source convention the density before the recurrence is

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2-1}}{\Gamma(s/2)} \frac{|\Gamma(s/4 + iy/2)|^2}{2\pi} dy.$$

Indeed $s/4 = 1/4 + \ell_s$, and each use of the classical recurrence $\Gamma(z + 1) = z\Gamma(z)$ [H22, eq. (5.5.1)] produces $|1/4 + t + iy/2|^2 = (y^2 + (2t + 1/2)^2)/4$; the source normalization therefore gives the exact multiplier

$$\frac{(2\pi)^{s/2} 2^{s/2-1-2\ell_s}}{\Gamma(s/2)} = \frac{(2\pi)^{s/2}}{\sqrt{2}\Gamma(s/2)}.$$

The empty product for $s = 1$ is one.

Let

$$\mathcal{M}(y) = |\mathcal{Q}_b(y)|^2 \prod_{t=0}^{\ell_s-1} (y^2 + (2t + 1/2)^2).$$

An opposite pair contributes $|y^2 - \zeta^2|^2$. The triangle inequality and its reverse therefore give, for real $|y| > R$,

$$(1 - R^2/y^2)^n \leq \frac{\mathcal{M}(y)}{|y|^{2n}} \leq (1 + R^2/y^2)^n. \quad (\text{EQ5})$$

Each Gamma factor is between y^2 and $y^2 + R^2$; its lower bound is also at least $y^2 - R^2$. This proves the exponent $n = Q_b + \ell_s$ in both directions. On the entire real line, including inside the root disk, the upper triangle inequality gives

$$\mathcal{M}(y) \leq (y^2 + R^2)^n. \quad (\text{EQ6})$$

A whole-coefficient inner-interval estimate. Assume

$$n \geq 100, \quad R \leq \epsilon_* n/2. \quad (\text{EQ7})$$

For every complex polynomial P of degree at most $2n$, define

$$H_0(P) = \int_{\mathbb{R}} |y|^{2n} |P(y)|^2 d\sigma(y), \quad H_{\mathcal{M}}(P) = \int_{\mathbb{R}} \mathcal{M}(y) |P(y)|^2 d\sigma(y).$$

The zero polynomial is immediate, so suppose $P \neq 0$. On $[n, 2n]$, the polynomials

$$\phi_d(y) = \sqrt{(2d + 1)/n} L_d(2y/n - 3), \quad 0 \leq d \leq 2n,$$

are an orthonormal basis for the Lebesgue polynomial norm, where L_d is the ordinary Legendre polynomial. The Rodrigues formula used here is the Legendre specialization of the classical formula [H21, eq. (18.5.5) and Table 18.5.1]. Their orthogonality and squared norm $2/(2d+1)$ on $[-1, 1]$ follow from Rodrigues' formula by d integrations by parts. The same formula gives the finite identity

$$L_d(z) = 2^{-d} \sum_{j=0}^d \binom{d}{j}^2 (z-1)^{d-j} (z+1)^j.$$

For $|z| \leq 4$, Vandermonde's identity and $\binom{2d}{d} \leq 4^d$ give

$$|L_d(z)| \leq (5/2)^d \binom{2d}{d} \leq 10^d.$$

If $|y| \leq \epsilon_* n$, then $|2y/n - 3| < 4$. Expanding P in this orthonormal basis and using the complex Cauchy–Schwarz inequality therefore proves

$$|P(y)|^2 \leq \frac{(2n+1)(4n+1)100^{2n}}{n} \int_n^{2n} |P(t)|^2 dt. \quad (\text{EQ8})$$

This bounds every coefficient direction at once.

On the same inner interval, (EQ6)–(EQ7) give

$$\mathcal{M}(y) \leq (5\epsilon_*^2 n^2/4)^n, \quad |y|^{2n} \leq (\epsilon_*^2 n^2)^n \leq (5\epsilon_*^2 n^2/4)^n.$$

On $[n, 2n]$, (EQ1) instead gives

$$H_0(P) \geq c_\Gamma n^{2n} e^{-\pi n} (1+2n)^{-1/2} \int_n^{2n} |P(t)|^2 dt. \quad (\text{EQ9})$$

Using the entire mass $\sqrt{2\pi}$ to bound the inner integration and cancelling (EQ9) proves each of the inequalities

$$\int_{|y| \leq \epsilon_* n} \mathcal{M}(y) |P(y)|^2 d\sigma(y) \leq \eta_n H_0(P), \quad \int_{|y| \leq \epsilon_* n} |y|^{2n} |P(y)|^2 d\sigma(y) \leq \eta_n H_0(P), \quad (\text{EQ10})$$

where

$$\eta_n = \frac{\sqrt{2\pi} (2n+1)(4n+1)\sqrt{1+2n}}{c_\Gamma n} 16^{-n}. \quad (\text{EQ11})$$

The base is valid because $(5/4)e^\pi 10^4 \epsilon_*^2 < 1/16$. For an entirely rational overestimate, use $\pi < 4$, $e < 3$, and $16(5/4)81 \cdot 10^4 < 2^{32}$. Also $\sqrt{2\pi}/c_\Gamma = \sqrt{\pi}e^{7/6} < 8$, whence $\eta_n \leq 240n^2 16^{-n} < 1/2$ already for $n \geq 4$. The last sequence decreases there since $((n+1)/n)^2 < 16$. Thus no inner half-interval, root factor, or coefficient direction has been removed.

The full form and all its original minima. For $0 \leq z < 1$, integration of $1/(1-z)$ gives $-\log(1-z) \leq z/(1-z)$, and $\log(1+z) \leq z$. Consequently (EQ5) implies on $|y| > \epsilon_* n$

$$e^{-B_{\text{root}}} |y|^{2n} \leq \mathcal{M}(y) \leq e^{B_{\text{root}}} |y|^{2n}, \quad B_{\text{root}} = \frac{nR^2}{\epsilon_*^2 n^2 - R^2}.$$

Combine this with (EQ10), retaining the integrals over both half-lines:

$$\boxed{\beta_s e^{-B_{\text{root}}} (1 - \eta_n) H_0(P) \leq \|\chi_b P\|_s^2 \leq \beta_s e^{B_{\text{root}}} (1 + \eta_n) H_0(P)}. \quad (\text{EQ12})$$

Here (EQ3) identifies the notation on the left and right with the original vertical-line coefficient space. For the upper bound the intermediate inequality is $H_{\mathcal{M}} \leq e^{B_{\text{root}}} H_{0,\text{outer}} + \eta_n H_0 \leq e^{B_{\text{root}}} (1 +$

$\eta_n)H_0$. For the lower bound it is $H_{\mathcal{M}} \geq e^{-B_{\text{root}}}H_{0,\text{outer}} \geq e^{-B_{\text{root}}}(1 - \eta_n)H_0$. This proves LRC14 with precisely its constants.

More generally fix any nonempty affine fibre $\mathcal{F}$ in the coefficient space of degree at most $2n$, including a fibre specified by all preceding relation constraints. Since (EQ12) holds at every point of this same fibre, taking its infimum proves

$$\beta_s e^{-B_{\text{root}}}(1 - \eta_n) \inf_{P \in \mathcal{F}} H_0(P) \leq \inf_{P \in \mathcal{F}} \|\chi_b P\|_s^2 \leq \beta_s e^{B_{\text{root}}}(1 + \eta_n) \inf_{P \in \mathcal{F}} H_0(P). \quad (\text{EQ13})$$

The forms are positive definite: a nonzero polynomial cannot vanish on an interval, and the weights are positive almost everywhere. Their finite coefficient Gram matrices are therefore positive definite. Completing a square proves attainment and uniqueness on every closed affine fibre, in particular the monic fibre. Equation (EQ13) is the exact quotient-norm comparison; it does not replace the old Gram inverse by a diagonal matrix.

For the actual discrete family, $R^2 = O_{\delta,\gamma}(a^2)$ and $n = a^2 + O(a)$. Moreover (EQ7) gives

$$B_{\text{root}} \leq \frac{4R^2}{3\epsilon_*^2 n} = O_{\delta,\gamma}(1). \quad (\text{EQ14})$$

This statement keeps $s = 1$ or a at both grids.

Parity and the exact retained-Gamma monic norm. Put $r = 2h + \varepsilon \geq 2$, $h \geq 1$, $\varepsilon \in \{0, 1\}$, and

$$\nu_r^0(n) = \min_{\deg P=r, P \text{ monic}} H_0(P), \quad \alpha = \frac{n + \varepsilon - 3/4}{h}, \quad \beta = \frac{\pi n}{2h}.$$

The even and odd polynomial subspaces are orthogonal for H_0 . The parity opposite to r has no leading term and contributes a nonnegative separate squared norm; its minimizer is zero. On the remaining parity fibre the exact monic bijection is

$$P(y) = n^r (y/n)^\varepsilon p(y^2/n^2), \quad p \in \mathbb{C}[x] \text{ monic}, \quad \deg p = h. \quad (\text{EQ15})$$

Define, for $y > 0$,

$$\mathfrak{r}(y) = \frac{\sigma(y)e^{\pi y/2}\sqrt{y}}{\sqrt{2}}, \quad h_{h,n}(\alpha, \beta) = \min_{\deg p=h, p \text{ monic}} \int_0^\infty |p(x)|^2 e^{-h(\beta\sqrt{x} - \alpha \log x)} \mathfrak{r}(n\sqrt{x}) dx. \quad (\text{EQ16})$$

The two real half-lines in (EQ15) contribute equally. Under $y = n\sqrt{x}$, their combined Jacobian is $n/\sqrt{x}$, and $\sigma(n\sqrt{x}) = \sqrt{2}n^{-1/2}x^{-1/4}e^{-\pi n\sqrt{x}/2}\mathfrak{r}(n\sqrt{x})$. It follows without a limiting step that

$$\boxed{\nu_r^0(n) = \sqrt{2}n^{2(n+r)+1/2}h_{h,n}(\alpha, \beta).} \quad (\text{EQ17})$$

In particular the shift is $-3/4$. The measures in (EQ16) have positive density, finite moments, and a positive definite coefficient Gram matrix, so their monic minima are attained uniquely as above.

Original equilibrium and the entire positive half-line. The accepted EIQ equilibrium for $V(x) = \beta\sqrt{x} - \alpha \log x$ has support $[A, B] = [u^2, v_*^2]$, probability density

$$H(x)\sqrt{(B-x)(x-A)}, \quad H(x) = \frac{\beta}{4\pi^2 x} \int_0^\infty \frac{\sqrt{t}}{(x+t)\sqrt{(A+t)(B+t)}} dt > 0. \quad (\text{EQ18})$$

Its logarithmic potential $P_{\text{eq}}(x) = \int \log|x-y| d\mu(y)$ and original Robin constant satisfy

$$D(x) = V(x) - 2P_{\text{eq}}(x) - \lambda \geq 0 \quad (0 < x < \infty), \quad D(x) = 0 \quad (A \leq x \leq B). \quad (\text{EQ19})$$

The endpoint equations and Robin identity are

$$uK(\kappa) = \frac{\alpha\pi}{\beta}, \quad v_*E(\kappa) = \frac{(\alpha+2)\pi}{\beta}, \quad \kappa^2 = 1 - u^2/v_*^2,$$

$$\lambda = 2(\alpha + 2) - 2\alpha \log((u + v_*)/2) - 2\log((B - A)/4). \quad (\text{EQ20})$$

These are exactly EIQ3–19 and ECL6, rather than a substituted potential. The independent quantile proof accompanying this fragment establishes LRC11–12 directly from (EQ18): for the h midpoint probability quantiles x_i ,

$$\sum_{i=1}^h \log |x - x_i| \leq hP_{\text{eq}}(x) + C_{\text{disc}} \quad (x \in \mathbb{R}), \quad C_{\text{disc}} = 24(B/A)^6 + 30(B/A)^2. \quad (\text{EQ21})$$

It applies in particular to $h = 1$.

From (EQ1),

$$0 < \mathfrak{r}(y) \leq r_+ = C_\Gamma/\sqrt{2}, \quad \mathfrak{r}(n\sqrt{x}) \geq r_- = \frac{c_\Gamma}{\sqrt{2}} \sqrt{\frac{nu}{1+nu}} \quad (A \leq x \leq B). \quad (\text{EQ22})$$

For the full tail, set

$$T_* = \max \left\{ 1, v_*, \left[\frac{8(\alpha + 2)}{\beta} \right]^2, \frac{4|\lambda|}{\beta} \right\}, \quad J_* = T_*^2 + \frac{8}{\beta^2}. \quad (\text{EQ23})$$

If $t \geq v_*$ and $y \in [A, B]$, then $0 \leq t^2 - y \leq t^2$. Hence $P_{\text{eq}}(t^2) \leq 2 \log t$, including $t = v_*$ by integrability of the endpoint logarithm. Thus

$$D(t^2) \geq \beta t - 2(\alpha + 2) \log t - |\lambda|.$$

For $t \geq 1$, $\log t \leq \sqrt{t}$, since $\max_{t \geq 1} (\log t)/\sqrt{t} = 2/e < 1$. The two nonconstant entries in (EQ23) imply for $t \geq T_*$

$$2(\alpha + 2) \log t \leq \beta t/4, \quad |\lambda| \leq \beta t/4, \quad D(t^2) \geq \beta t/2.$$

Because $h \geq 1$ and (EQ19) holds all the way to zero,

$$\int_0^\infty e^{-hD(x)} dx \leq T_*^2 + \int_0^\infty 2te^{-\beta t/2} dt = J_*. \quad (\text{EQ24})$$

This proves the displayed LRC15 cutoff itself. No extra $\log 2$ is needed: the stronger inequality $t^2 - y \leq t^2$ at $t \geq v_*$ replaces the looser estimate $t^2 + y \leq 2t^2$.

The arcsine lower constant, including all complex roots. Put $c_{\text{cap}} = (B - A)/4$ and let $d\nu = dx/(\pi\sqrt{(B - x)(x - A)})$ be arcsine probability. For every complex z ,

$$\int \log |x - z| d\nu(x) \geq \log c_{\text{cap}}, \quad (\text{EQ25})$$

with equality for $z \in [A, B]$. Here is a direct proof. Write $x = (A + B)/2 + d \cos \theta$, $d = (B - A)/2$, and factor $Z^2 - 2(z - (A + B)/2)Z/d + 1 = (Z - z_1)(Z - z_2)$. On $|Z| = 1$, $|x - z| = (d/2)|Z - z_1||Z - z_2|$, while $z_1 z_2 = 1$. The circle mean of $\log |Z - z_j|$ is $\log \max(1, |z_j|)$: integrate the convergent logarithmic power series inside the circle, factor out $|z_j|$ outside it, and use integrability of $|\log |\theta||$ for the boundary limit. Both means are nonnegative; both vanish exactly when the two roots are on the circle, which occurs for $z \in [A, B]$. Symmetry in $\cos \theta$ identifies this circle mean with the arcsine integral.

Factor any complex monic degree- h polynomial root by root. Equation (EQ25) gives $\int \log |p| d\nu \geq h \log c_{\text{cap}}$. Integrating the Euler equality (EQ19), interchanging its logarithmic integrals, and applying equality in (EQ25) gives $\int V d\nu = 2 \log c_{\text{cap}} + \lambda$. All the logarithms are integrable: the equilibrium density is bounded, and arcsine endpoint singularities are integrable against a logarithm. The entropy is exactly

$$\int \log(d\nu/dx) d\nu = -\log(\pi c_{\text{cap}}).$$

Indeed the same angular substitution reduces this to $\pi^{-1} \int_0^\pi \log \sin \theta \, d\theta = -\log 2$; the last identity follows by adding its sine and cosine integrals on $[0, \pi/2]$ and doubling the angle. The corresponding finite circle-weight Jensen inequality is recorded by Szegő [H24, §12.3, eq. (12.3.4), pp. 300–301]; the preceding angular map and entropy computation give its exact interval constants here. Probability Jensen applied to $|p|^2 e^{-hV}/(d\nu/dx)$ now proves

$$\int_A^B |p(x)|^2 e^{-hV(x)} dx \geq \pi c_{\text{cap}} e^{-h\lambda}. \quad (\text{EQ26})$$

Zeros cause only integrable logarithmic singularities, so this applies to every complex monic polynomial. Multiply by (EQ22) and then take the same monic minimum.

For the upper bound use the actual quantile polynomial $p_h(x) = \prod_{i=1}^h (x - x_i)$. Equations (EQ19)–(EQ21) give pointwise on the entire positive half-line

$$|p_h(x)|^2 e^{-hV(x)} \leq e^{2C_{\text{disc}}} e^{-h\lambda} e^{-hD(x)}.$$

Integrating with (EQ22)–(EQ24) proves

$$\boxed{c_- e^{-h\lambda} \leq h_{h,n}(\alpha, \beta) \leq c_+ e^{-h\lambda}, \quad c_- = r_- \pi (B - A)/4, \quad c_+ = r_+ J_* e^{2C_{\text{disc}}}.} \quad (\text{EQ27})$$

This is LRC15 with exactly the stated constants.

Explicit uniform constants when the degree starts at one. The following proof does not assume an upper bound on α . It applies on the whole wedge

$$\alpha \geq 1/2, \quad \pi/3 \leq c := \beta/\alpha \leq \pi. \quad (\text{EQ28})$$

Write $s_0 \in (0, 1)$ for the unique solution

$$\frac{E(\sqrt{1-s_0^2})}{s_0 K(\sqrt{1-s_0^2})} = 5. \quad (\text{EQ29})$$

Existence and uniqueness follow from the endpoint monotonicity in EIQ6; they can also be checked from the derivative just below and the limits 1 at $s = 1$ and $+\infty$ at $s = 0$. For the latter limit, $E \geq 1$ and $sK \rightarrow 0$ by dominated convergence in its defining integral. Use the abbreviations $K_s = K(\sqrt{1-s^2})$, $E_s = E(\sqrt{1-s^2})$, $f(s) = E_s/(sK_s)$. The endpoint equation is

$$f(s) = 1 + 2/\alpha, \quad s = u/v_*.$$

Direct differentiation under the integral and integration of the derivative of

$$\frac{\sin \theta \cos \theta}{\sqrt{\cos^2 \theta + s^2 \sin^2 \theta}}$$

give

$$-f'(s) = \frac{(K_s - E_s)(E_s - s^2 K_s)}{(1-s^2)s^2 K_s^2} > 0. \quad (\text{EQ30})$$

Here $K_s - E_s$ is the integral of $(1-s^2) \sin^2 \theta / \sqrt{\cos^2 \theta + s^2 \sin^2 \theta}$; $E_s - s^2 K_s$ is the same integral with $\cos^2 \theta$. Each is between $(1-s^2)\pi/4$ and $(1-s^2)\pi/(4s)$, whereas $\pi/2 \leq K_s \leq \pi/(2s)$. Since (EQ28) implies $s \geq s_0$, these inequalities yield

$$\frac{1-s^2}{4} \leq -f'(s) \leq \frac{1-s^2}{4s_0^4}.$$

Integration from s to 1 and

$$\frac{1}{16} \leq \frac{s+2}{12(1+s)^2} \leq \frac{1}{6} \quad (0 \leq s \leq 1)$$

give the explicit comparison

$$\frac{(1-s^2)^2}{16} \leq \frac{2}{\alpha} \leq \frac{(1-s^2)^2}{6s_0^4}. \quad (\text{EQ31})$$

In particular this estimate includes $\alpha \rightarrow \infty$ without an asymptotic assumption.

Define the fixed, explicitly evaluable positive numbers

$$u_0 = \frac{2s_0}{\pi}, \quad v_0 = \frac{30}{\pi s_0}, \quad C_0 = \frac{2\sqrt{3}s_0^4}{\pi^2}, \quad C_1 = \frac{900\sqrt{2}}{\pi^2 s_0^2}.$$

The original endpoint formulas, the bounds on K_s, E_s , and $1 + 2/\alpha \leq 5$ prove

$$u \geq u_0, \quad v_* \leq v_0, \quad B/A \leq s_0^{-2}, \quad C_0 \leq c_{\text{cap}}\sqrt{\alpha} \leq C_1. \quad (\text{EQ32})$$

For the last inequality use $c_{\text{cap}} = v_*^2(1-s^2)/4$, (EQ31), and $u_0 \leq v_* \leq v_0$. Put

$$L_w = \max\{|\log u_0|, |\log v_0|\}, \quad L_c = \max\{|\log C_0|, |\log C_1|\}, \quad L_\lambda = \frac{3}{\pi}(12 + 2L_w + 4L_c).$$

For $\alpha \geq 1/2$, $|\log \alpha|/\alpha \leq 2$. Equation (EQ32) then gives $|\log c_{\text{cap}}|/\alpha \leq 2L_c + 1$. The exact Robin formula (EQ20) therefore proves

$$|\lambda|/\beta \leq L_\lambda.$$

Since $\beta \geq \pi/6$, the following are uniform upper bounds for $T_*, J_*, C_{\text{disc}}$:

$$\bar{T} = \max\{1, v_0, (120/\pi)^2, 4L_\lambda\}, \quad \bar{J} = \bar{T}^2 + 288/\pi^2, \quad \bar{C}_{\text{disc}} = 24s_0^{-12} + 30s_0^{-4}. \quad (\text{EQ33})$$

For $n \geq 100$ a uniform lower bound in (EQ22) is

$$\underline{r} = \frac{c_\Gamma}{\sqrt{2}} \sqrt{\frac{100u_0}{1+100u_0}} > 0.$$

Consequently (EQ27) gives the fully explicit uniform interval

$$\log(\underline{r}\pi C_0) - \frac{1}{2} \log \alpha \leq \log h_{h,n}(\alpha, \beta) + h\lambda \leq \log\left(\frac{C_\Gamma}{\sqrt{2}} \bar{J} e^{2\bar{C}_{\text{disc}}}\right). \quad (\text{EQ34})$$

For the exact LRC parameters with $2 \leq r = 2h + \varepsilon \leq 2n$, one has (EQ28), $1 \leq h \leq n$, and $h \geq (r+1)/4$. Since $n + \varepsilon - 3/4 \leq n + 1/4 \leq 5n/4$, this proves

$$\alpha \leq 5n/(r+1) \quad (n \geq 100). \quad (\text{EQ35})$$

It follows from (EQ34)–(EQ35) that

$$|\log h_{h,n}(\alpha, \beta) + h\lambda| \leq C_{\text{unif}} + \frac{1}{2} \log^+ \frac{n}{r+1}, \quad (\text{EQ36})$$

where, for example, one may take

$$C_{\text{unif}} = \max \left\{ |\log(\underline{r}\pi C_0)| + \frac{1}{2} \log 5, \left| \log\left(\frac{C_\Gamma}{\sqrt{2}} \bar{J} e^{2\bar{C}_{\text{disc}}}\right) \right| \right\}.$$

This verifies the scalar uniformity needed in LRC18 even for $h = 1$.

The scalar ranks $r = 0, 1$. The unique reference monic minimizers are $1, y$ by parity, so these norms equal $\mu_M = \int_{\mathbb{R}} |y|^{2M} d\sigma(y)$ with $M = n + r$. Put

$$G_M = \int_0^\infty y^{2M-1/2} e^{-\pi y/2} dy = \frac{\Gamma(2M+1/2)}{(\pi/2)^{2M+1/2}}.$$

The upper bound in (EQ1) immediately gives $\mu_M \leq 2C_\Gamma G_M$. For the lower bound restrict only this estimate, not the definition, to $y \geq 1$, where $(1+y)^{-1/2} \geq y^{-1/2}/\sqrt{2}$. The missing proportion satisfies, by one Gamma integration by parts,

$$\frac{\int_0^1 y^{2M-1/2} e^{-\pi y/2} dy}{G_M} \leq \frac{\int_0^\infty y^{2M-3/2} e^{-\pi y/2} dy}{G_M} = \frac{\pi}{4M-1} < \frac{1}{2} \quad (M \geq 4).$$

Hence

$$(c_\Gamma/\sqrt{2})G_M \leq \mu_M \leq 2C_\Gamma G_M. \quad (\text{EQ37})$$

The actual-root scalar minima are then compared by (EQ13), so these ranks need no singular equilibrium substitution.

3.4 The separate power-exponential reference and its moving centre

The second intake uses the original-coordinate monic minimum

$$\mathfrak{n}(Q, b) = \min_{\deg p=b, p \text{ monic}} \int_{\mathbb{R}} |y|^{2Q} |p(y)|^2 e^{-\pi|y|/2} dy. \quad (\text{EQ38})$$

This is a different reference measure from (EQ16). The following exact map proves the relation needed here, including its different quarter shift and its normalization. Let $b = 2d + \varepsilon \geq 2$, $d \geq 1$, $\varepsilon \in \{0, 1\}$, and retain the positive scale q appearing in the original upper grid. Orthogonality of the parity subspaces for the even weight proves that the unique monic minimizer has parity b . The monic bijection is

$$p(y) = q^b (y/q)^\varepsilon P(y^2/q^2), \quad P \in \mathbb{C}[x] \text{ monic}, \quad \deg P = d.$$

The two half-lines have combined Jacobian $q/\sqrt{x}$ under $y = q\sqrt{x}$. Therefore, exactly,

$$\mathfrak{n}(Q, b) = q^{2(Q+b)+1} \min_{\deg P=d, P \text{ monic}} \int_0^\infty |P(x)|^2 e^{-dV(x)} dx, \quad V(x) = \beta\sqrt{x} - \alpha \log x, \\ \alpha = (Q + \varepsilon - 1/2)/d, \quad \beta = \pi q/(2d). \quad (\text{EQ39})$$

The exponent is $-1/2$, rather than $-3/4$, because no residual Gamma density is inside (EQ38). The coefficient Gram matrices are positive definite and all moments finite, so the minima are attained uniquely. The argument covers $d = 1$.

Let $[u^2, v_0^2]$ be the EIQ support at these exact parameters,

$$W = v_0^2 - u^2, \quad c_{\text{cap}} = W/4, \quad M = \frac{\alpha}{2\pi u^2}, \quad \lambda = 2(\alpha + 2) - 2\alpha \log((u + v_0)/2) - 2 \log c_{\text{cap}}.$$

The EIQ positive formula gives $h_{\text{EIQ}}(x) \leq \alpha/(xuv_0)$. To see the constant directly, replace $1/(x+t)$ by $1/t$ in its positive integral: the endpoint mass identity EIQ9 is

$$\frac{\beta}{2\pi} \int_0^\infty \frac{dt}{\sqrt{t(u^2+t)(v_0^2+t)}} = \frac{\alpha}{uv_0}.$$

Thus

$$\rho(x) = \frac{\sqrt{(v_0^2 - x)(x - u^2)}}{2\pi} h_{\text{EIQ}}(x) \leq \frac{\alpha}{2\pi u\sqrt{x}} \leq M \quad (u^2 \leq x \leq v_0^2), \quad (\text{EQ40})$$

where $(v_0^2 - x)(x - u^2) \leq v_0^2 x$ was used.

The precise clipped-quantile estimate. Take the d midpoint probability quantiles x_i of this equilibrium and put $P_d(x) = \prod_i (x - x_i)$. The empirical distribution function differs from the equilibrium distribution function by at most $1/(2d)$, and hence by at most $1/d$. For a fixed real x , define on the support

$$f_x(y) = \log \max\{|x - y|, 1/d\}.$$

The function is continuous and of bounded variation. If x lies inside the support, there are two monotone pieces, each having variation at most $\log(1 + dW)$; if x lies outside it, the sole monotone piece has that bound. The latter statement follows from $\max(r + W, 1/d)/\max(r, 1/d) \leq 1 + dW$, $r \geq 0$. Integration by parts against the signed difference of the two probability measures has zero endpoint term and gives

$$\sum_{i=1}^d f_x(x_i) \leq d \int f_x d\mu + 2 \log(1 + dW).$$

The clipping error, using (EQ40), satisfies

$$d \int (f_x(y) - \log |x - y|) d\mu(y) \leq dM \int_{-1/d}^{1/d} \log \frac{1/d}{|t|} dt = 2M.$$

Since each nodal logarithm is no larger than its clipped value, including the value $-\infty$ at a root, this proves for every real x

$$\log |P_d(x)| \leq dP_{\text{eq}}(x) + 2M + 2 \log(1 + dW). \quad (\text{EQ41})$$

No unproved smoothness of a clipped kernel is used.

Set

$$C = 2 \log(1 + v_0^2) + |\lambda|, \quad T_* = \max\{1, v_0, [8(\alpha + 2)/\beta]^2, 4C/\beta\}, \quad J_* = T_*^2 + 8/\beta^2. \quad (\text{EQ42})$$

The global EIQ inequality is $D = V - 2P_{\text{eq}} - \lambda \geq 0$ on $(0, \infty)$. Because $C \geq |\lambda|$, the proof of (EQ23)–(EQ24) applies with this larger cutoff. It gives $\int_0^\infty e^{-dD(x)} dx \leq J_*$. Squaring (EQ41), multiplying by e^{-dV} , and integrating yields

$$\min_{\deg P=d, P \text{ monic}} \int_0^\infty |P|^2 e^{-dV} dx \leq J_* e^{-d\lambda + 4M} (1 + dW)^4.$$

The arcsine proof (EQ25)–(EQ26), with the residual Gamma factor absent, gives the lower bound $\pi c_{\text{cap}} e^{-d\lambda}$. This is the same finite Jensen inequality [H24, §12.3, eq. (12.3.4), pp. 300–301], applied to the power-exponential weight in (EQ39). Together with (EQ39) these prove exactly

$$H^-(Q, b) \leq \log \mathbf{n}(Q, b) \leq H^+(Q, b), \quad (\text{EQ43})$$

$$H^-(Q, b) = [2(Q + b) + 1] \log q - d\lambda + \log(\pi c_{\text{cap}}),$$

$$H^+(Q, b) = [2(Q + b) + 1] \log q - d\lambda + \log J_* + 4M + 4 \log(1 + dW).$$

This is the second intake's equation (9), with its stated constants. The bounded quantile theorem (EQ21) additionally permits the stronger upper endpoint

$$H^\sharp(Q, b) = [2(Q + b) + 1] \log q - d\lambda + \log J_* + 2C_{\text{disc}}. \quad (\text{EQ44})$$

Both upper endpoints may be retained and intersected. No new root/Gamma comparison is presumed by this scalar improvement.

Exact scale cancellation and the all-degree error. Define the same centre as the second intake,

$$\mathcal{H}(Q, b) = 2(Q + b) \log Q - \frac{b}{2} \lambda(2Q/b, \pi Q/b). \quad (\text{EQ45})$$

The EIQ dilation identity, including its original Robin constant, is

$$\lambda(\alpha, \beta c_0) = \lambda(\alpha, \beta) + 2(\alpha + 2) \log c_0. \quad (\text{EQ46})$$

One may prove it without differentiating an asymptotic remainder: the map $x \mapsto c_0^{-2}x$ pushes the unique equilibrium at (α, β) to the unique equilibrium at $(\alpha, \beta c_0)$. At a corresponding point, the logarithmic potential changes by $-2 \log c_0$, and the external field changes by $2\alpha \log c_0$. Subtracting twice the potential therefore gives precisely (EQ46). The minimizer is carried correctly because the logarithmic-energy functional changes by a constant independent of the probability measure.

Set $\alpha_0 = 2Q/b$, $\beta_0 = \pi q/b$. Taking $c_0 = q/Q$ in (EQ46) proves the exact cancellation

$$2(Q + b) \log q - \frac{b}{2} \lambda(\alpha_0, \beta_0) = \mathcal{H}(Q, b). \quad (\text{EQ47})$$

In particular the q -versus- Q scale is not an $O(j)$ error. It cancels identically.

For the parity shift, the exact multiplied differences are

$$A = d(\alpha - \alpha_0) = \varepsilon - \frac{1}{2} + \frac{\varepsilon Q}{b}, \quad B = d(\beta - \beta_0) = \frac{\varepsilon \pi q}{2b}. \quad (\text{EQ48})$$

On the straight parameter segment $(\alpha_\theta, \beta_\theta) = (\alpha_0, \beta_0) + \theta(\alpha - \alpha_0, \beta - \beta_0)$, the exact ECL derivatives are

$$\lambda_\alpha = -2 \log((u + v_0)/2), \quad \lambda_\beta = 2(\alpha + 2)/\beta. \quad (\text{EQ49})$$

These include all moving-support terms. On the wedge (EQ28) the first has absolute value at most $2L_w$, and the second at most $30/\pi$. Consequently the fundamental theorem of calculus gives the finite bound

$$\left| d\lambda(\alpha, \beta) - \frac{b}{2} \lambda(\alpha_0, \beta_0) \right| \leq 2L_w |A| + \frac{30}{\pi} |B| + \frac{\varepsilon}{2} |\lambda(\alpha_0, \beta_0)|. \quad (\text{EQ50})$$

This argument requires no smallness of the displacement and applies when $b = 2$ and $d = 1$.

Here are sufficient original-domain checks for the two actual grids. For $j \geq 257$,

$$q = (j + 1)^2, \quad q' = (j - 7)^2, \quad \ell_s \in \{0, (j - 1)/4\}, \quad Q_\pm = q^{(\prime)} + \ell_s.$$

Direct expansion gives

$$\frac{9}{10}q \leq Q_\pm \leq \frac{11}{10}q. \quad (\text{EQ51})$$

For example $q' - (9/10)q = (j^2 - 142j + 489)/10 > 0$ at 257 and increases thereafter; also $\ell_s \leq j/4 \leq q/10$. For the upper degrees $2 \leq b = r \leq q + 32j$, and for the lower degrees $b = \Delta - v + r$, $0 \leq v \leq 80$, $2 \leq r \leq q + 32j$, one has

$$2 \leq b \leq q + 48j - 48 \leq \frac{6}{5}q. \quad (\text{EQ52})$$

The final inequality follows since $q - 240j + 240 = j^2 - 238j + 241 > 0$ at 257 and increases thereafter. The lower degree is positive because $\Delta - v = 16j - 48 - v > 0$. Equations (EQ51)–(EQ52) put both actual parity endpoints and both ideal endpoints in the wedge (EQ28). For instance

$$\alpha \geq 2(0.9q - 0.5)/(1.2q) > 1/2, \quad \frac{\beta}{\alpha} = \frac{\pi q}{2(Q + \varepsilon - 1/2)} \in [\pi/3, \pi].$$

These inequalities follow from $q \geq (258)^2$ and (EQ51). The ideal ratio is $\pi q/(2Q)$ in the same interval. Along their straight segment $\alpha_\theta \geq 1/2$, while $\beta_\theta/\alpha_\theta$ is a positive weighted average of the endpoint ratios, so the entire segment remains in (EQ28).

On this domain, (EQ32)–(EQ33) and (EQ48)–(EQ50) give

$$d\lambda(\alpha, \beta) = \frac{b}{2}\lambda(\alpha_0, \beta_0) + O(q/b). \quad (\text{EQ53})$$

The constants are fixed numbers expressible in the explicitly defined $s_0, u_0, v_0, C_0, C_1, L_w, L_\lambda$. Indeed $|A| \leq 1/2 + Q/b$, $|B| \leq \pi q/(2b)$, and $|\lambda(\alpha_0, \beta_0)| \leq L_\lambda \pi q/b$. No constant depends on the parity, the row, or j .

Furthermore $\alpha = O(q/b)$, $u \geq u_0$, so $M = O(q/b)$. Equation (EQ32) gives $|\log c_{\text{cap}}| \leq L_c + \frac{1}{2}|\log \alpha| = O(\log q)$. The modified tail constant (EQ42) remains uniformly bounded: relative to (EQ33), its only additional entry is at most

$$\frac{8 \log(1 + v_0^2)}{\beta} \leq \frac{48}{\pi} \log(1 + v_0^2).$$

Thus $1 \leq J_* \leq J_0$ for an explicit fixed J_0 . Finally $dW \leq 2bC_1/\sqrt{\alpha} = O(q)$. Substitute these facts, (EQ47), and (EQ53) into (EQ43):

$$\boxed{H^\pm(Q, b) = \mathcal{H}(Q, b) + O(q/b + \log q).} \quad (\text{EQ54})$$

This proves the second intake's equation (12) over every actual degree $2 \leq r \leq q + 32j$, including the fixed small degrees. It does not require r to exceed the old growing prefix.

The actual original-root bounds denoted $b^\pm$ in the second intake are a separate full-form input to its equation (8). Equations (EQ43)–(EQ54) establish exactly the scalar Jensen, quantile, tail, parity and centre ingredients used after that full-form comparison. When using instead the proved retained-Gamma comparison (EQ12), the exact map is (EQ17), not (EQ39); its stronger all-degree finite interval is (EQ27), with uniform error (EQ36).

4 Complete root–Gamma bounds for the two attained-tail monic fibres

Throughout this section retain the original parameters

$$0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad j = 4l + 1 \geq 257, \quad s \in \{1, j\},$$

and the original fixed conductor order v , with $0 \leq v \leq \Delta$. Put

$$\begin{aligned} q &= (j+1)^2, \quad q' = (j-7)^2, \quad \Delta = 16j - 48, \quad m = \Delta - v, \quad c_+ = j/2, \quad c_- = j/2 - 4, \\ D &= \sqrt{\delta^2 + \gamma^2}, \quad A = jD, \quad \epsilon = 2^{-32}, \quad T = \epsilon q, \quad A \leq T/2, \quad 0 \leq r \leq q + 32j. \end{aligned} \quad (\text{ATG1})$$

The letter A here is a real root-radius bound, not the original conductor operator. No parameter or root is changed by this notation. Define

$$\ell_s = (s-1)/4, \quad M_s = (2\pi)^{s/2}, \quad \beta_s = \frac{M_s}{\sqrt{2}\Gamma(s/2)}, \quad \vartheta = \pi/2, \quad h = 1/q. \quad (\text{ATG2})$$

The integers used as monic degrees are $b_+ = r$ and $b_- = m + r$; the two root counts are $n_+ = q$ and $n_- = q'$; and $Q_\pm = n_\pm + \ell_s$.

Here are all finite constants in the comparison:

$$\begin{aligned}
C_L &= \frac{\pi}{\Gamma(3/4)^2}, & C_U &= \frac{\Gamma(1/4)^2}{2\pi}, & K_q &= \frac{\sqrt{2\pi}}{C_L} \frac{(2q+1)^2}{q} e^{-2q}, \\
E_j &= \frac{qA^2}{T^2 - A^2} + \log(1 + K_q), \\
S_\ell &= \sum_{a=0}^{\ell-1} (2a + \tfrac{1}{2})^2 = \frac{\ell(16\ell^2 - 12\ell - 1)}{12}, \\
\rho_0 &= 0, & \rho_\ell &= \frac{S_\ell}{T^2} + \log(1 + K_q) \quad (\ell > 0), \\
L_h &= C_L(\sin h)^{3/2}, & U_h &= C_U(\sin h)^{-1/2}, \\
L_{b,Q} &= b + \sqrt{b(b+2Q)}, & K_{b,Q} &= 2Q + 1 + 2L_{b,Q}, \\
b_{n,b,s}^- &= -E_j + \log L_h - K_{b,n+\ell_s} \log(1 + h/\vartheta), \\
b_{n,b,s}^+ &= E_j + \rho_{\ell_s} + \log U_h + K_{b,n+\ell_s} \log \frac{1}{1 - h/\vartheta}.
\end{aligned} \tag{ATG3}$$

The E_j in both signs is exactly the common GPA root error, with the original symbol k replaced by its attained-tail index j . In particular, the lower-root application still uses the unchanged common outer scale q , cutoff T , and rate displacement h .

We prove below, for $\zeta \in \{+, -\}$, every $b \leq 2q$, and every $f \in \mathbb{C}[S]$ of degree at most b , the whole-space inequalities

$$\begin{aligned}
\beta_s e^{b_{n_\zeta, b, s}^-} \int_{\mathbb{R}} |f(c_\zeta + iy)|^2 |y|^{2(n_\zeta + \ell_s)} e^{-\vartheta|y|} dy &\leq \int_{\mathbb{R}} |\chi_{j_\zeta}(c_\zeta + iy) f(c_\zeta + iy)|^2 dm_s(y) \\
&\leq \beta_s e^{b_{n_\zeta, b, s}^+} \int_{\mathbb{R}} |f(c_\zeta + iy)|^2 |y|^{2(n_\zeta + \ell_s)} e^{-\vartheta|y|} dy,
\end{aligned} \tag{ATG4}$$

where $j_+ = j$, $j_- = j - 8$, and the original source is

$$dm_s(y) = M_s \frac{2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy. \tag{ATG5}$$

No restriction on the zeros, parity, coefficients, or extremality of f occurs in (ATG4).

4.1 The exact original root and coefficient maps

The complete original root polynomials, including every multiplicity, are

$$\begin{aligned}
P_j(y) &= \prod_{a,t=0}^j [y - (2t - j)\gamma + i(2a - j)\delta], \\
P_{j-8}(y) &= \prod_{a,t=0}^{j-8} [y - (2t - j + 8)\gamma + i(2a - j + 8)\delta].
\end{aligned} \tag{ATG6}$$

Their exact original-coordinate identities are $\chi_j(c_+ + iy) = i^q P_j(y)$ and $\chi_{j-8}(c_- + iy) = i^{q'} P_{j-8}(y)$. The respective involutions $(a, t) \mapsto (j - a, j - t)$ and $(a, t) \mapsto (j - 8 - a, j - 8 - t)$ pair every root with its negative. All roots have modulus at most A , and both degrees are even. For fixed b , the map

$$(\Pi_{c,b} f)(y) = i^{-b} f(c + iy), \quad \Pi_{c,b}^{-1} p(S) = i^b p((S - c)/i) \tag{ATG7}$$

is an exact linear isomorphism of the degree-at-most- b coefficient spaces. Its matrix entry in row a , column t is $\binom{t}{a} c^{t-a} i^{a-b}$ for $a \leq t$, and zero otherwise. It maps the original monic degree- b affine fibre bijectively onto the monic degree- b fibre in y . The factors i^{-b} and i^{n_ζ} have absolute value one; they therefore disappear only after the squared norms are formed, without changing any norm or minimum.

4.2 Full source mass and the positive Gamma recurrence

The defining Euler integral is the classical formula recorded by R. A. Askey and R. Roy [H22, eq. (5.2.1)]. For $a > 0$, two such Euler integrals, the substitution $t = e^u$ in their beta integral, and Fourier inversion give

$$\frac{\Gamma(a+ix)\Gamma(a-ix)}{\Gamma(2a)} = \int_{\mathbb{R}} e^{ixu} (2 \cosh(u/2))^{-2a} du, \quad \int_{\mathbb{R}} |\Gamma(a+ix)|^2 dx = 2\pi\Gamma(2a)2^{-2a}. \quad (\text{ATG8})$$

To justify the inversion absolutely, rotate Euler's integration ray through $\text{sgn}(x)\theta$, with $0 < \theta < \pi/2$. The large connecting arc is bounded by a polynomial times an exponential with negative real exponent; the small arc tends to zero because $a > 0$. Taking the modulus on the rotated ray proves

$$|\Gamma(a+ix)| \leq e^{-\theta|x|} \Gamma(a)(\cos \theta)^{-a}. \quad (\text{ATG9})$$

Thus the integrable Fourier transform in (ATG8) may be inverted at $u = 0$. With $a = s/4$ and $y = 2x$, (ATG8) proves $\int dm_s = M_s$, including its full original mass.

Let $d\sigma(y) = |\Gamma(1/4 + iy/2)|^2 dy / (2\pi)$. Integration by parts in Euler's integral gives $\Gamma(z+1) = z\Gamma(z)$, the classical functional equation [H22, eq. (5.5.1)]. Iterating exactly ℓ_s times gives

$$dm_s(y) = \beta_s \prod_{a=0}^{\ell_s-1} (y^2 + (2a + \frac{1}{2})^2) d\sigma(y). \quad (\text{ATG10})$$

Indeed each squared recurrence factor contributes $(y^2 + (2a + 1/2)^2)/4$; since $s/2 = 2\ell_s + 1/2$, the remaining constant is precisely $\beta_s/(2\pi)$ before the Gamma square. For $s = 1$, $\ell_s = 0$ and $\beta_1 = 1$, so $\int d\sigma = \sqrt{2\pi}$.

The reflection identity used here is Euler's reflection formula, as recorded by Askey and Roy [H22, eq. (5.5.3)]. For completeness, the reflection product needed below follows directly from the beta integral $\Gamma(z)\Gamma(1-z) = \int_0^\infty t^{z-1}/(1+t) dt$, $0 < \Re z < 1$. On the keyhole contour with argument in $(0, 2\pi)$, the radial integrals give the multiplier $1 - e^{2\pi iz}$; the residue at -1 is $e^{i\pi(z-1)}$; the two circular contributions tend to zero as $O(r^{\Re z})$ and $O(R^{\Re z-1})$. The residue theorem therefore evaluates that integral as $\pi/\sin(\pi z)$. At $z = 1/4 + iy/2$ it yields

$$|\Gamma(1/4 + iy/2)|^2 |\Gamma(3/4 - iy/2)|^2 = \frac{2\pi^2}{\cosh(\pi y)}. \quad (\text{ATG11})$$

The unrotated Euler integral bounds the second squared factor by $\Gamma(3/4)^2$, and consequently

$$\frac{d\sigma}{dy} \geq C_L e^{-\pi|y|}, \quad \frac{dm_s}{dy} \geq \beta_s q^{2\ell_s} C_L e^{-2\pi q} \quad (q \leq y \leq 2q). \quad (\text{ATG12})$$

Furthermore, without suppressing either original mass,

$$\frac{M_s}{\beta_s q^{2\ell_s}} = \sqrt{2} \Gamma(2\ell_s + 1/2) q^{-2\ell_s} \leq \sqrt{2\pi}. \quad (\text{ATG13})$$

To see the inequality, expand the Gamma value as $\sqrt{\pi} \prod_{a=0}^{2\ell_s-1} (a + 1/2)$ and bound each factor by $s/2 \leq q$.

4.3 A bound on every polynomial direction, including the inner region

For every $p \in \mathbb{C}[y]$ of degree at most b and $0 < T \leq q/2$,

$$\sup_{|y| \leq T} |p(y)|^2 \leq \frac{(b+1)^2 10^{2b}}{q} \int_q^{2q} |p(u)|^2 du. \quad (\text{ATG14})$$

Here is a coefficient-space proof. The Rodrigues polynomials L_a are the classical Legendre polynomials; the formula and norm are recorded by T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt [H21, eq. (18.5.5), Table 18.5.1, and the Legendre and shifted-Legendre rows of Table 18.3.1]. In precisely those conventions, the polynomials $L_a(x) = (2^a a!)^{-1} (d/dx)^a (x^2 - 1)^a$ have squared norm $2/(2a + 1)$ on $[-1, 1]$ and are orthogonal to lower degrees, by a integrations by parts. The boundary derivatives of orders less than a vanish. Their norm follows by pairing the leading coefficient $(2a)!/(2^a (a!)^2)$ against x^a and using $\int_0^1 t^a (1 - t)^a dt = (a!)^2/(2a + 1)!$, itself obtained by repeated integration by parts. Leibniz's rule gives $L_a(x) = 2^{-a} \sum_{t=0}^a \binom{a}{t}^2 (x - 1)^{a-t} (x + 1)^t$. For $|x| \leq 4$, its modulus is at most $(5/2)^a \sum_t \binom{a}{t}^2 \leq 10^a$. Expand $p(qz)$ in the orthonormal basis $\sqrt{2a + 1} L_a(2z - 3)$ on $1 \leq z \leq 2$. For $|z| \leq 1/2$, Cauchy-Schwarz and $\sum_{a=0}^b (2a + 1) = (b + 1)^2$ give (ATG14), after changing variables back to y .

Let P be either polynomial (ATG6), with degree $n \in \{q, q'\}$. For $|y| \geq T$, pairing roots $\omega, -\omega$ gives

$$\left| \log \frac{|P(y)|^2}{|y|^{2n}} \right| \leq -n \log(1 - A^2/T^2) \leq \frac{nA^2}{T^2 - A^2}. \quad (\text{ATG15})$$

This follows from $1 - u \leq |1 - z| \leq 1 + u$ for $|z| \leq u < 1$, and $-\log(1 - u) \leq u/(1 - u)$. For all real y , the same pairing gives $|P(y)|^2 \leq (y^2 + A^2)^n$. Set $B = \int_q^{2q} |y|^{2n} |p(y)|^2 dm_s(y)$. By (ATG12)–(ATG14), each of

$$\int_{|y| < T} |P(y)p(y)|^2 dm_s(y), \quad \int_{|y| < T} |y|^{2n} |p(y)|^2 dm_s(y)$$

is at most κB , where

$$\kappa = \frac{M_s}{\beta_s q^{2\ell_s} C_L} \frac{(b + 1)^2}{q} e^{2\pi q} 10^{2b} \left(\frac{T^2 + A^2}{q^2} \right)^n. \quad (\text{ATG16})$$

The bounds $n \geq q/2$, $b \leq 2q$, $A \leq T/2$ imply

$$\kappa \leq \frac{\sqrt{2\pi}}{C_L} \frac{(2q + 1)^2}{q} \exp\{q(2\pi + 4 \log 10 + \tfrac{1}{2} \log(5/4) + \log \epsilon)\} \leq K_q. \quad (\text{ATG17})$$

For an exact elementary check of the last exponent, $\pi < 4$, $\log 10 < 4 \log 2$, $\log(5/4) < 1/4$, and $\log 2 > 2/3$ give $2\pi + 4 \log 10 + \frac{1}{2} \log(5/4) - 32 \log 2 < 8 - 16 \log 2 + 1/8 < -61/24 < -2$. The inequalities for π and $\log 2$ follow respectively from $\pi/4 = \int_0^1 (1 + x^2)^{-1} dx$ and $\log 2 = 2 \int_0^{1/3} (1 - t^2)^{-1} dt$.

Let I_{out} be the reference integral with $|y|^{2n}$ outside $(-T, T)$. Since $B \leq I_{\text{out}}$, both inner integrals are bounded by $K_q I_{\text{out}}$. The full reference norm lies between I_{out} and $(1 + K_q) I_{\text{out}}$. The full actual norm lies between $e^{-nA^2/(T^2 - A^2)} I_{\text{out}}$ and $e^{nA^2/(T^2 - A^2)} (1 + K_q) I_{\text{out}}$. Thus for every coefficient direction, including the zero polynomial,

$$e^{-E_j} \int |p|^2 |y|^{2n} dm_s \leq \int |pP|^2 dm_s \leq e^{E_j} \int |p|^2 |y|^{2n} dm_s. \quad (\text{ATG18})$$

We also prove the full-form recurrence comparison. Put $Q = n + \ell_s$. Pointwise, $|y|^{2n} \prod_a (y^2 + (2a + 1/2)^2) \geq |y|^{2Q}$. For $|y| \geq T$ its logarithmic ratio to $|y|^{2Q}$ is at most S_{ℓ_s}/T^2 . On $|y| < T$ its numerator is at most $(T^2 + (j/2)^2)^Q$. Apply (ATG14), the $s = 1$ case of (ATG12), and $\int d\sigma = \sqrt{2\pi}$. Since $j/2 \leq A \leq T/2$, $Q \geq q/2$, and the base $(5/4)\epsilon^2$ is less than one, the calculation (ATG16)–(ATG17) bounds the entire inner numerator by K_q times the reference integral on $[q, 2q]$. The full upper ratio is therefore at most $e^{S_{\ell_s}/T^2} + K_q \leq e^{\rho_{\ell_s}}$. At $\ell_s = 0$ the comparison is the exact identity. In all cases,

$$\beta_s \int |p|^2 |y|^{2Q} d\sigma \leq \int |p|^2 |y|^{2n} dm_s \leq \beta_s e^{\rho_{\ell_s}} \int |p|^2 |y|^{2Q} d\sigma. \quad (\text{ATG19})$$

In particular, no inner interval has been discarded in either comparison.

4.4 Global density bounds and their exact rate-change map

The classical inputs are again Euler's integral and Euler's reflection identity [H22, eqs. (5.2.1), (5.5.3)]; the complete inequalities following from them are proved here. Apply (ATG9) with $\theta = \pi/2 - h$, first at $a = 1/4$. After squaring and dividing by 2π , it gives $d\sigma/dy \leq U_h e^{-(\vartheta-h)|y|}$. At $a = 3/4$, (ATG9) bounds the second Gamma square in (ATG11) by $\Gamma(3/4)^2 (\sin h)^{-3/2} e^{-(\vartheta-h)|y|}$. Dividing (ATG11) by that bound, then using $\cosh(\pi y) \leq e^{\pi|y|}$, proves on the whole real line

$$L_h e^{-(\vartheta+h)|y|} \leq \frac{d\sigma}{dy}(y) \leq U_h e^{-(\vartheta-h)|y|}. \quad (\text{ATG20})$$

Write $\|p\|_{Q,\kappa}^2 = \int |p(y)|^2 |y|^{2Q} e^{-\kappa|y|} dy$ and $\mathbf{A}p = yp'(y)$ on polynomials of degree at most b . We claim $\|\mathbf{A}\| \leq L_{b,Q}$ for each $\kappa > 0$. On the positive half-line, set $x = \kappa y$ and $\alpha = 2Q$. The classical Laguerre Rodrigues formula, series, and squared norms are recorded by Koornwinder, Wong, Koekoek, Swarttouw and Reinhardt [H21, Table 18.3.1; eq. (18.5.5), Table 18.5.1; eq. (18.5.12)]. In those exact conventions, the polynomials

$$\mathcal{L}_a^{(\alpha)}(x) = \frac{x^{-\alpha} e^x}{a!} \left(\frac{d}{dx} \right)^a (e^{-x} x^{a+\alpha}) = \sum_{t=0}^a \frac{(-1)^t}{t!} \binom{a+\alpha}{a-t} x^t \quad (\text{ATG21})$$

are orthogonal to lower degrees and have squared norm $\Gamma(a+\alpha+1)/a!$ in $x^\alpha e^{-x} dx$. These assertions follow by a integrations by parts; every boundary term vanishes, as the power at zero before the final integration is at least $\alpha+1$ and the exponential controls infinity. The coefficient formula in (ATG21) gives $x(\mathcal{L}_a^{(\alpha)})' = a\mathcal{L}_a^{(\alpha)} - (a+\alpha)\mathcal{L}_{a-1}^{(\alpha)}$. In the orthonormal basis, this is the sum of a diagonal matrix of norm b and a one-step shift of norm $\sqrt{b(b+2Q)}$. The same estimate applies to $p(-y)$ on the negative half-line; adding the two squared estimates proves the claim on the full line. The pullback $x = \kappa y$ preserves the operator and contributes only the common positive factor κ^{-2Q-1} to squared norms.

The exact dilation map is $p(y) \mapsto p(e^t y) = \exp(t\mathbf{A})p$, with diagonal monomial matrix $\text{diag}(1, e^t, \dots, e^{bt})$. The finite matrix exponential bound, followed by the same bound for its inverse, gives $e^{-L_{b,Q}|t|} \|p\|_{Q,\vartheta} \leq \|p(e^t \cdot)\|_{Q,\vartheta} \leq e^{L_{b,Q}|t|} \|p\|_{Q,\vartheta}$. With $u = \kappa/\vartheta$, the exact substitution is

$$\|p\|_{Q,\kappa}^2 = u^{-(2Q+1)} \|p(u^{-1} \cdot)\|_{Q,\vartheta}^2. \quad (\text{ATG22})$$

Retaining that scalar first and then bounding it gives

$$e^{-K_{b,Q}|\log(\kappa/\vartheta)|} \|p\|_{Q,\vartheta}^2 \leq \|p\|_{Q,\kappa}^2 \leq e^{K_{b,Q}|\log(\kappa/\vartheta)|} \|p\|_{Q,\vartheta}^2. \quad (\text{ATG23})$$

Combining (ATG18)–(ATG20) and (ATG23), at rates $\vartheta+h$ and $\vartheta-h$, proves (ATG4) with precisely the constants (ATG3).

4.5 Domain, explicit error control, and monic minima

All uses of the preceding inequalities are in their proved domains. Indeed $2q' - q = j^2 - 30j + 97 > 0$ for $j \geq 257$, so $q' \geq q/2$; $Q_\pm \leq q + (j-1)/4 \leq 2q$; and

$$r \leq q + 32j \leq 2q, \quad 0 \leq m + r \leq q + 48j - 48 \leq 2q. \quad (\text{ATG24})$$

For the last two upper inequalities, subtract the left sides from $2q$: the resulting lower bounds are respectively $j^2 - 30j + 1$ and $j^2 - 46j + 49$, both positive for $j \geq 257$. Also $j/2 \leq A$ because $\gamma > 2$, and $0 < h < \vartheta$. The nonnegativity of $m + r$ uses the original $m = \Delta - v \geq 0$; it is not inferred merely from the displayed numerical domain.

Set $B_q = \log(1 + K_q)$ and $G_q = \log q + \log(\pi/2)$. Then the following fully explicit bounds apply simultaneously to both signs, both source orders, both root counts, and all degrees $b \leq 2q$:

$$\begin{aligned} |b_{n,b,s}^-| &\leq \frac{4D^2}{3\epsilon^2} + B_q + |\log C_L| + \frac{3}{2}G_q + \frac{17}{\vartheta-1}, \\ |b_{n,b,s}^+| &\leq \frac{4D^2}{3\epsilon^2} + 2B_q + \frac{1}{16\epsilon^2 j} + |\log C_U| + \frac{1}{2}G_q + \frac{17}{\vartheta-1}. \end{aligned} \quad (\text{ATG25})$$

To verify these, $A^2 \leq T^2/4$ and $j^2/q \leq 1$ imply $qA^2/(T^2 - A^2) \leq 4D^2/(3\epsilon^2)$. Further, $S_{\ell_s} \leq \ell_s j^2/4 \leq j^3/16$ and $q \geq j^2$. Since $b, Q \leq 2q$, $K_{b,Q} \leq (8 + 4\sqrt{3})q + 1 \leq 16q + 1$. Both absolute rate logarithms are at most $h/(\vartheta - h)$; their products with $K_{b,Q}$ are at most $(16q + 1)/(\vartheta q - 1) \leq 17/(\vartheta - 1)$. Finally $2h/\pi \leq \sin h \leq h$, so $|\log \sin h| \leq G_q$. The exact K_q is bounded, since $K_q \leq 9\sqrt{2\pi}C_L^{-1}qe^{-2q} \leq 9\sqrt{2\pi}C_L^{-1}e^{-2}$ for $q \geq 1$. This proves the uniform $O_{\delta,\gamma}(\log q)$ claim without absorbing β_s into the error.

The two monic minima are defined with their original centres by

$$\begin{aligned} \nu_{r,s}^+ &= \min_{f \text{ monic}, \deg f=r} \int |\chi_j(c_+ + iy)f(c_+ + iy)|^2 dm_s(y), \\ \nu_{m+r,s}^- &= \min_{f \text{ monic}, \deg f=m+r} \int |\chi_{j-8}(c_- + iy)f(c_- + iy)|^2 dm_s(y). \end{aligned} \quad (\text{ATG26})$$

Each minimum exists uniquely: all moments are finite by (ATG9), and the Gram form is positive definite because a nonzero polynomial cannot vanish on a real set of positive measure. Its restriction to the finite-dimensional lower-coefficient vector is consequently strictly positive definite and coercive. Define $\mathbf{n}(Q, b) = \min_{p \text{ monic}, \deg p=b} \|p\|_{Q,\vartheta}^2$. The exact affine-fibre bijections (ATG7) allow minimization of both sides of (ATG4) over exactly these fibres. Therefore

$$\boxed{\nu_{r,s}^+ \leq \beta_s e^{b_{q,r,s}^+} \mathbf{n}(q + \ell_s, r), \quad \nu_{m+r,s}^- \geq \beta_s e^{b_{q',m+r,s}^-} \mathbf{n}(q' + \ell_s, m + r).} \quad (\text{ATG27})$$

Both opposite inequalities also hold with their corresponding b^- and b^+ . The positive original mass cancels only upon forming the matched same-source ratio:

$$\frac{\nu_{r,s}^+}{\nu_{m+r,s}^-} \leq e^{b_{q,r,s}^+ - b_{q',m+r,s}^-} \frac{\mathbf{n}(q + \ell_s, r)}{\mathbf{n}(q' + \ell_s, m + r)}. \quad (\text{ATG28})$$

No relation Gram, coefficient direction, source order, or root coefficient is changed in this passage. The proof holds on all of (ATG1); restricting to $\lfloor j^{3/2}/\log j \rfloor < r \leq q + 32j$ preserves every constant and makes the claimed tail uniformity immediate.

5 The attained upper fibre and the complete native-row profile

5.1 Original objects and exact meaning of every cutoff

Keep the original simple quartet, its complete nonreduced conductor, fixed period and logarithm branch. In particular retain

$$\begin{aligned} k &= 4l + 1 \geq 9, \quad j = k + 4, \quad q = (j + 1)^2, \quad q' = (j - 7)^2, \quad \Delta = q - q' = 16j - 48, \\ c &= j/2, \quad c' = j/2 - 4, \quad v = \text{ord}_0 E_A, \quad L = q - v, \quad m = \Delta - v > 0, \quad s \in \{1, j\}, \end{aligned}$$

$$E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, \quad \beta_{d,a,b} = d/2 + (2a - d)\delta + i(2b - d)\gamma, \quad (\text{ATA1})$$

$$\mu_h = E_A^{(h)}(0), \quad \mu_v \neq 0, \quad \chi_j(S) = \prod_{a,b=0}^j (S - \beta_{j,a,b}), \quad \chi_-(S') = \chi_{j-8}(S').$$

The actual conductor has $0 \leq v \leq 80$ by NIB1; no value of v is replaced by zero. For either centre $d = c, c'$ use precisely

$$\|P\|_{s,d}^2 = \int_{\mathbb{R}} |P(d + iy)|^2 dm_s(y), \quad dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy. \quad (\text{ATA2})$$

Write D_n^{nat} for NTG18's form at native polynomial degree n . If a target cutoff N is used, write instead $D_N = D_{N-v}^{\text{nat}}$, as in FTR3. Thus the same row has native degree $L + r$ and target cutoff $q + r$. These conventions give $D_{L+r}^{\text{nat}} = D_{q+r}$, including the unchanged centre c' .

The original relations of NTG2 are

$$(\mathcal{T}_A P)(S') = \sum_{a,b=0}^8 a_{ab} P(S' + \beta_{8,a,b}), \quad g_r = \mathcal{T}_A(\chi_j S^r), \quad \ell_r = \mu_v \binom{q+r}{v}, \quad u_r = g_r / \ell_r. \quad (\text{ATA3})$$

They are monic of degree $L+r$ and belong to $\chi_- \mathbb{C}[S']_{\leq m+r}$. All ℓ_r are nonzero. The full lower-ideal minimum and the attained relation minimum are

$$\begin{aligned} \nu_{m+r,s}^- &= \min_{P \text{ monic, deg } P=m+r} \|\chi_- P\|_{s,c'}^2, \\ \eta_{r,s} &= \min_{z \in \mathbb{C}^r} \left\| u_r + \sum_{t=0}^{r-1} z_t u_t \right\|_{s,c'}^2, \quad \gamma_{r,s} = \log(\eta_{r,s} / \nu_{m+r,s}^-). \end{aligned} \quad (\text{ATA4})$$

For $r = 0$ the second minimum has its unique empty coefficient vector. Both minima exist uniquely: in a finite polynomial coefficient space the relevant Gram is strictly positive, because a nonzero polynomial cannot vanish on a real interval and the original density is positive. Completing its positive quadratic coefficient square gives the minimum. NTG8 and NIB5–9 give these same minima, with no change of fibre. Since every member of the second affine fibre belongs to the first monic lower-ideal fibre, $\eta_{r,s} \geq \nu_{m+r,s}^- > 0$.

5.2 An affine bijection before the conductor and its attained slack

Define the original upper minimum by

$$\nu_{r,s}^+ = \min_{P \text{ monic, deg } P=r} \|\chi_j P\|_{s,c'}^2, \quad P_{r,s}^+ \text{ its unique minimizer.} \quad (\text{ATA5})$$

For $P(S) = S^r + \sum_{t < r} p_t S^t$, linearity gives literally

$$\frac{\mathcal{T}_A(\chi_j P)}{\ell_r} = u_r + \sum_{t < r} p_t \frac{\ell_t}{\ell_r} u_t, \quad p_t = z_t \frac{\ell_r}{\ell_t}. \quad (\text{ATA6})$$

These coefficient maps are mutually inverse affine maps between the complete upper monic fibre and the complete original relation fibre in ATA4. Consequently

$$\eta_{r,s} = |\ell_r|^{-2} \min_{P \text{ monic, deg } P=r} \|\mathcal{T}_A(\chi_j P)\|_{s,c'}^2. \quad (\text{ATA7})$$

Let $E_r = \text{span}(u_0, \dots, u_{r-1})$, with $E_0 = 0$, and let Π_r be its orthogonal projection in ATA2. Put

$$w_r = \mathcal{T}_A(\chi_j P_{r,s}^+) / \ell_r, \quad \tau_{r,s} = \|w_r\|_{s,c'}^2.$$

Equation ATA6 gives $w_r - u_r \in E_r$, hence $(I - \Pi_r)w_r = (I - \Pi_r)u_r$. For every $e \in E_r$,

$$\|w_r + e\|_{s,c'}^2 = \|(I - \Pi_r)u_r\|_{s,c'}^2 + \|\Pi_r w_r + e\|_{s,c'}^2.$$

Taking its exact minimum and then $e = 0$ proves

$$\eta_{r,s} = \|(I - \Pi_r)u_r\|_{s,c'}^2, \quad \tau_{r,s} = \eta_{r,s} + \|\Pi_r w_r\|_{s,c'}^2. \quad (\text{ATA8})$$

For example if $G_{<r} = (\langle u_a, u_b \rangle)_{a,b < r}$, $h_a = \langle u_a, w_r \rangle$, then $\Pi_r w_r = \sum_{a < r} u_a (G_{<r}^{-1} h)_a$. Thus this projection uses the entire original Gram inverse. This is the complex Hermitian quadratic minimum underlying the classical Schur calculation of Boyd and Vandenberghe [H23, Appendix A.5.5, p. 650]; the displayed coefficient map and square completion give its full specialization.

5.3 A polynomial translation bound for the actual conductor

The unchanged NTG3 polynomials and squared norms are

$$p_0 = 1, \quad p_1 = y, \quad p_{n+1} = yp_n - n(n-1+s/2)p_{n-1}, \quad h_{n,s} = (2\pi)^{s/2}n!(s/2)_n. \quad (\text{ATA9})$$

Their generating function is

$$F_s(y, z) = \sum_{n \geq 0} \frac{p_{n,s}(y)}{n!} z^n = (1+z^2)^{-s/4} e^{y \arctan z}. \quad (\text{ATA10})$$

Indeed multiplication of the recurrence by $z^n/n!$ and summation gives $(1+z^2)\partial_z F_s = (y-(s/2)z)F_s$ and $F_s(y, 0) = 1$; the right side of ATA10 solves that equation. Its Taylor coefficients therefore agree recursively, and the right side is analytic for $|z| < 1$. The human-authored NIST DLMF generating formula [H21, eq. (18.23.7)] gives the same identity under $p_{n,s}(y) = n!P_n^{(s/4)}(y/2; \pi/2)$. The previously verified DLMF dictionary also supplies exactly ATA9's norm, including $dy = 2dx$ and $(2\pi)^{s/2}$; see NTG3 and LRP3.

For a complex shift z_0 expand $e^{z_0 \arctan z} = \sum_{a \geq 0} e_a(z_0)z^a$. Coefficient comparison gives

$$p_{n,s}(y+z_0) = \sum_{a=0}^n \frac{n!}{a!} e_{n-a}(z_0) p_{a,s}(y). \quad (\text{ATA11})$$

Fix an integer $d \geq 1$ and $0 \leq a \leq d$. On $|z| = d/(d+1)$, the power series of $\arctan$ gives $|\arctan z| \leq \operatorname{artanh}(d/(d+1)) = \frac{1}{2} \log(2d+1)$. Cauchy's coefficient estimate and $(1+1/d)^d \leq e$ give

$$|e_a(z_0)| \leq e(2d+1)^{|z_0|/2}. \quad (\text{ATA12})$$

In the original orthonormal coordinates the remaining entry factor is

$$\frac{n!}{a!} \sqrt{\frac{h_{a,s}}{h_{n,s}}} = \sqrt{\prod_{b=a+1}^n \frac{b}{b-1+s/2}} \leq \sqrt{2d+1} \quad (0 \leq a \leq n \leq d, s \geq 1). \quad (\text{ATA13})$$

For the inequality, $s \geq 1$ bounds the product by $\prod_{b=a+1}^n b/(b-1/2)$, and each omitted factor $b/(b-1/2)$ exceeds one. Its enlargement from $b=1$ to n equals $4^n / \binom{2n}{n} \leq 2n+1$: the middle binomial coefficient is the maximum among the $2n+1$ coefficients with sum 4^n . At $n=0$ the product equals one. The translation matrix has at most $(d+1)^2$ entries. Its operator norm is bounded by its Frobenius norm, so ATA12–13 prove, for every polynomial of degree at most d ,

$$\|p(\cdot + z_0)\|_s \leq e(d+1)(2d+1)^{(|z_0|+1)/2} \|p\|_s. \quad (\text{ATA14})$$

The centres in ATA1 give the exact shift

$$z_{ab} = -i(\beta_{8,a,b} - 4) = (2b-8)\gamma - i(2a-8)\delta, \quad c' + iy + \beta_{8,a,b} = c + i(y + z_{ab}).$$

Thus the triangle inequality in the unchanged measure, applied to each literal conductor summand, gives

$$\begin{aligned} \mathfrak{C}_A(d) &= e(d+1)\sqrt{2d+1} \sum_{a,b=0}^8 |a_{ab}|(2d+1)^{|z_{ab}|/2}, \\ \|\mathcal{T}_A P\|_{s,c'} &\leq \mathfrak{C}_A(d) \|P\|_{s,c} \quad (\deg P \leq d), \\ \eta_{r,s} &\leq \tau_{r,s} \leq \frac{\mathfrak{C}_A(q+r)^2}{|\ell_r|^2} \nu_{r,s}^+. \end{aligned} \quad (\text{ATA15})$$

For the fixed actual conductor the finite sum has fixed coefficients and exponents. It has a nonzero term since $\mu_v \neq 0$; hence $|\log \mathfrak{C}_A(d)| = O_{\text{actual}}(\log(d+1))$. Likewise the exact finite product in $\binom{q+r}{v}$ gives $|\log |\ell_r|| = O_{\text{actual}}(\log q)$ uniformly on $0 \leq r \leq q+32j$. No inverse-conductor bound is used.

5.4 Receiving the complete finite analytic endpoint

Use the actual root-comparison domain

$$j \geq 257, \quad j\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q, \quad R_j < r \leq q + 32j, \quad R_j = \lfloor j^{3/2}/\log j \rfloor. \quad (\text{ATA16})$$

The additive GPA reconstruction ATG3–28 proves the complete monic comparisons with $Q_+ = q + (s-1)/4$ and $Q_- = q' + (s-1)/4$:

$$\nu_{r,s}^+ \leq \beta_s e^{b_{q,r,s}^+} \mathbf{n}(Q_+, r), \quad \nu_{m+r,s}^- \geq \beta_s e^{b_{q',m+r,s}^-} \mathbf{n}(Q_-, m+r), \quad \beta_s = \frac{(2\pi)^{s/2}}{\sqrt{2}\Gamma(s/2)}.$$

The complete power-exponential norm proof EQ43–54 in Section 3.4 gives its finite endpoints $H^-(Q, b)$ and $H^+(Q, b)$. Define the actual endpoint by the full expression

$$U_{j,s,r} = 2 \log \mathfrak{C}_A(q+r) - 2 \log |\ell_r| + b_{q,r,s}^+ - b_{q',m+r,s}^- + H^+(Q_+, r) - H^-(Q_-, m+r). \quad (\text{ATA17})$$

Here every factor belongs to the same original source order s . The mass β_s cancels only in this matched ratio. Substitution in ATA15 proves the two finite inequalities

$$0 \leq \gamma_{r,s} \leq U_{j,s,r}, \quad 0 \leq \log(\tau_{r,s}/\eta_{r,s}) \leq U_{j,s,r} - \gamma_{r,s}. \quad (\text{ATA18})$$

In particular this explicitly defined $U_{j,s,r}$ is nonnegative. The exact centre calculation EQ45–54 and AER11–14 proves uniformly on ATA16

$$U_{j,s,r} = \Delta \mathfrak{f}(r/q) + O_{\text{actual}}(q/r + \log q), \quad \mathfrak{f}(t) = \log \frac{1 + \rho_t}{1 - \rho_t}, \quad \frac{\rho_t K(\sqrt{1 - \rho_t^2})}{E(\sqrt{1 - \rho_t^2})} = \frac{1}{1 + t}. \quad (\text{ATA19})$$

The same calculation proves $\mathfrak{f} > 0$, decreasing and smooth on $(0, \infty)$, $\mathfrak{f}(t) = -\frac{1}{2} \log t + O(\sqrt{t})$ at zero, and $\sup_{0 < t \leq 4} t|\mathfrak{f}'(t)| < \infty$. The exact integral calculation PCB1–8, with the original EIQ energy $F = -E_{\min}$ and logarithmic moment $\mathcal{L}$, gives

$$2 \int_0^1 \mathfrak{f}(t) dt = 4 \log(4/\pi) - 10 - 2F(2, \pi) + 4\mathcal{L}(2, \pi) = C_\partial. \quad (\text{ATA20})$$

The right side is precisely the previously defined ECL16 coefficient.

5.5 A finite discrepancy and weighted convergence of the actual rows

For the original four-endpoint return retain $\vartheta_0 = \vartheta_q = 1$ and $\vartheta_r = 2$ for $0 < r < q$. HJR21–26, with its degree parameter set to j , supply the exact total and its finite lower endpoint

$$\begin{aligned} \mathcal{T}_{j,s} &= \sum_{r=0}^q \vartheta_r \gamma_{r,s}, \\ \mathcal{T}_{j,s}^- &= \max\{0, \mathcal{F}_{\partial,j}^{(s)} - e_{j,s}^- - \mathcal{E}_{j,s}^{\text{FEX}} - E_{j,s}^{\text{tr}} - (v - t_0)\mathcal{L}_s\} \leq \mathcal{T}_{j,s}, \\ \mathcal{T}_{j,s}^- &= \Delta q C_\partial + o(jq), \quad \mathcal{T}_{j,s} = \Delta q C_\partial + o(jq). \end{aligned} \quad (\text{ATA21})$$

Every error and original source restriction in this displayed endpoint is the HJR25 expression, not a new independently chosen error. The lower asymptotic follows from its displayed finite errors and the same outer-return asymptotic used in HJR26; ATA20 gives $C_\partial > 0$, so its maximum with zero does not change that asymptotic. NIB14 and NIB18–20 give the explicit prefix endpoint

$$E_{j,s}^{\text{pre}} = \sum_{r=0}^{R_j} \vartheta_r E_{j,s,r}^{\text{NIB}} = O_{\text{actual}}(j^{5/2} + j^3/\log j) = o(jq). \quad (\text{ATA22})$$

Define the finite number

$$\mathfrak{D}_{j,s} = \sum_{r=R_j+1}^q \vartheta_r U_{j,s,r} - \mathcal{T}_{j,s}^- + E_{j,s}^{\text{pre}}. \quad (\text{ATA23})$$

All tail sums in this section end at q , even though ATA19 is proved on the larger range for the separate displacement application. Subtracting the actual prefix from the exact total gives

$$\begin{aligned} 0 &\leq \sum_{r=R_j+1}^q \vartheta_r (U_{j,s,r} - \gamma_{r,s}) \\ &= \sum_{r=R_j+1}^q \vartheta_r U_{j,s,r} - \mathcal{T}_{j,s} + \sum_{r=0}^{R_j} \vartheta_r \gamma_{r,s} \leq \mathfrak{D}_{j,s}. \end{aligned} \quad (\text{ATA24})$$

To evaluate its size, fix one constant for the uniform remainder in ATA19. Its summed absolute value is at most

$$C \sum_{r=R_j+1}^q (q/r + \log q) = O_{\text{actual}}(q \log q) = o(jq).$$

For a decreasing integrable positive function,

$$\int_{1/q}^{1+1/q} \mathfrak{f}(t) dt \leq \frac{1}{q} \sum_{r=1}^q \mathfrak{f}(r/q) \leq \int_0^1 \mathfrak{f}(t) dt.$$

Both bounds have the same limit. The removed first R_j terms have sum at most $q \int_0^{R_j/q} \mathfrak{f}(t) dt = o(q)$, and the single endpoint correction from its weight one is $O(1)$. Therefore ATA19–20 imply

$$\sum_{r=R_j+1}^q \vartheta_r U_{j,s,r} = \Delta q C_\partial + o(jq).$$

Together with ATA21–22 this proves $\mathfrak{D}_{j,s} = o(jq)$. For each row set $a_r = \Delta \mathfrak{f}(r/q)$; since $U_{j,s,r} \geq \gamma_{r,s}$, the triangle inequality gives $|\gamma_{r,s} - a_r| \leq U_{j,s,r} - \gamma_{r,s} + |U_{j,s,r} - a_r|$. Consequently

$$\begin{aligned} \sum_{r=R_j+1}^q \vartheta_r |\gamma_{r,s} - \Delta \mathfrak{f}(r/q)| &= o(jq), \\ 0 &\leq \sum_{r=R_j+1}^q \vartheta_r \log(\tau_{r,s}/\eta_{r,s}) \leq \mathfrak{D}_{j,s} = o(jq). \end{aligned} \quad (\text{ATA25})$$

The second line follows directly from ATA18 and ATA24; it preserves the nonzero projection term in ATA8. This convergence asserts a weighted total of absolute differences. It assigns no pointwise lower estimate to a prescribed row.

5.6 Correlated determinant and every sublinear row set

NTG12–15 gives the compatible original columns x_r and $\Omega_r = I_m + \sum_{t=0}^r x_t x_t^*$, $\Omega_{-1} = I_m$. For integers $-1 \leq R' < Q' \leq q$ and $Y = [x_{R'+1}, \dots, x_{Q'}]$, elimination of the entire earlier relation Gram, as in NTG23–25, gives exactly

$$\log \det(I + Y^* \Omega_{R'}^{-1} Y) = \log \frac{\det \Omega_{Q'}}{\det \Omega_{R'}} = \sum_{r=R'+1}^{Q'} \gamma_{r,s}. \quad (\text{ATA26})$$

For completeness its graph-coordinate block Gram is $I + Y^* Y - Y^* X_{R'} (I + X_{R'}^* X_{R'})^{-1} X_{R'}^* Y$; multiplication verifies $I - X_{R'} (I + X_{R'}^* X_{R'})^{-1} X_{R'}^* = \Omega_{R'}^{-1}$. The leading later triangular block H_{22}

remains on both sides of this Gram and cancels only against $\prod_{r=R'+1}^{Q'} \nu_{m+r,s}^-$ in its determinant. The earlier/late block H_{12} disappears by the exact earlier minimum, while all pairings in $Y^* \Omega_{R'}^{-1} Y$ remain.

ATA22 and ATA25 imply uniform convergence of cumulative sums:

$$\sup_{0 \leq n \leq q} \left| \frac{1}{jq} \sum_{r=0}^n \gamma_{r,s} - 16 \int_0^{n/q} \mathfrak{f}(t) dt \right| \longrightarrow 0. \quad (\text{ATA27})$$

Here the true prefix has the upper bound ATA22. Its comparison profile contributes at most $\Delta q \int_0^{R_j/q} \mathfrak{f} = o(jq)$. The remaining total absolute error is bounded by ATA25 because $\vartheta_r \geq 1$. Uniform Riemann-sum convergence follows from the same decreasing-function bounds just proved, applied at every upper endpoint. Finally $\Delta/j \rightarrow 16$. For $R'/q \rightarrow a$, $Q'/q \rightarrow b$, $0 \leq a \leq b \leq 1$, subtraction of the two cumulative limits in ATA27 and ATA26 therefore proves

$$\frac{1}{jq} \log \det(I + Y^* \Omega_{R'}^{-1} Y) \longrightarrow 16 \int_a^b \mathfrak{f}(t) dt. \quad (\text{ATA28})$$

Adding the cases $Q' = q - 1, q$ gives

$$\frac{1}{jq} \log \frac{\det \Omega_{q-1} \det \Omega_q}{(\det \Omega_{R'})^2} \longrightarrow 32 \int_a^1 \mathfrak{f}(t) dt. \quad (\text{ATA29})$$

Define a finite budget for each original set $\mathcal{J} \subset \{0, \dots, q\}$ by

$$\mathcal{E}_s(\mathcal{J}) = \sum_{\substack{r \in \mathcal{J} \\ r \leq R_j}} \vartheta_r E_{j,s,r}^{\text{NIB}} + \sum_{\substack{r \in \mathcal{J} \\ r > R_j}} \vartheta_r U_{j,s,r}. \quad (\text{ATA30})$$

If $h = |\mathcal{J}|$, monotonicity of $\mathfrak{f}$ and the uniform remainder yield the explicit upper estimate

$$0 \leq \mathcal{E}_s(\mathcal{J}) \leq E_{j,s}^{\text{pre}} + 2\Delta q \int_0^{h/q} \mathfrak{f}(t) dt + O_{\text{actual}}(q \log q). \quad (\text{ATA31})$$

Indeed the largest sum of h values of a decreasing sequence is its first h terms; $\sum_{r=1}^h \mathfrak{f}(r/q)$ is at most the displayed integral times q . The absolute remainder on any subset is at most its whole-range sum. Thus whenever $h = o(q)$, $\mathcal{E}_s(\mathcal{J}) = o(jq)$, uniformly over the locations of those h rows, by absolute continuity of the integral at zero. For $R'_j = \lfloor q/\log j \rfloor$, the expansion at zero gives

$$\mathcal{E}_s(\{0, \dots, R'_j\}) = O_{\text{actual}} \left(j^{5/2} + \frac{j^3 \log \log j}{\log j} + q \log q \right) = o(jq). \quad (\text{ATA32})$$

In particular the $j^3/\log j$ term of ATA22 has been absorbed in $j^3 \log \log j / \log j$ for sufficiently large j , not discarded.

At every row the exact NTG22 update satisfies $\Omega_r = \Omega_{r-1} + x_r x_r^*$ and $x_r^* \Omega_{r-1}^{-1} x_r = e^{\gamma_{r,s}} - 1$. Cauchy–Schwarz in the Ω_{r-1} metric gives $x_r x_r^* \preceq (e^{\gamma_{r,s}} - 1) \Omega_{r-1}$. After the unchanged conjugation and A map of NTG20, it follows that

$$D_{q+r-1} \preceq D_{q+r} \preceq e^{\gamma_{r,s}} D_{q+r-1} \preceq e^{U_{j,s,r}} D_{q+r-1}. \quad (\text{ATA33})$$

For any original restriction or quotient pair, extend its actual kernel and value frames to the whole original coefficient space. Schur factorization decomposes the full determinant ratio into the restriction, the specified quotient, and its exterior quotient. Every factor is at least one, because an increasing metric increases both restriction and a minimum over the identical affine value fibre. The four-copy ambient logarithmic ratio is exactly $4\gamma_{r,s}$. Therefore each original CPQ quotient increment is between zero and $4\gamma_{r,s}$, with no quotient-rank multiplier. The original two chains in

NIB16 give the signed row bound $4\gamma_{r,s}$ as well. Summing those same labelled occurrences over $\mathcal{J}$ proves

$$\begin{aligned} 0 &\leq S_K^{\mathcal{J}}, S_R^{\mathcal{J}} \leq \mathcal{T}^{\mathcal{J}} \leq \mathcal{E}_s(\mathcal{J}), \\ |\mathcal{L}_e^{\mathcal{J}}|, |\mathcal{P}_e^{\mathcal{J}}| &\leq 4\mathcal{E}_s(\mathcal{J}), \quad 0 \leq V_X^{\mathcal{J}} \leq 4\mathcal{E}_s(\mathcal{J}) \quad (X = b, \text{rem}, s, A), \\ 4S_K^{\mathcal{J}} &= \mathcal{L}_e^{\mathcal{J}} + V_b^{\mathcal{J}} + V_{\text{rem}}^{\mathcal{J}} + V_s^{\mathcal{J}} + V_A^{\mathcal{J}}, \quad S_K^{\mathcal{J}} + S_R^{\mathcal{J}} = \mathcal{T}^{\mathcal{J}}. \end{aligned} \quad (\text{ATA34})$$

All quantities in ATA34 are negligible at scale jq on every $o(q)$ -row set. This assertion concerns the identical row sums; it does not commute, independently optimize, or identify the eight programme flags. Native-to-transported metric errors and both threshold comparison errors remain those in the exact receiving formulas FTR/IPR/STF.

5.7 The independent proper source and all four displaced cutoffs

Apply the preceding result separately at degree k , with its own source orders $s_0 = 1, k$, centre $c'_k = k/2 - 4$, root count $Q = q_k = (k+1)^2$ and root-comparison domain. Put

$$b = q_{k+4} - q_k = 8k + 24, \quad \Delta_k = 16k - 48.$$

The proper form in PRD7–10 is

$$G_N = J_0^* D_{k,N-v}^{(s_0)} J_0, \quad Q_N = \text{Sch}(G_N; Z_E, Z_R), \quad g(N) = \log \det Q_N. \quad (\text{ATA35})$$

Its original frames J_0, Z_E, Z_R are fixed. Its four target-displaced cutoffs in PRD22 are $Q+b-1, Q+b, 2Q+2b-1, 2Q+2b$ in the N convention, and its own standard cutoffs are $Q-1, Q, 2Q-1, 2Q$. Their exact difference of returns is consequently

$$\begin{aligned} \delta V_Q &= g(2Q+2b-1) - g(2Q-1) + g(2Q+2b) - g(2Q) \\ &\quad - g(Q+b-1) + g(Q-1) - g(Q+b) + g(Q). \end{aligned} \quad (\text{ATA36})$$

Schur factorization in the single degree- k ambient source, with the actual restriction J_0 and actual quotient Z_E , proves

$$0 \leq g(Q+r) - g(Q+r-1) \leq \gamma_{k,s_0,r}. \quad (\text{ATA37})$$

This is the same frame-extension argument used before ATA34, applied in one copy. For a zero-dimensional value space its determinant is one and the assertion also holds. Expanding ATA36 by these increments gives its two negative intervals $r = 0, \dots, b-1$ and $r = 1, \dots, b$, and its two positive intervals $r = Q, \dots, Q+2b-1$ and $r = Q+1, \dots, Q+2b$. Since $2b = 16k + 48 \leq 32k$ for $k \geq 3$, both positive intervals lie in the extended range proved in ATA16–19 at degree k . Their lower endpoints exceed R_k , so ATA18 applies there. NIB14 applies at every low row, irrespective of its location relative to R_k . Therefore the following finite directed bounds hold:

$$\begin{aligned} -E_Q^- &\leq \delta V_Q \leq E_Q^+, \\ E_Q^- &= \sum_{r=0}^{b-1} E_{k,s_0,r}^{\text{NIB}} + \sum_{r=1}^b E_{k,s_0,r}^{\text{NIB}}, \\ E_Q^+ &= \sum_{r=Q}^{Q+2b-1} U_{k,s_0,r} + \sum_{r=Q+1}^{Q+2b} U_{k,s_0,r}. \end{aligned} \quad (\text{ATA38})$$

The low intervals contain $2b$ counted rows, each bounded by $O_{\text{actual}}(k \log k)$ by NIB15 for $r \leq b = O(k)$. Thus $E_Q^- = O_{\text{actual}}(k^2 \log k)$. The two high intervals contain $4b$ counted rows. On them $r/Q = 1 + O(b/Q)$ and the derivative of $\mathfrak{f}$ is bounded on a fixed neighbourhood of one. Therefore ATA19 gives

$$\begin{aligned} E_Q^+ &= 4b\Delta_k \mathfrak{f}(1) + O_{\text{actual}}(\Delta_k b^2/Q + b \log Q) \\ &= 4b(16k-48)\mathfrak{f}(1) + O_{\text{actual}}(k \log Q) = O_{\text{actual}}(k^2). \end{aligned} \quad (\text{ATA39})$$

No extra four-copy coefficient occurs in ATA37–39. The coefficient $4b$ counts the actual two high intervals of length $2b$. It follows that $\delta V_Q = o(kq_k)$. This estimates precisely the displaced-cutoff correction and leaves the proper source's own standard-window return in its exact degree- k form. In particular the separate target-prefix identity remains

$$(V_A^{>R} - V_Q) - (V_A - V_Q) = -V_A^{\leq R}. \quad (\text{ATA40})$$

Neither this identity nor the profile determines the directional allocation split, the signed eight-flag coefficient, or the transferred arithmetic sign of the original projected action.

6 Literal covariance, logarithmic endpoint term, and the complete action

The packet degree, source order and four cutoffs in this section are exactly those of TAC1 and TAC1a. Thus

$$\begin{aligned} a &\equiv 1 \pmod{4}, \quad s \in \{1, a\}, \quad q = (a+1)^2, \quad Q = (a-7)^2, \\ \Delta &= q - Q = 16a - 48, \quad 0 \leq v \leq 80, \quad m = \Delta - v, \quad \ell_s = (s-1)/4. \end{aligned}$$

The lower packet has degree $a-8$ and retains the same order s . The actual quartet, conductor unit, period and branch remain fixed. For either $b = a$ or $b = a-8$ write

$$Q_b = (b+1)^2, \quad n = Q_b + \ell_s.$$

The finite root-comparison guards of LRC13 are retained. They imply the row-domain bound needed for the entire lower window:

$$m + q \leq 2(Q + \ell_s) \iff 3\Delta - v - 2\ell_s \leq q. \quad (\text{LRS1})$$

Indeed the root radius in LRC13 is at least $1/2$, so $\mathcal{R}_b \leq 2^{-17}n$ forces $n \geq 65536$. At $b = a-8$, an original $a \leq 41$ would instead give $n = (a-7)^2 + \ell_s \leq 34^2 + 10 = 1166$. Consequently $a \geq 45$, where $q - 3\Delta = a^2 - 46a + 145 > 0$. This proves LRS1 from the stated finite guards, without adding a hypothesis or substituting a different row window.

6.1 The exact determinant and the summable row error

Let χ_b be the original monic root polynomial and let $\nu_{r,s}^{(b)}$ be the squared norm of its degree- r monic relation polynomial in $|\chi_b|^2 dm_s$. The source norms, including the original mass, are those of TAC1a:

$$h_{d,s} = (2\pi)^{s/2} d! (s/2)_d.$$

In the source coefficient basis

$$1, y, \dots, y^{Q_b-1}, \quad \chi_b, \chi_b y, \dots, \chi_b y^{D-Q_b},$$

the coefficient change is triangular with determinant of modulus one. The same assertion holds in the original S coordinate: the change $S = b/2 + iy$ has only unit-modulus leading factors. Let G denote the full Gram matrix in this basis and let G_{rel} be its lower right block. The matrix of evaluation is $[V_b \ 0]$, with the original Vandermonde V_b . Since the evaluation covariance is $[V_b \ 0]G^{-1}[V_b \ 0]^*$, the block inverse formula gives

$$\det K_b(D, s) = |\det V_b|^2 \frac{\det G_{\text{rel}}}{\det G}.$$

Here $\det G = \prod_{d=0}^D h_{d,s}$ and monic orthogonalization of the full relation space gives $\det G_{\text{rel}} = \prod_{r=0}^{D-Q_b} \nu_{r,s}^{(b)}$. Consequently

$$\det K_b(D, s) = |\det V_b|^2 \frac{\prod_{r=0}^{D-Q_b} \nu_{r,s}^{(b)}}{\prod_{d=0}^D h_{d,s}}, \quad D \geq Q_b - 1. \quad (\text{LRS2})$$

The numerator product is empty at $D = Q_b - 1$. This proof includes every cross block through the Schur minimum. Only on subtracting two cutoffs does the fixed Vandermonde cancel. In particular

$$\tau_{b,s,r} := \log \det K_b(Q_b + r, s) - \log \det K_b(Q_b + r - 1, s) = \log \nu_{r,s}^{(b)} - \log h_{Q_b+r,s}. \quad (\text{LRS3})$$

For $r = 2h + \epsilon \geq 2$, $\epsilon \in \{0, 1\}$, set

$$\alpha = \frac{n + \epsilon - 3/4}{h}, \quad \beta = \frac{\pi n}{2h}, \quad M = n + r.$$

The norm comparison LRC14–15 (EQ12–13 and EQ27 of the full equilibrium norm proof), proved for these parameters and the complete polynomial space of degree at most $2n$, gives

$$\begin{aligned} Z_r + \log c_- - B_{\text{root}} + \log(1 - \eta_n) &\leq \tau_{b,s,r} \leq Z_r + \log c_+ + B_{\text{root}} + \log(1 + \eta_n), \\ Z_r &= \log \beta_s + \frac{1}{2} \log 2 + (2M + \frac{1}{2}) \log n - h\lambda(\alpha, \beta) - \log h_{Q_b+r,s}. \end{aligned} \quad (\text{LRS4})$$

The constants $c_{\pm}, B_{\text{root}}, \eta_n$ are the finite ones in that norm proof; no original Gram determinant occurs in their definition. For $r = 0, 1$ use its scalar-moment interval, with centre

$$\log \beta_s + \log \frac{\Gamma(2(n+r) + 1/2)}{(\pi/2)^{2(n+r)+1/2}} - \log h_{Q_b+r,s}.$$

Here is the precise uniformity needed to sum LRS4. Throughout $2 \leq r \leq 2n$, $n \geq 100$, its equilibrium parameters satisfy $\alpha \geq 1/2$ and $\beta/\alpha \in [\pi/3, \pi]$. The uniform endpoint proof EQ28–36 accompanying LRC15 gives fixed constants $c_0, c_1, w_0, w_1 > 0$ such that

$$c_0 \leq c_{\text{cap}} \sqrt{\alpha} \leq c_1, \quad w_0 \leq (u + v_*)/2 \leq w_1, \quad |\log c_{\pm}| \leq C(1 + \log^+ \alpha). \quad (\text{LRS5})$$

These assertions include $\alpha \rightarrow \infty$; they do not follow merely by compactness of a bounded parameter region. The original EIQ5 endpoint equations and their convergent small-modulus expansions give $u, v_* \rightarrow 2/(\beta/\alpha)$ and $c_{\text{cap}} \sqrt{\alpha} \rightarrow 4\sqrt{2}/(\beta/\alpha)^2$. The explicit inequalities in the preceding uniform endpoint proof give the positive constants in LRS5 over the entire stated range.

The Gamma cancellation before Stirling is exact. Since $s/2 = 2\ell_s + 1/2$ and $\beta_s \sqrt{2} = (2\pi)^{s/2}/\Gamma(s/2)$,

$$\frac{h_{Q_b+r,s}}{\beta_s \sqrt{2}} = \Gamma(M - \ell_s + 1) \Gamma(M + \ell_s + 1/2). \quad (\text{LRS6})$$

For real $x > 0$, write

$$\log \Gamma(x) = (x - \frac{1}{2}) \log x - x + \frac{1}{2} \log(2\pi) + R(x), \quad 0 < R(x) < \frac{1}{12x}.$$

This is the positive-real first-remainder form of DLMF 5.11.1 and 5.11(ii)[H22, §5.11(i), equation (5.11.1), and §5.11(ii)], checked at <https://dlmf.nist.gov/5.11#ii>. Taylor's formula for $f(x) = (x - 1/2) \log x - x$ has $f'(M) = \log M - 1/(2M)$ and $f''(x) = 1/x + 1/(2x^2)$. If $M \geq 100$ and $\ell_s \leq \sqrt{M}$, the two shifts $1 - \ell_s$ and $\ell_s + 1/2$ keep $x \geq M - \sqrt{M}$, where $f''(x) \leq 2/M$. Their sum is $3/2$ and the sum of their squares is $2\ell_s^2 - \ell_s + 5/4 \leq 2(\ell_s + 1)^2$. The two Taylor remainders, their linear $-3/(4M)$ term and the two Stirling remainders are therefore bounded in absolute value by $8(\ell_s + 1)^2/M$. Hence

$$\log \Gamma(M - \ell_s + 1) + \log \Gamma(M + \ell_s + 1/2) = (2M + \frac{1}{2}) \log M - 2M + \log(2\pi) + O\left(\frac{(\ell_s + 1)^2}{M}\right). \quad (\text{LRS7})$$

We also give the Robin displacement calculation explicitly. ECL11 gives $\lambda_{\alpha} = -2 \log w$, where $w = (u + v_*)/2$. The exact dilation $x \mapsto c^{-2}x$ shows $\lambda(\alpha, c\beta) = \lambda(\alpha, \beta) + 2(\alpha + 2) \log c$; its

interaction and logarithmic-potential terms give the displayed coefficient. Together with ECL6 this gives

$$\lambda - \alpha\lambda_\alpha - \beta\lambda_\beta = -2\log c_{\text{cap}}. \quad (\text{LRS8})$$

At fixed n , define $G(h, N) = h\lambda(N/h, \pi n/(2h))$. Exact differentiation yields

$$\partial_h G = -2\log c_{\text{cap}}, \quad \partial_N G = -2\log w.$$

On the straight interpolation between $(h, N) = ((r - \epsilon)/2, n + \epsilon - 3/4)$ and $(r/2, n)$, the bounds $\alpha \geq 1/2$, $\beta/\alpha \in [\pi/3, \pi]$ still hold. Furthermore $h \geq r/3$, $|h - r/2| \leq 1/2$ and $|N - n| \leq 3/4$. LRS5 consequently gives

$$h\lambda(\alpha, \beta) = \frac{r}{2}\lambda(2n/r, \pi n/r) + O\left(1 + \log^+ \frac{n}{r+1}\right). \quad (\text{LRS9})$$

This proves the claimed logarithmic dependence without replacing the unbounded- α family by a bounded one.

Set, with the original EIQ/ECL Robin convention,

$$\psi(t) = (1+t)(1 - \log(1+t)) - \frac{t}{4}\lambda(2/t, \pi/t), \quad t > 0, \quad \psi(0) = a_0 = \log(4/\pi). \quad (\text{LRS10})$$

Substitute LRS6–9 in LRS4. The remaining difference of the main Gamma terms is

$$2M(1 - \log(M/n)) - (r/2)\lambda(2n/r, \pi n/r) - \frac{1}{2}\log(M/n) - \log(2\pi).$$

Its first two terms equal $2n\psi(r/n)$, and the last two are bounded for $r \leq 2n$. The scalar moment estimates at $r = 0, 1$ give the same bound: Stirling makes their centre $2(n+r)a_0 - \frac{1}{2}\log(n+r) + O(1 + (\ell_s + 1)^2/(n+r))$, and LRS14 below treats $\psi(r/n)$. Thus, including both small ranks,

$$\tau_{b,s,r} = 2n\psi(r/n) + O_{\text{actual}}\left(1 + \log^+ \frac{n}{r+1} + \frac{(\ell_s + 1)^2}{n+r}\right), \quad 0 \leq r \leq 2n. \quad (\text{LRS11})$$

The bound is summable with cost $O_{\text{actual}}(n)$: indeed $\sum_{r=0}^{2n} \log^+(n/(r+1)) \leq n$ by comparison with $\int_0^n \log(n/x) dx = n$, and $\sum_{r=0}^{2n} (\ell_s + 1)^2/(n+r) \leq 3(\ell_s + 1)^2 = O(n)$. The constants are uniform for $b = a, a-8$ and $s = 1, a$ on the actual family. In particular no $\frac{1}{2}\log n$ has been charged once per row.

6.2 The exact primitive and both endpoint expansions

Let $k(t)$ be the unique EIQ modulus at $(\alpha, \beta) = (2/t, \pi/t)$. Write $r_k = \sqrt{1-k^2}$ only in the formulas involving this modulus. EIQ5 gives $v_* = 2(1+t)/E(k)$ and $u = r_k v_*$. Applying ECL6 in LRS10 and cancelling its explicitly displayed terms gives the exact identity

$$\psi(t) = \log \frac{1+r_k}{E(k)} + t \log \frac{k}{E(k)}, \quad \psi'(t) = -\log(1+t) + \frac{1}{2}\log c_{\text{cap}} = \log \frac{k}{E(k)}. \quad (\text{LRS12})$$

The derivative follows directly from LRS8, so no derivative of an unstated minimizing selection is involved. Since $E(k) > 1 > k$ for $0 < k < 1$, ψ decreases on $(0, \infty)$.

The convergent elliptic series obtained by expanding the integrands in EIQ2 give

$$\frac{E(k)}{\sqrt{1-k^2}K(k)} = 1 + \frac{k^4}{16} + O(k^6).$$

The endpoint equation makes its left side $1+t$. Therefore $k = 2t^{1/4}(1 + O(\sqrt{t}))$ as $t \downarrow 0$. LRS12 first gives $\psi(0) = \log(4/\pi)$ and then

$$\psi'(t) = \frac{1}{4}\log t + a_0 + O(\sqrt{t}). \quad (\text{LRS13})$$

Integrating from zero, where $\int_0^t \log x \, dx = t \log t - t$, proves

$$\psi(t) = a_0 + \frac{t}{4} \log t + (a_0 - \frac{1}{4})t + O(t^{3/2}). \quad (\text{LRS14})$$

In the original opposite-energy convention $F = -E_{\min}$, EIQ30–31 and ECL14 give $F_\alpha = \mathcal{L}$, $\beta F_\beta = -2(\alpha + 1)$ and $F = -\lambda/2 - (\alpha + 1) + \alpha \mathcal{L}/2$. Thus

$$\frac{d}{dt} \left[\frac{t^2}{2} F(2/t, \pi/t) \right] = tF - \frac{t}{2} (\alpha F_\alpha + \beta F_\beta) = -\frac{t}{2} \lambda(2/t, \pi/t).$$

It follows that the continuous primitive with value zero at zero is

$$\mathfrak{F}(t) = \frac{t^2}{4} F(2/t, \pi/t) + \frac{(1+t)^2}{2} \left(\frac{3}{2} - \log(1+t) \right) - \frac{3}{4}, \quad \mathfrak{F}' = \psi. \quad (\text{LRS15})$$

The asserted value at zero follows either from ECL14 and the bounded support, which give $F(2/t, \pi/t) = O(1/t + |\log t|)$, or directly by integrating the continuous LRS14 and taking the displayed limit. Put

$$a_1 = \psi(1), \quad C_B = 4\mathfrak{F}(1) = 9 - 8 \log 2 + F(2, \pi).$$

In ECL16 the original outer coefficient is $C_\partial = 4a_0 - 10 - 2F(2, \pi) + 4\mathcal{L}(2, \pi)$. Since $a_1 = 2(1 - \log 2) - \lambda(2, \pi)/4$ and $F(2, \pi) = -\lambda(2, \pi)/2 - 3 + \mathcal{L}(2, \pi)$, substitution gives

$$C_\partial = 2C_B - 8a_1 + 4a_0. \quad (\text{LRS16})$$

6.3 The displaced covariance and its finite error interval

For integers $b_0 \geq 0, w \geq 1, b_0 + w \leq 2n$, define

$$\mathcal{J}(n; b_0, w) = 4n^2 [\mathfrak{F}((b_0 + w)/n) - \mathfrak{F}(b_0/n)].$$

The reference row sum with weights $1, 2, \dots, 2, 1$ differs from this quantity by at most

$$A(n; b_0, w) = 2n[\psi(b_0/n) - \psi((b_0 + w)/n)]. \quad (\text{LRS17})$$

Indeed on each unit interval the integral of the decreasing function $2n\psi(x/n)$ lies between its two endpoint values; its difference from their average is at most half their difference. Sum these inequalities and multiply by two. This proof applies also at zero, where ψ' is unbounded but ψ is continuous and monotone.

For a completely finite interval, let $e_r^\pm$ be the two bounds in LRS4 (or their scalar-moment counterparts) minus $2n\psi(r/n)$. For $\vartheta_0 = \vartheta_w = 1$ and $\vartheta_j = 2$ inside, the actual weighted sum S satisfies

$$\mathcal{J} + \sum_{j=0}^w \vartheta_j e_{b_0+j}^- - A \leq S \leq \mathcal{J} + \sum_{j=0}^w \vartheta_j e_{b_0+j}^+ + A. \quad (\text{LRS18})$$

This supplies directed finite errors, while LRS11 and LRS17 show that their sizes are $O_{\text{actual}}(q)$ on both original windows.

For the literal lower-grid covariance the cutoff $D = N - v$ has $D - Q = m - 1, m, m + q - 1, m + q$ at $N = q - 1, q, 2q - 1, 2q$. Telescoping LRS3 therefore gives exactly the weights just used, with $n = Q + \ell_s, b_0 = m, w = q$. Consequently

$$\mathcal{C}_{a,s}^- = \mathcal{J}(Q + \ell_s; m, q) + O_{\text{actual}}(q). \quad (\text{LRS19})$$

No lower natural window or order $a - 8$ has been inserted.

Set $n = Q + \ell_s$ and $H = m + q - n = 2\Delta - v - \ell_s$. Smoothness near one and integration of LRS14 at zero give

$$\begin{aligned}\mathfrak{F}(1 + H/n) &= C_B/4 + (H/n)a_1 + O(H^2/n^2), \\ \mathfrak{F}(m/n) &= a_0m/n + \frac{m^2}{8n^2} \log(m/n) + (a_0/2 - 3/16)m^2/n^2 + O((m/n)^{5/2}).\end{aligned}$$

Here $m, H = O_{\text{actual}}(a)$ and $n \asymp q \asymp a^2$. Every displayed nonlogarithmic quadratic term is $O(q)$; the last lower-end error after multiplication by $4n^2$ is $O(m^{5/2}/\sqrt{n}) = O(a^{3/2})$. Therefore

$$\mathcal{C}_{a,s}^- = C_B n^2 + 4n(Ha_1 - ma_0) + \frac{m^2}{2} \log(n/m) + O_{\text{actual}}(q). \quad (\text{LRS20})$$

Because $m = 16a + O_{\text{actual}}(1)$ and $n = a^2 + O(a)$,

$$\frac{m^2}{2} \log(n/m) = 128a^2 \log a + O_{\text{actual}}(a^2 + a \log a) = 128q \log a + O_{\text{actual}}(q).$$

Expanding $n = q - \Delta + \ell_s$ in the other terms of LRS20 gives

$$C_B q^2 + q\Delta(-2C_B + 8a_1 - 4a_0) + q\ell_s(2C_B - 4a_1) + O_{\text{actual}}(q).$$

Use LRS16 and $\Delta = 16a - 48$. We obtain

$$\begin{aligned}\mathcal{C}_{a,1}^- &= C_B q^2 - 16C_\partial a q + 128q \log a + O_{\text{actual}}(q), \\ \mathcal{C}_{a,a}^- &= C_B q^2 + (-16C_\partial + C_B/2 - a_1)aq + 128q \log a + O_{\text{actual}}(q).\end{aligned} \quad (\text{LRS21})$$

For the upper grid the original four cutoffs instead give $n = q + \ell_s, b_0 = 0, w = q$. Taylor expansion at $q/n = 1 - \ell_s/n$ yields

$$\mathcal{B}_a^{(s)} = C_B q^2 + \ell_s q(2C_B - 4a_1) + O_{\text{actual}}(q). \quad (\text{LRS22})$$

6.4 The original prefix and the correlated native tail

At an integer cutoff $0 \leq R < q$ define

$$\mathcal{P}(n; b_0, R) = 4n^2[\mathfrak{F}((b_0 + R)/n) - \mathfrak{F}(b_0/n)] + 2n\psi((b_0 + R)/n),$$

and retain the actual scalar centre

$$\mathfrak{P}_{a,s}(R) = \mathcal{P}(q + \ell_s; 0, R) - \mathcal{P}(Q + \ell_s; m, R). \quad (\text{LRS23})$$

For a determinant $f(N)$ the exact prefix operation is $\text{pre}_R f = 2f(q+R) - f(q) - f(q-1)$. Writing consecutive differences $\tau_r = f(q+r) - f(q+r-1)$ gives $\text{pre}_R f = \tau_0 + 2\sum_{r=1}^R \tau_r$. Thus the last $+2n\psi$ in LRS23 is necessary, including at $R = 0$. The scalar-prefix errors have finite intervals from LRS18 with the final row's weight increased from one to two. They are $O_{\text{actual}}(q)$ uniformly in every $R < q$.

At every endpoint the complete TAC6/TAC10 factorization, with the original conductor and inverse-dual maps, gives

$$\mathcal{B}_a^{(s)} = \mathcal{C}_{a,s}^- + \mathcal{R}e^C + \mathcal{T}_{a,s} + F_{L_a}^{(s)} + \Xi_{a,s}.$$

The all-cutoff exterior transport bounds LRC8–10 and LT26–28 apply to the same restrictions and attained quotients; their complete proof gives $\mathcal{R}e^C, \Xi_{a,s} = O_{\text{actual}}(q)$. LRP10–13 give $0 \leq F_{L_a}^{(s)} = O_{\text{actual}}(q)$ for the original nonzero quotient metric. Subtract LRS21 from LRS22 in this exact factorization to obtain

$$\mathcal{T}_{a,s} = 16C_\partial a q - 128q \log a + O_{\text{actual}}(q), \quad s = 1, a. \quad (\text{LRS24})$$

The two logarithmic terms have the opposite signs forced by the same equality.

For specificity, if an endpoint error satisfies $-L_N \leq e_N \leq U_N$, then the directed prefix and tail estimates are

$$\begin{aligned} -2L_{q+R} - U_q - U_{q-1} &\leq \text{pre}_R e \leq 2U_{q+R} + L_q + L_{q-1}, \\ -L_{2q-1} - L_{2q} - 2U_{q+R} &\leq \text{tail}_R e \leq U_{2q-1} + U_{2q} + 2L_{q+R}, \end{aligned} \quad (\text{LRS25})$$

where $\text{tail}_R e = e_{2q-1} + e_{2q} - 2e_{q+R}$. These follow term by term, with no change of signs, and are uniform $O_{\text{actual}}(q)$ by the all-cutoff bounds. The low-quotient prefix is between zero and its full return: TAC9–10 proves the quotient covariance increases with N , so each consecutive log-determinant difference is nonnegative. Its prefix and tail partition its nonnegative full weighted sum. Apply the exact endpoint factorization to pre_R , use this low bound, LRS25 and the two scalar-prefix comparisons. This proves

$$\mathcal{T}_{a,s}^{\leq R} = \mathfrak{P}_{a,s}(R) + O_{\text{actual}}(q), \quad 0 \leq R < q. \quad (\text{LRS26})$$

Subtract it from LRS24, preserving the same actual determinants, to get

$$\mathcal{T}_{a,s,R} = 16C_{\delta} a q - 128q \log a - \mathfrak{P}_{a,s}(R) + O_{\text{actual}}(q). \quad (\text{LRS27})$$

We give the growing-prefix leading term with an error calculation. Take $R = \lfloor a^{3/2}/\log a \rfloor$ and $n = Q + \ell_s$; then $m/R \rightarrow 0$, $R/n \rightarrow 0$ and $n_+ = n + \Delta$. From LRS13–14, uniformly for $0 < t \leq (R+m)/n$,

$$\psi(t) - t\psi'(t) = a_0 - t/4 + O(t^{3/2}).$$

Differentiate the integral expression for $\mathcal{P}(n; 0, R)$ in n and integrate from n to $n + \Delta$. This bounds its change by $O(\Delta R + \Delta R^2/n + \Delta)$. For the displacement at fixed n , exact integration gives

$$\begin{aligned} \mathcal{P}(n; 0, R) - \mathcal{P}(n; m, R) &= -4 \int_0^R \int_x^{x+m} \psi'(u/n) du dx \\ &\quad - 2 \int_R^{R+m} \psi'(u/n) du. \end{aligned}$$

Insert LRS13. Its double logarithmic integral is $mR \log(n/R) + O(mR)$: for $0 \leq z \leq m \leq R$,

$$\int_0^R \log((x+z)/R) dx = (R+z) \log(1+z/R) - z \log(z/R) - R = O(R).$$

The constant term costs $O(mR)$, the square-root remainder costs $O(mR\sqrt{(R+m)/n})$, and the final single integral costs $O(m(1 + \log(n/R)))$. All these errors and the preceding change in n are $o(a^{5/2})$. Since

$$mR \log(n/R) = \frac{16a^{5/2}}{\log a} \left[\frac{1}{2} \log a + \log \log a + O(1) \right] (1 + o(1)),$$

LRS26 proves

$$\mathcal{T}_{a,s}^{\leq R} \sim 8a^{5/2}. \quad (\text{LRS28})$$

For the precision of LRS27 the full LRS23 centre must still be used; LRS28 alone does not bound its omitted terms by $O(q \log a)$.

6.5 The exact complete-action signs and the logarithmic window

TAC13–15 retain the signed arithmetic correction and give

$$J_a = \mathcal{B}_a^{(s)} + \delta_{a,s}^{\text{ar}} + W_a - P_{a,-} - P_{a,+}. \quad (\text{LRS29})$$

Here $P_{a,\pm}$ are exactly the nonnegative penalties of TAC14. In the simple-quartet regime the original CAI28–35 estimates give

$$W_a - P_{a,-} - P_{a,+} = 4q \log q + (8 \log 2 - 4)q + O_h(a + \log q). \quad (\text{LRS30})$$

The subscript h here denotes the fixed original quartet polynomial of CAI1, $h(z) = \prod_{\zeta \in \{1/2 \pm \delta \pm i\gamma\}} (z - \zeta)$, not the parity index $h = \lfloor r/2 \rfloor$ above. For a direct check of the coefficient, CAI31's exact comparison window is

$$\mathcal{W}_q^\sigma = 2 \log \frac{(4q)!}{(2q)!} - 4q \log 2 + \log \frac{2q-1}{2(4q-1)}.$$

Positive-real Stirling gives $\mathcal{W}_q^\sigma = 4q \log q + (8 \log 2 - 4)q - \log 2 + O(1/q)$; the original directed errors CAI29 and penalties CAI28 give LRS30. Since $4q \log q = 8q \log a + 8q \log(1 + 1/a)$ and the last term is $O(a)$, LRS22 at $s = 1$ and LRS29–30 prove

$$J_a - \delta_{a,1}^{\text{ar}} = C_B q^2 + 8q \log a + O_{\text{actual},h}(q). \quad (\text{LRS31})$$

CAI5/TAC13 define $\mathcal{H}_a = \mathcal{B}_a^{(1)} - \mathcal{B}_a^{(a)}$ and $\delta_{a,a}^{\text{ar}} - \delta_{a,1}^{\text{ar}} = \mathcal{H}_a$. Consequently the extra $(C_B/2 - a_1)aq$ term at order $s = a$ cancels with that exact source correction, with the indicated negative sign. There is also an independent check against the original Gamma-source coefficient. At $(\alpha, \beta) = (2, \pi)$, ECL14 and AMT48 give

$$2C_B - 4a_1 = 10 - 8 \log 2 + 2F + \lambda = 4 - 8 \log 2 + 2\mathcal{L} = \mathcal{C}_\Gamma.$$

Thus the order difference in LRS22 is $\ell_s q \mathcal{C}_\Gamma + O(q)$, exactly opposite to AKS46's $\mathcal{H}_a = -\ell_a q \mathcal{C}_\Gamma + O(q)$. Equivalently subtract the two exact TAC10 identities and use the same correction to recover LRP18,

$$\Delta \mathcal{C} + \Delta \mathcal{T} + \Delta \Xi + \mathcal{H}_a = -\Delta F_L.$$

Neither $C_B q^2$ nor $\delta_{a,1}^{\text{ar}} = \delta_a^\sigma$ has been removed from the original action or its exterior benchmark.

7 The finite envelope from degree two and a quantitative full-row profile

Retain exactly ATA1–3 and the actual domain $j \geq 257$, $j\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q$. In particular $j = k + 4 \equiv 1 \pmod{4}$, $s \in \{1, j\}$, $0 \leq v \leq 80$, $m = 16j - 48 - v > 0$, and the original quartet, conductor, period and branch are fixed. The following proof extends ATA19 to every integer $2 \leq r \leq q + 32j$. Its finite endpoint is still the identical ATA17 expression: no trial, measure, conductor coefficient, lower minimum or original flag is changed.

Write $\lambda(\alpha, \beta)$ for the EIQ Euler constant of $V(x) = \beta\sqrt{x} - \alpha \log x$. For its support $[u^2, v_0^2]$ write $w = (u + v_0)/2$ and $c_{\text{cap}} = (v_0^2 - u^2)/4$. The exact Robin identities established by EQ20, EQ46 and EQ49 in the accompanying analytic norm proof are

$$\begin{aligned} \lambda &= 2(\alpha + 2) - 2\alpha \log w - 2 \log c_{\text{cap}}, & \lambda_\alpha &= -2 \log w, & \lambda_\beta &= \frac{2(\alpha + 2)}{\beta}, \\ \lambda(\alpha, t\beta) &= \lambda(\alpha, \beta) + 2(\alpha + 2) \log t & (t > 0). \end{aligned} \quad (\text{AER1})$$

The last identity also follows directly by the measure and support dilation $x \mapsto t^{-2}x$: it multiplies w by t^{-1} and c_{cap} by t^{-2} in the displayed first formula.

For the power-exponential minimum

$$\mathfrak{n}(Q, b) = \min_{p \text{ monic}, \deg p = b} \int_{\mathbb{R}} |p(y)|^2 |y|^{2Q} e^{-\pi|y|/2} dy$$

write $b = 2d + \varepsilon$, $\varepsilon \in \{0, 1\}$. Every integer $b \geq 2$ has $d \geq 1$, so the exact parity map and the substitution $y = q\sqrt{x}$ give

$$\mathbf{n}(Q, b) = q^{2(Q+b)+1} \min_{P \text{ monic, deg } P=d} \int_0^\infty |P(x)|^2 e^{-dV_{\alpha, \beta}(x)} dx, \quad \alpha = \frac{Q + \varepsilon - 1/2}{d}, \quad \beta = \frac{\pi q}{2d}. \quad (\text{AER2})$$

The finite Jensen and clipped-quantile proof EQ38–43 gives both endpoints

$$H^\pm(Q, b) = [2(Q + b) + 1] \log q - d\lambda(\alpha, \beta) + R^\pm(Q, b), \quad (\text{AER3})$$

where, with exactly the same parameters as in AER2,

$$R^- = \log(\pi c_{\text{cap}}), \quad R^+ = \log J_* + 4M + 4\log(1 + dW), \quad W = v_0^2 - u^2, \quad M = \frac{\alpha}{2\pi u^2}, \quad (\text{AER4})$$

$$C = 2\log(1 + v_0^2) + |\lambda|, \quad T_* = \max\{1, v_0, [8(\alpha + 2)/\beta]^2, 4C/\beta\}, \quad J_* = T_*^2 + 8/\beta^2.$$

The proof of these finite endpoints requires $d \geq 1$, with no large- d or small- α restriction. This is the exact point which permits the present extension down to $b = 2$.

For $Q \in \{q + \ell_s, q' + \ell_s\}$, $\ell_s = (s - 1)/4$, and $2 \leq b \leq 2q$, the parameters of AER2 lie in one fixed wedge

$$\alpha \geq a_0 > 0, \quad 0 < b_0 \leq \beta/\alpha \leq b_1 < \infty. \quad (\text{AER5})$$

The constants are independent of j, s, b : use $q/2 \leq Q \leq 2q$, $1 \leq d \leq q$ and $Q - 1/2 \leq Q + \varepsilon - 1/2 \leq Q + 1/2$. On the actual narrower wedge these ratios stay in a fixed neighbourhood of $\pi/2$, although only AER5 is needed here. The EIQ endpoint equations and their small-modulus expansion prove uniformly on AER5

$$0 < u_0 \leq u < v_0 \leq v_1 < \infty, \quad c_0(1 + \alpha)^{-1/2} \leq W \leq c_1, \quad (\text{AER6})$$

$$|\lambda| \leq C_0(1 + \alpha), \quad |\lambda_\alpha| + |\lambda_\beta| \leq C_1, \quad M \leq C_2\alpha, \quad 0 < J_* \leq C_3.$$

For clarity, the endpoint equation is $\sqrt{1 - \kappa^2}K(\kappa)/E(\kappa) = \alpha/(\alpha + 2)$. As $\alpha \rightarrow \infty$, its Taylor expansion gives $2/\alpha = \kappa^4/16 + O(\kappa^6)$, $u, v_0 = 2\alpha/\beta + o(1)$, and $v_0^2 - u^2 \asymp \alpha^{-1/2}$ uniformly for β/α in the compact interval in AER5. The remaining bounded- α region is compact, and the smooth positive endpoints have positive lower bounds there. These facts prove the first line of AER6. The remaining lines follow from AER1, $\log(1 + \alpha) = O(1 + \alpha)$, and AER4. In particular a value $\alpha \asymp q$ at $b = 2$ is allowed. Consequently

$$|R^-(Q, b)| = O(\log q), \quad |R^+(Q, b)| = O(q/b + \log q). \quad (\text{AER7})$$

Define the smooth centre

$$\mathcal{H}(Q, b) = 2(Q + b) \log Q - \frac{b}{2} \lambda(2Q/b, \pi Q/b). \quad (\text{AER8})$$

At fixed scale q , put $\alpha_0 = 2Q/b$, $\beta_0 = \pi q/b$. The exact multiplied parity shifts are

$$d(\alpha - \alpha_0) = \varepsilon - \frac{1}{2} + \varepsilon Q/b, \quad d(\beta - \beta_0) = \frac{\varepsilon \pi q}{2b}. \quad (\text{AER9})$$

The straight parameter segment between these pairs stays in a fixed wedge of the form AER5. AER6, AER9 and $d = b/2 - \varepsilon/2$ give

$$d\lambda(\alpha, \beta) = \frac{b}{2} \lambda(\alpha_0, \beta_0) + O(q/b).$$

For example the coefficient change $\varepsilon\lambda/2$ is $O(1 + Q/b)$ by AER6, and the derivative terms have that same order. There is no requirement that the parameter displacement itself tend to zero. By the exact dilation identity in AER1,

$$2(Q + b) \log q - \frac{b}{2} \lambda(2Q/b, \pi q/b) = \mathcal{H}(Q, b).$$

Combining this identity with AER3 and AER7 proves on the whole range $2 \leq b \leq 2q$

$$H^\pm(Q, b) = \mathcal{H}(Q, b) + O(q/b + \log q). \quad (\text{AER10})$$

The additional $\log q$ in AER3 is retained in this error.

Differentiate the explicit centre AER8 using AER1. The result is

$$\partial_Q \mathcal{H} = 2 \log Q + 2 \log w, \quad \partial_b \mathcal{H} = 2 \log Q + \log c_{\text{cap}}, \quad (\partial_Q - \partial_b) \mathcal{H} = \mathfrak{f}(b/Q). \quad (\text{AER11})$$

The last equality uses $w^2/c_{\text{cap}} = (u + v_0)/(v_0 - u)$, so the function is exactly the previously defined elliptic $\mathfrak{f}$. These are derivatives of an exact smooth formula, not derivatives of asymptotic errors.

For $Q_+ = q + \ell_s$, $Q_- = q' + \ell_s$, integrate AER11 along the straight line with constant $Q + b = Q_+ + r$:

$$\mathcal{H}(Q_+, r) - \mathcal{H}(Q_-, r + \Delta) = \int_{Q_-}^{Q_+} \mathfrak{f}\left(\frac{Q_+ + r - x}{x}\right) dx. \quad (\text{AER12})$$

Let $t_x = (Q_+ + r - x)/x$ and $t_0 = r/q$. All of these arguments remain in $(0, 4]$ on the retained range and numerical domain. The bound $|\mathfrak{f}'(t)| \leq C/t$ gives $|\mathfrak{f}(t_x) - \mathfrak{f}(t_0)| \leq C|\log(t_x/t_0)|$. Put $h_x = Q_+ - x \in [0, \Delta]$. The exact factorization is

$$\frac{t_x}{t_0} = (1 + h_x/r)(Q_+/x)(q/Q_+).$$

Using $\log(1 + y) \leq y$ for every $y \geq 0$ and $\log(Q_+/q) \leq \ell_s/q$ gives

$$|\log(t_x/t_0)| \leq h_x/r + h_x/x + \ell_s/q.$$

This estimate is valid even when Δ/r is large. Integrating it and using $x \geq Q_- \geq q/2$, $\Delta = O(j)$, $\ell_s = O(j)$ proves

$$\mathcal{H}(Q_+, r) - \mathcal{H}(Q_-, r + \Delta) = \Delta \mathfrak{f}(r/q) + O(\Delta^2/r + \Delta^2/q + \Delta \ell_s/q) = \Delta \mathfrak{f}(r/q) + O(q/r + 1). \quad (\text{AER13})$$

The actual lower degree is $m + r = r + \Delta - v$. On the interval between this degree and $r + \Delta$, AER6 and AER11 give $|\partial_b \mathcal{H}| = O(\log q)$; its length is the fixed original integer v . Replacing $r + \Delta$ by $m + r$ therefore costs $O_{\text{actual}}(\log q)$. Both degrees are at least two on the stated domain. The monic comparison proof ATG1–28 applies to every degree at most $2q$, with $b^\pm = O_{\text{actual}}(\log q)$; ATA15 gives the same order for both conductor logarithms. Insert AER10 and AER13 into the unchanged finite endpoint ATA17 to obtain

$$0 \leq \gamma_{r,s} \leq U_{j,s,r} = \Delta \mathfrak{f}(r/q) + O_{\text{actual}}(q/r + \log q), \quad 2 \leq r \leq q + 32j. \quad (\text{AER14})$$

The exact optimized-trial inequality $0 \leq \log(\tau_{r,s}/\eta_{r,s}) \leq U_{j,s,r} - \gamma_{r,s}$ holds on this same range by ATA18.

7.1 Quantitative full-row consequence on the actual total

The independently derived native-return identity LRS24 in the lower covariance calculation, at these same source orders and degree, is

$$\mathcal{T}_{j,s} = 16C_\partial j q - 128q \log j + O_{\text{actual}}(q) = \Delta q C_\partial + O_{\text{actual}}(q \log q). \quad (\text{AER15})$$

The second equality uses $\Delta = 16j - 48$ and retains its exact constant $48C_\partial q$ in the $O(q)$ term. Here $C_\partial = 2 \int_0^1 \mathfrak{f}$ is the exact mass identity ATA20, and $\mathcal{T}$ is the same original HJR weighted total. Thus this is a quantitative input about that original total, not an independently selected normalization of the rows.

For the two exceptional small degrees put

$$E_{01} = E_{j,s,0}^{\text{NIB}} + 2E_{j,s,1}^{\text{NIB}}, \quad \mathfrak{D}_{j,s}^{\text{all}} = \sum_{r=2}^q \vartheta_r U_{j,s,r} - \mathcal{T}_{j,s} + E_{01}. \quad (\text{AER16})$$

NIB15 gives $E_{01} = O_{\text{actual}}(j \log q)$. Exactly as in ATA24,

$$0 \leq \sum_{r=2}^q \vartheta_r (U_{j,s,r} - \gamma_{r,s}) \leq \mathfrak{D}_{j,s}^{\text{all}}. \quad (\text{AER17})$$

For the decreasing function $\mathfrak{f}$, its Riemann-sum error is bounded by the missing integral on $[0, 1/q]$ plus the last interval of length $1/q$. The expansion at zero bounds this by $O((\log q)/q)$. Removing the $r = 1$ term and reducing the $r = q$ weight from two to one costs $O(\log q)$ before multiplication by Δ . Summing AER14 therefore gives

$$\sum_{r=2}^q \vartheta_r U_{j,s,r} = \Delta q C_{\partial} + O_{\text{actual}}(q \log q).$$

Combining this equality with AER15–17 proves $0 \leq \mathfrak{D}_{j,s}^{\text{all}} = O_{\text{actual}}(q \log q)$. The triangle inequality used in ATA25 now yields

$$\sum_{r=2}^q \vartheta_r |\gamma_{r,s} - \Delta \mathfrak{f}(r/q)| = O_{\text{actual}}(q \log q). \quad (\text{AER18})$$

At $r = 1$, NIB15 and $\mathfrak{f}(1/q) = O(\log q)$ give $|\gamma_{1,s} - \Delta \mathfrak{f}(1/q)| \leq E_{j,s,1}^{\text{NIB}} + \Delta \mathfrak{f}(1/q) = O_{\text{actual}}(j \log q)$. The exact row $r = 0$ is a separate nonnegative atom, with $0 \leq \gamma_{0,s} \leq E_{j,s,0}^{\text{NIB}} = O_{\text{actual}}(j \log q)$; the undefined value $\mathfrak{f}(0)$ is never introduced. Thus

$$\boxed{\gamma_{0,s} + \sum_{r=1}^q \vartheta_r |\gamma_{r,s} - \Delta \mathfrak{f}(r/q)| = O_{\text{actual}}(q \log q).} \quad (\text{AER19})$$

The optimized-trial slack is controlled on its proved range by

$$0 \leq \sum_{r=2}^q \vartheta_r \log(\tau_{r,s}/\eta_{r,s}) \leq \mathfrak{D}_{j,s}^{\text{all}} = O_{\text{actual}}(q \log q). \quad (\text{AER20})$$

No estimate for $\tau_{0,s}$ or $\tau_{1,s}$ is inferred from the NIB bound for $\eta_{0,s}$ or $\eta_{1,s}$.

The exact determinant identity ATA26 and this total absolute error give the uniform cumulative estimate

$$\sup_{0 \leq n \leq q} \left| \frac{1}{jq} \sum_{r=0}^n \gamma_{r,s} - 16 \int_0^{n/q} \mathfrak{f}(t) dt \right| = O_{\text{actual}} \left(\frac{\log q}{j} \right). \quad (\text{AER21})$$

The difference $\Delta/j - 16 = -48/j$ contributes $O(1/j)$, and the uniform Riemann error contributes $O((\log q)/q)$, both absorbed by the displayed bound. No rate for moving endpoint convergence $R'/q \rightarrow a$ or $Q'/q \rightarrow b$ is inferred without specifying their own rates.

Finally, for $\mathcal{J} \subset \{0, \dots, q\}$ with $h = |\mathcal{J}|$, use NIB for rows zero and one and AER14 for every larger row. Its corresponding finite envelope budget obeys

$$\mathcal{E}_s^{\text{all}}(\mathcal{J}) \leq E_{01} + 2\Delta q \int_0^{h/q} \mathfrak{f}(t) dt + O_{\text{actual}}(q \log q). \quad (\text{AER22})$$

The same four-copy and quotient identities ATA34 apply to this budget. It is $o(jq)$ for every $h = o(q)$, and, at $h \leq 1 + q/\log j$, it is $O_{\text{actual}}(j^3 \log \log j / \log j + q \log q)$. Every independent proper-source displacement in ATA35–40 keeps its original degree- k metric and its coefficient $4b\Delta_k$.

8 The exact profile–covariance dictionary and quantitative joint return

All objects below retain the original simple quartet, its period, its conductor coefficients, and the two source orders $s = 1, a$. Put $a \equiv 1 \pmod{4}$, $a \geq 257$, $q = (a + 1)^2$, $Q = (a - 7)^2$, $\Delta = q - Q = 16a - 48$, $m = \Delta - v$, and $\ell_s = (s - 1)/4$. The original full polynomial and the attained old-relation minimum are those of NTG; none is replaced by a diagonal norm. This section connects the two web derivations received on 18 September 2026. Their original statements and their reviewed corrections are retained in the accompanying source records. The equilibrium identities used below were proved for these exact conventions in EIQ, ECL6, ECL14–17, and GRE6–8.

Every finite comparison below is on the intersection of the two original root-comparison domains. Explicitly, retain $0 < \delta < 1/2$, $\gamma > 2$, $v = \text{ord}_0 E_A \in \{0, \dots, 80\}$, $\mu_v \neq 0$, and impose

$$\begin{aligned} a\sqrt{\delta^2 + \gamma^2} &\leq 2^{-33}q, & n_b &= (b + 1)^2 + \ell_s \geq 100, \\ \max\{b\sqrt{\delta^2 + \gamma^2}, 2\ell_s + 1/2\} &\leq 2^{-17}n_b & (b = a, a - 8). \end{aligned} \quad (\text{PCB0})$$

These are the stated attained-envelope and retained-Gamma covariance guards. They hold eventually for every fixed actual quartet and both orders $s = 1, a$. In this domain $Q > 0$, $m = \Delta - v > 0$ and $0 < R < q$; the lower row cutoffs obey the original whole-polynomial degree bounds. The scalar identities PCB1–6 hold for every $t > 0$ independently of these finite-family guards.

8.1 An exact dictionary of the two scalar functions

For $t > 0$, let $u(t), v_*(t)$ be the endpoints for $\alpha = 2/t$, $\beta = \pi/t$, and set

$$w = (u + v_*)/2, \quad c = (v_*^2 - u^2)/4, \quad \rho = u/v_*, \quad k = (1 - \rho^2)^{1/2}. \quad (\text{PCB1})$$

The actual EIQ equations give $uK(k) = 2$ and $v_*E(k) = 2(1 + t)$. In particular $\rho K(k)/E(k) = 1/(1 + t)$; this is precisely the implicit parameter in the attained-tail response. Retain the actual Robin constant

$$\lambda = 2(\alpha + 2) - 2\alpha \log w - 2 \log c, \quad \lambda_\alpha = -2 \log w, \quad \lambda_\beta = 2(\alpha + 2)/\beta. \quad (\text{PCB2})$$

These are GRE6–8, including the endpoint motion and the original Robin sign. Thus $\lambda - \alpha\lambda_\alpha - \beta\lambda_\beta = -2 \log c$. Differentiating the explicit function

$$\psi(t) = (1 + t)(1 - \log(1 + t)) - \frac{t}{4}\lambda(2/t, \pi/t) \quad (\text{PCB3})$$

by the chain rule proves

$$\psi'(t) = -\log(1 + t) + \frac{1}{2} \log c = \log(k/E(k)), \quad 2\psi(t) - 2(1 + t)\psi'(t) = \log \frac{w^2}{c} = \log \frac{1 + \rho}{1 - \rho} =: \mathfrak{f}(t). \quad (\text{PCB4})$$

For the middle equality substitute PCB2: the terms $2(1 + t)$ cancel $t(\alpha + 2)$ exactly because $t\alpha = 2$. The remaining expression is $2 \log w - \log c$. This proves the exact map between the covariance function and the attained-row profile, rather than identifying them by their integrated constants.

Write $a_0 = \psi(0) = \log(4/\pi)$, $a_1 = \psi(1)$, and $C_B = 4 \int_0^1 \psi(t) dt$. The convergent elliptic-integral expansions at $k = 0$, or the explicit expansion in the covariance proof, give

$$\psi(t) = a_0 + \frac{1}{4}t \log t + (a_0 - \frac{1}{4})t + O(t^{3/2}), \quad \mathfrak{f}(t) = -\frac{1}{2} \log t + O(\sqrt{t}). \quad (\text{PCB5})$$

In particular the improper integral of $\mathfrak{f}$ exists. Integrating PCB4 on $[\epsilon, 1]$ and then sending ϵ to zero gives

$$2 \int_0^1 \mathfrak{f}(t) dt = 8 \int_0^1 \psi(t) dt - 8a_1 + 4a_0 = 2C_B - 8a_1 + 4a_0 = C_\partial. \quad (\text{PCB6})$$

The final equality is the original ECL16 coefficient expressed by its Robin/energy primitive. All constants, including the factor two for the two-endpoint return, are therefore the same in the two derivations.

8.2 A quantitative improvement obtained by using both results

The native row is $\gamma_{r,s} = \log(\eta_{r,s}/\nu_{m+r,s}^-) \geq 0$, where η is the complete attained preceding-relation minimum. Its original weights are $\vartheta_0 = \vartheta_q = 1$ and $\vartheta_r = 2$ for $0 < r < q$. Set $R = \lfloor a^{3/2}/\log a \rfloor$ and $B = \{R+1, \dots, q\}$. Every sum in this subsection has this finite range, not the larger range available to the envelope for the proper-source displacement.

The audited covariance row estimate and the original finite endpoint factorization give, uniformly in this range,

$$\left| \sum_{r \in B} \vartheta_r \gamma_{r,s} - \sum_{r \in B} \vartheta_r \{G(q + \ell_s, r) - G(Q + \ell_s, m + r)\} \right| \leq E_{a,s}^{\text{nat}}(R) = O_{\text{actual}}(q), \quad (\text{PCB7})$$

$$G(n, r) = 2n\psi(r/n).$$

Here E^{nat} is a finite certified allowance: sum the absolute row errors from the covariance norm intervals over the two grids, and add the absolute endpoint allowances for the conductor and analytic quotient at $q + R, 2q - 1, 2q$, together with the low-factor tail bound. Those row errors are bounded by $C_{\text{actual}}\{1 + \log^+(n/(r+1)) + (\ell_s + 1)^2/(n+r)\}$; the lower row uses $m + r$. Their sums are $O(q)$. The transport endpoint allowances are $O(q)$ at every rank, and the low-factor tail is between zero and its proved $O(q)$ full return. This proves PCB7 by the exact TAC factorization, with no replacement of the actual tail by independent endpoint choices.

There is a direct finite-difference identity behind the leading profile:

$$(\partial_n - \partial_r)G(n, r) = \mathfrak{f}(r/n), \quad G(q + \ell_s, r) - G(Q + \ell_s, r + \Delta) = \int_{Q+\ell_s}^{q+\ell_s} \mathfrak{f}\left(\frac{q + \ell_s + r - x}{x}\right) dx. \quad (\text{PCB8})$$

It follows by differentiating $G(x, q + \ell_s + r - x)$ and using PCB4. The function $t\mathfrak{f}'(t)$ is bounded on $0 < t \leq 4$: away from zero this follows from the nonzero derivative in the EIQ implicit endpoint equation; at zero the convergent expansion in powers of $\sqrt{t}$ in PCB5 gives $t\mathfrak{f}'(t) = -1/2 + O(\sqrt{t})$. Also $|\psi'(t)| \leq C(1 + |\log t|)$ on that interval by PCB4–5. For $x \in [Q + \ell_s, q + \ell_s]$, the logarithmic difference of the argument in PCB8 from r/q is bounded by $C\{\Delta/r + (\Delta + \ell_s)/q\}$. The mean-value integral for $\mathfrak{f}$ with respect to $\log t$ now proves

$$|G(q + \ell_s, r) - G(Q + \ell_s, m + r) - \Delta \mathfrak{f}(r/q)| \leq C_{\text{actual}}(q/r + \log q), \quad r \in B. \quad (\text{PCB9})$$

The difference between $m + r$ and $r + \Delta$ is exactly the fixed integer v and costs at most $2v \sup |\psi'| = O_{\text{actual}}(\log q)$. The other terms use $\Delta^2 = O(q)$ and $\Delta \ell_s = O(q)$. Summing PCB9 gives a finite allowance $E_{a,s}^{\text{cmp}}(R) = O_{\text{actual}}(q \log q)$.

Let $U_{a,s,r}$ be the original un-intersected conductor/optimized-trial upper endpoint in the attained-tail proof, with its full conductor factor, original mass cancellation and lower-ideal minimum. A smaller endpoint obtained by intersecting with NIB may bound the attained row but need not bound this particular trial; it is not used for the trial-slack statements. Put

$$E_{a,s}^U(R) = \sum_{r \in B} \vartheta_r |U_{a,s,r} - \Delta \mathfrak{f}(r/q)|, \quad E_{a,s}^T(R) = E_{a,s}^{\text{nat}}(R) + E_{a,s}^{\text{cmp}}(R). \quad (\text{PCB10})$$

The finite envelope gives $0 \leq \gamma \leq U$ and $E^U = O_{\text{actual}}(q \log q)$. Combining PCB7–10 gives

$$0 \leq \sum_{r \in B} \vartheta_r (U_{a,s,r} - \gamma_{r,s}) \leq E_{a,s}^U(R) + E_{a,s}^T(R), \quad (\text{PCB11})$$

$$\sum_{r \in B} \vartheta_r |\gamma_{r,s} - \Delta \mathfrak{f}(r/q)| \leq 2E_{a,s}^U(R) + E_{a,s}^T(R) = O_{\text{actual}}(q \log q).$$

Indeed subtract the actual tail sum from the upper-endpoint sum for the first inequality. For the second use pointwise $|\gamma - h| \leq (U - \gamma) + |U - h|$, valid whenever $\gamma \leq U$. This uses the actual one-sided envelope, not a pointwise lower estimate. If $\tau_{r,s}$ denotes the optimized upper-fibre trial norm, the exact projection identity from the attained-tail proof further gives

$$0 \leq \sum_{r \in B} \vartheta_r \log(\tau_{r,s}/\eta_{r,s}) \leq E_{a,s}^U(R) + E_{a,s}^T(R) = O_{\text{actual}}(q \log q). \quad (\text{PCB12})$$

Thus the two independently derived results together replace the earlier unquantified $o(aq)$ tail-profile and trial-slack conclusion by a proved $O(q \log q)$ allowance on the unchanged retained tail. Relative to aq this is $O_{\text{actual}}(\log a/a)$. It remains on the scale relevant to the final exterior benchmark; no sign at that scale has been inferred.

The estimate receives every original subinterval of B , since restricting an absolute-error sum can only decrease it. It gives no individual allocation coefficient: the exact identities still read $S_K + S_R = \mathcal{T}$ and $4S_K = \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A$. The independent degree- k proper source, original arithmetic correction and projected pairing remain in their own proved receiving maps.

8.3 The full row range and a uniform rate for actual mixed determinants

The attained-envelope proof also applies for every integer $2 \leq r \leq q + 32a$. Its finite norm estimate uses $h = \lfloor r/2 \rfloor \geq 1$, and the error $O(q/r + \log q)$ explicitly permits r fixed. No expansion in a small Δ/r is used: in PCB8, $\log(1+x) \leq x$ bounds the logarithmic argument change for all $x \geq 0$. The parity shifts in the exact Robin parameters cost $O(q/r)$, and the original root comparison holds before minimization over the full stated polynomial range. These details are proved in the attained-envelope section; they extend its initially stated tail range without changing its objects.

Retain row zero as its actual finite atom $\gamma_{0,s}$, rather than assigning a value to $\mathfrak{f}(0)$. The existing NIB estimates give

$$0 \leq \gamma_{0,s}, \gamma_{1,s} \leq C_{\text{actual}} a \log q, \quad h_r := \Delta \mathfrak{f}(r/q) \quad (1 \leq r \leq q). \quad (\text{PCB13})$$

Because $\mathfrak{f}$ is positive and decreasing and $\mathfrak{f}(t) \leq C(1 + \log(1/t))$ on $(0, 1]$, comparison on each mesh interval, including the first improper integral, gives

$$\left| \sum_{r=1}^q \vartheta_r h_r - \Delta q C_{\partial} \right| \leq C \Delta \log q. \quad (\text{PCB14})$$

For clarity, the lower rectangle sum $q^{-1} \sum_{r=1}^q \mathfrak{f}(r/q)$ is below the integral. All discrepancies after the first cell telescope to at most $\mathfrak{f}(1/q)/q$, and the first cell costs $\int_0^{1/q} \mathfrak{f}(t) dt = O(\log q/q)$. Multiplication by $2\Delta q$, followed by subtraction of the single extra endpoint value $\Delta \mathfrak{f}(1)$, proves PCB14.

The now evaluated native total, with its original sign, is

$$\mathcal{T}_{a,s} = 16C_{\partial} a q - 128q \log a + O_{\text{actual}}(q) = \Delta q C_{\partial} + O_{\text{actual}}(q \log q). \quad (\text{PCB15})$$

The second equality uses $\Delta = 16a - 48$; the omitted term is exactly $48C_{\partial} q$ together with the displayed $-128q \log a$, and no favorable sign is selected. PCB13–15 imply $|\sum_{r=2}^q \vartheta_r (\gamma_{r,s} - h_r)| = O(q \log q)$. The all-row envelope has $\sum_{r=2}^q \vartheta_r |U_{a,s,r} - h_r| = O(q \log q)$, since $\sum_{r=2}^q q/r = O(q \log q)$. Apply the two elementary inequalities used in PCB11 to this whole finite range, and then restore row one using PCB13. The result is

$$\boxed{\sum_{r=1}^q \vartheta_r |\gamma_{r,s} - \Delta \mathfrak{f}(r/q)| = O_{\text{actual}}(q \log q), \quad 0 \leq \gamma_{0,s} = O_{\text{actual}}(a \log q).} \quad (\text{PCB16})$$

The optimized-trial slack has the same $O(q \log q)$ bound over $2 \leq r \leq q$. No trial-slack assertion for rows zero or one is inferred from NIB, which bounds the attained row rather than that particular trial.

Let $0 \leq p \leq r \leq q$ be any original indices and $Y = [x_{p+1}, \dots, x_r]$. The complete NTG determinant identity gives

$$\log \det(I + Y^* \Omega_p^{-1} Y) = \sum_{j=p+1}^r \gamma_{j,s}.$$

Restricting the absolute sum in PCB16 bounds the difference from $\Delta \sum_{j=p+1}^r \mathfrak{f}(j/q)$ by $O(q \log q)$, uniformly in both indices. The same monotone integral comparison as PCB14 costs at most $O(\Delta \log q)$ uniformly, even if $p = 0$. Since $\Delta/a = 16 - 48/a$, we obtain

$$\sup_{0 \leq p \leq r \leq q} \left| \frac{\log \det(I + Y^* \Omega_p^{-1} Y)}{aq} - 16 \int_{p/q}^{r/q} \mathfrak{f}(t) dt \right| = O_{\text{actual}}(\log a/a). \quad (\text{PCB17})$$

Every old inverse and every later cross pairing in this determinant is retained. The two-endpoint tail has the corresponding coefficient 32; the last single row costs $O(q \log q)$ by PCB16 and so obeys the same uniform rate. These are quantitative versions of the actual mixed determinant profile, not pointwise lower estimates on every row.

Finally, for any set $I \subset \{0, \dots, q\}$ of cardinality $d \geq 1$, decreasingness of $\mathfrak{f}$ places its largest d positive-index values first. The bound in PCB14's proof or $\sum_{j=1}^d \log(q/j) \leq d\{1 + \log(q/d)\}$ therefore gives

$$0 \leq \mathcal{T}_{a,s}^I \leq C_{\text{actual}}\{ad(1 + \log(q/d)) + q \log q\}. \quad (\text{PCB18})$$

For $d = q + 1$ use the same statement with d replaced by q , plus the row-zero allowance in PCB13. For the empty set the return is exactly zero. All original two allocations and four CPQ complements receive this bound with their established coefficients one and four, respectively, through the existing exact flag maps. In particular every $d = o(q)$ set is $o(aq)$. The signed allocation or projected arithmetic pairing still requires its own calculation; PCB18 supplies no replacement sign.

8.4 Exact derivative and uniform bounds for the elliptic profile

For $0 < \rho < 1$, retain the original modulus $k = \sqrt{1 - \rho^2}$ and define

$$D_\rho(\theta) = \cos^2 \theta + \rho^2 \sin^2 \theta, \quad K = K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{D_\rho(\theta)}}, \quad E = E(k) = \int_0^{\pi/2} \sqrt{D_\rho(\theta)} d\theta.$$

Put

$$A = \int_0^{\pi/2} \frac{\sin^2 \theta}{\sqrt{D_\rho(\theta)}} d\theta, \quad B = \int_0^{\pi/2} \frac{\cos^2 \theta}{\sqrt{D_\rho(\theta)}} d\theta, \quad \mathfrak{q}(\rho) = \frac{\rho K}{E}.$$

All these functions are C^∞ on $(0, 1)$: on each compact subinterval, D_ρ has a positive lower bound, and every derivative of each integrand has a uniform integrable bound. Directly,

$$A + B = K, \quad K - E = (1 - \rho^2)A, \quad E - \rho^2 K = (1 - \rho^2)B, \quad E_\rho = \rho A.$$

The function $\sin \theta \cos \theta / \sqrt{D_\rho(\theta)}$ vanishes at both integration endpoints and has derivative

$$\frac{\cos^4 \theta - \rho^2 \sin^4 \theta}{D_\rho(\theta)^{3/2}}.$$

Its integral is zero. Since

$$\rho^2 \sin^2 \theta - \cos^2 \theta D_\rho(\theta) = \rho^2 \sin^4 \theta - \cos^4 \theta,$$

this proves

$$\int_0^{\pi/2} \frac{\sin^2 \theta}{D_\rho(\theta)^{3/2}} d\theta = \frac{B}{\rho^2}, \quad K_\rho = -\frac{B}{\rho}.$$

Consequently the derivative relevant to the implicit equation is exactly

$$\mathfrak{q}'(\rho) = \frac{K - B}{E} - \frac{\rho^2 K A}{E^2} = \frac{(1 - \rho^2)AB}{E^2} = \frac{(K - E)(E - \rho^2 K)}{(1 - \rho^2)E^2} > 0.$$

In particular this derivative never vanishes at an interior parameter. As $\rho \uparrow 1$, dominated convergence gives $K, E \rightarrow \pi/2$ and $\mathfrak{q}(\rho) \rightarrow 1$. To check the other endpoint without an elliptic asymptotic formula, Cauchy–Schwarz and $u = \tan \theta$ give

$$K^2 \leq \frac{\pi}{2} \int_0^{\pi/2} \frac{d\theta}{D_\rho(\theta)} = \frac{\pi}{2} \int_0^\infty \frac{du}{1 + \rho^2 u^2} = \frac{\pi^2}{4\rho}.$$

Also $E \geq \int_0^{\pi/2} \cos \theta d\theta = 1$. Thus $0 < \mathfrak{q}(\rho) \leq (\pi/2)\sqrt{\rho} \rightarrow 0$ as $\rho \downarrow 0$. Strict monotonicity and these two limits show that $\mathfrak{q}$ is a C^∞ bijection from $(0, 1)$ to $(0, 1)$ with C^∞ inverse. Therefore the original equation

$$\mathfrak{q}(\rho(t)) = \frac{1}{1+t}, \quad t > 0,$$

defines a unique C^∞ function $\rho(t)$, strictly decreasing, and $f(t) = \log((1 + \rho(t))/(1 - \rho(t)))$ is C^∞ . Differentiating the exact equation gives

$$\rho'(t) = -\frac{\mathfrak{q}(\rho(t))^2}{\mathfrak{q}'(\rho(t))}, \quad f'(t) = -\frac{2\mathfrak{q}(\rho(t))^2}{(1 - \rho(t)^2)\mathfrak{q}'(\rho(t))}.$$

Using $t = (1 - \mathfrak{q})/\mathfrak{q}$, this becomes

$$tf'(t) = -\frac{2\rho K(E - \rho K)}{(K - E)(E - \rho^2 K)} < 0.$$

Every K, E, ρ in this expression has its original value at t .

Here is a uniform bound requiring no numerical elliptic evaluation. Set $e_\rho = E/K$. The already proved inequality $\mathfrak{q} < 1$ gives $e_\rho > \rho$. For any positive x, y , $(x - y)(x^{-1/2} - y^{-1/2}) \leq 0$. Apply this with $x = D_\rho(\theta)$ and $y = D_\rho(\phi)$ and integrate over $[0, \pi/2]^2$. Expanding the four terms yields

$$2\frac{\pi}{2}E - 2K \int_0^{\pi/2} D_\rho(\theta) d\theta \leq 0.$$

Since the last integral is $\pi(1 + \rho^2)/4$, this proves

$$\rho < e_\rho \leq \frac{1 + \rho^2}{2} < 1.$$

The polynomial $p(x) = (1 - x)(x - \rho^2)$ has derivative $1 + \rho^2 - 2x \geq 0$ on $[\rho, (1 + \rho^2)/2]$. It follows that

$$(1 - e_\rho)(e_\rho - \rho^2) \geq p(\rho) = \rho(1 - \rho)^2, \quad 2\rho(e_\rho - \rho) \leq \rho(1 - \rho)^2.$$

Substitution in the exact derivative proves

$$0 < -tf'(t) = \frac{2\rho(e_\rho - \rho)}{(1 - e_\rho)(e_\rho - \rho^2)} \leq 1 \quad (t > 0).$$

In particular, for any $0 < t_1 < t_2$, the fundamental theorem of calculus gives the finite-difference estimate

$$0 < f(t_1) - f(t_2) \leq \log(t_2/t_1).$$

For completeness, all constants in the small- t assertion can also be bounded directly from these integrals. Since $D_\rho \leq 1$, $A, B \geq \pi/4$ and $E \leq \pi/2$, whence $AB/E^2 \geq 1/4$. On the other hand, $AB \leq (A+B)^2/4 = K^2/4$ and $E > \rho K$, whence $AB/E^2 \leq 1/(4\rho^2)$. Thus

$$\frac{1-\rho^2}{4} \leq \mathfrak{q}'(\rho) \leq \frac{1-\rho^2}{4\rho^2}.$$

Write $d_\rho = 1 - \rho$ and $b_\rho = 1 - \mathfrak{q}(\rho)$. Integration from ρ to 1 gives the exact bounds

$$\frac{d_\rho^2(3-d_\rho)}{12} \leq b_\rho \leq \frac{d_\rho^2}{4\rho}.$$

Also $K \geq E$, so $\mathfrak{q} \geq \rho$ and

$$\frac{d_\rho^2(3-d_\rho)}{12} \leq t = \frac{b_\rho}{\mathfrak{q}} \leq \frac{d_\rho^2}{4\rho^2}.$$

If $\rho \leq 1/2$, the lower bound for b_ρ at $\rho = 1/2$ and monotonicity imply $t \geq b_\rho \geq 5/96 > 1/20$. Hence $0 < t \leq 1/20$ implies $0 < d_\rho < 1/2$ and

$$d_\rho \leq \sqrt{24/5} \sqrt{t}.$$

Set $W = t(1+\rho)^2/d_\rho^2$. The preceding bounds give

$$W \geq \frac{(2-d_\rho)^2(3-d_\rho)}{12} = 1 - \frac{4}{3}d_\rho + \frac{7}{12}d_\rho^2 - \frac{1}{12}d_\rho^3 \geq 1 - \frac{4}{3}d_\rho, \quad W \leq \left(\frac{1+\rho}{2\rho}\right)^2.$$

Using $\log(1+x) \leq x$ for $x \geq 0$ and $\log(1-x) \geq -x/(1-x)$ for $0 \leq x < 1$, with $d_\rho \leq 1/2$, yields

$$-2d_\rho \leq \frac{1}{2} \log W = f(t) + \frac{1}{2} \log t \leq d_\rho.$$

Thus, with its original constant term retained,

$$\left| f(t) + \frac{1}{2} \log t \right| \leq 2\sqrt{24/5} \sqrt{t} < \frac{22}{5} \sqrt{t} \quad (0 < t \leq 1/20).$$

To obtain the derivative remainder independently of differentiating an O -term, set $T = -tf'(t) = 2\mathfrak{q}b_\rho/((1-\rho^2)\mathfrak{q}')$. The same bounds give

$$T \geq \frac{2\rho^3(2+\rho)}{3(1+\rho)^2} \geq \frac{\rho^3}{2}, \quad T \leq \frac{2}{\rho(1+\rho)^2}.$$

The second lower inequality follows from $4(2+\rho) - 3(1+\rho)^2 = (1-\rho)(3\rho+5) \geq 0$. For $1/2 \leq \rho < 1$, the lower estimate implies $T - 1/2 \geq -3d_\rho/2$, since $(1-d_\rho)^3 \geq 1 - 3d_\rho$. For the upper estimate,

$$\frac{2}{\rho(1+\rho)^2} - \frac{1}{2} = \frac{d_\rho(\rho^2 + 3\rho + 4)}{2\rho(1+\rho)^2} \leq \frac{32}{9}d_\rho < 4d_\rho.$$

Consequently

$$\left| tf'(t) + \frac{1}{2} \right| \leq 4\sqrt{24/5} \sqrt{t} < 9\sqrt{t} \quad (0 < t \leq 1/20),$$

equivalently $f'(t) = -1/(2t) + O(t^{-1/2})$ with the displayed explicit constant. These arguments prove both $f(t) = -\frac{1}{2} \log t + O(\sqrt{t})$ and its derivative assertion directly, without differentiating an unproved remainder estimate.

Finally, when $0 < t \leq 4$, one has $\mathfrak{q} \geq 1/5$; the previously proved $\mathfrak{q} \leq (\pi/2)\sqrt{\rho}$ gives $\rho \geq 4/(25\pi^2)$. The lower bound $T \geq \rho^3/2$ therefore supplies, if a strictly positive uniform lower constant is needed,

$$\frac{32}{15625\pi^6} \leq -tf'(t) \leq 1 \quad (0 < t \leq 4).$$

A completely explicit comparison allowance. Put $n_+ = q + \ell_s$, $n_- = Q + \ell_s$ and $P_1 = -\psi'(1) > 0$. The derivative calculation gives $0 < -t\mathfrak{f}'(t) \leq 1$ for all $t > 0$. Differentiating the exact identity PCB4 therefore gives

$$0 < \psi''(t) = -\frac{\mathfrak{f}'(t)}{2(1+t)} \leq \frac{1}{2t(1+t)}.$$

Moreover $\psi'(t) = \log(k/E(k)) < 0$, since $0 < k < 1 < E(k)$. For $0 < t \leq 1$, integrate the preceding upper bound from t to 1; for $t \geq 1$, use that ψ' is increasing and negative. Thus

$$|\psi'(t)| \leq P_1 + \frac{1}{2} \log^+(1/t), \quad t > 0.$$

For $0 \leq y \leq \Delta$, write $z_y = (r+y)/(q+\ell_s-y)$. The logarithmic difference satisfies

$$\left| \log \frac{z_y}{r/q} \right| \leq \log(1+y/r) + \left| \log \frac{q}{q+\ell_s-y} \right| \leq \frac{y}{r} + \frac{\ell_s+y}{Q}.$$

The last inequality uses $|\log A - \log B| \leq |A-B|/\min(A,B)$, $q+\ell_s-y \geq Q+\ell_s \geq Q > 0$, and $\log(1+x) \leq x$ for $x \geq 0$. Because $|t\mathfrak{f}'(t)| \leq 1$, integration on logarithmic coordinates and then on $0 \leq y \leq \Delta$ proves

$$\left| \int_0^\Delta \mathfrak{f}(z_y) dy - \Delta \mathfrak{f}(r/q) \right| \leq \frac{\Delta^2}{2r} + \frac{\Delta \ell_s}{Q} + \frac{\Delta^2}{2Q}.$$

The fixed displacement has its original sign:

$$G(n_-, r+\Delta) - G(n_-, r+m) = 2 \int_{r+m}^{r+\Delta} \psi'(u/n_-) du \leq 0.$$

Since $m = \Delta - v \geq 0$, its absolute value is at most $2vP_1 + v \log^+(n_-/r)$. Consequently, for every integer $r \geq 1$,

$$|G(n_+, r) - G(n_-, r+m) - \Delta \mathfrak{f}(r/q)| \leq e_{a,s}^{\text{cmp}}(r) := \frac{\Delta^2}{2r} + \frac{\Delta \ell_s}{Q} + \frac{\Delta^2}{2Q} + 2vP_1 + v \log^+(n_-/r).$$

This compares the same two Robin centres; it does not replace either original monic minimum. Its finite weighted sum can therefore be used as

$$E_{a,s}^{\text{cmp}}(R) = \sum_{r=R+1}^q \vartheta_r e_{a,s}^{\text{cmp}}(r).$$

The harmonic term is at most $\Delta^2 \sum_{r=R+1}^q 1/r$; the two constant rational terms have sum at most $2q(\Delta \ell_s/Q + \Delta^2/(2Q)) = O(q)$. For the logarithmic term, put $N = \lfloor n_- \rfloor = n_-$ and use

$$\sum_{r=1}^q \log^+(n_-/r) \leq \sum_{r=1}^N \log(N/r) = N \log N - \log(N!) \leq N,$$

where $\log(N!) \geq \int_1^N \log x dx = N \log N - N + 1$. The fixed v term is therefore $O(q)$. Because $\Delta^2 = O(q)$, $\Delta \ell_s = O(q)$, $Q \asymp q$, $n_- \asymp q$ and $0 \leq v \leq 80$, the finite allowance is $O_{\text{actual}}(q \log q)$ for every $0 \leq R < q$, and has the sharper form $O_{\text{actual}}(q\{1 + \log(q/(R+1))\})$.

9 Propagation into the earlier receiving equations

This section uses the unchanged domain PCB0, source orders $s = 1, a$, $q = (a+1)^2$, $Q = (a-7)^2$, $m = q - Q - v$, and the original complete coefficient spaces and attained quotient metrics. It records the current receiving equations beside the earlier formulas. In particular, historical statements that a term had only an $o(aq)$ estimate are strengthened by the estimates below; their original proofs remain valid.

9.1 The conductor and covariance receiver

The original WCF26 and TAC6 square, including its ordinary transpose, identifies the transported covariance with the lower-root covariance and its actual endpoint error. The determinant phase and the two complementary exterior maps are proved in LT1–28 and XIM1–23. Their full-rank estimates imply

$$\mathcal{R}e^C = O_{\text{actual}}(q), \quad \Xi_{a,s} = O_{\text{actual}}(q), \quad F_{L_a}^{(s)} = O_{\text{actual}}(q), \quad (\text{DRU1})$$

where the last estimate is the unchanged LRP10–13 bound. Each symbol is the actual term in the exact identity

$$\mathcal{B}_a^{(s)} = \mathcal{C}_{a,s}^- + \mathcal{R}e^C + F_{L_a}^{(s)} + \Xi_{a,s} + \mathcal{T}_{a,s}. \quad (\text{DRU2})$$

The directed finite intervals remain those in the transport proof; DRU1 records their asymptotic sizes and does not choose their signs. Substitution of the independently proved LRS21–22 in DRU2 gives

$$\mathcal{T}_{a,s} = 16C_{\partial}aq - 128q \log a + O_{\text{actual}}(q), \quad s = 1, a. \quad (\text{DRU3})$$

Thus the lower-root $+128q \log a$ term and the tail's negative term are related by the original covariance equality. No diagonal replacement of an attained norm is used in this substitution.

9.2 Every original prefix and the correlated tail

For the FTR11 operations, keep their exact coefficients:

$$\text{pre}_R h = 2h(q + R) - h(q) - h(q - 1), \quad \text{tail}_R h = h(2q - 1) + h(2q) - 2h(q + R). \quad (\text{DRU4})$$

They sum to the same complete four-endpoint return. Applying these linear operations to the endpoint covariance identity underlying DRU2, and using the directed LRS25 intervals, proves, uniformly for every integer $0 \leq R < q$,

$$\mathcal{T}_{a,s}^{\leq R} = \mathfrak{P}_{a,s}(R) + O_{\text{actual}}(q), \quad \mathcal{T}_{a,s,R} = 16C_{\partial}aq - 128q \log a - \mathfrak{P}_{a,s}(R) + O_{\text{actual}}(q). \quad (\text{DRU5})$$

Here $\mathfrak{P}$ is the complete LRS23 expression, including its last single-row term. The old growing-prefix statement is thereby refined: at $R = \lfloor a^{3/2} / \log a \rfloor$ its leading size is $8a^{5/2}$, while the full $\mathfrak{P}$ is retained whenever $O(q)$ precision is used. A claim that this prefix itself is negligible at scale $q \log a$ would contradict this evaluated size. Its occurrence in the complete action is controlled by the exact partition in DRU4–5.

The FTR9–10 native tail is exactly

$$\tau_{\text{total}} = \log \frac{\det \Omega_{q-1} \det \Omega_q}{(\det \Omega_R)^2} = \mathcal{T}_{a,s,R}. \quad (\text{DRU6})$$

For every original pair of row cutoffs $0 \leq p \leq r \leq q$, the complete Gram update gives

$$\log \det(I + Y^* \Omega_p^{-1} Y) = 16aq \int_{p/q}^{r/q} \mathfrak{f}(t) dt + O_{\text{actual}}(q \log q), \quad Y = [x_{p+1}, \dots, x_r]. \quad (\text{DRU7})$$

The error is uniform in both cutoffs by PCB16–17. Indeed the determinant equals the sum of the same attained rows, whose full absolute discrepancy is $O(q \log q)$; the decreasing integral comparison costs $O(a \log q)$, and replacing Δ by $16a$ costs $O(q)$. Thus the old inverse and all cross pairings remain in the evaluated determinant. DRU5 gives the finer correlated total with its $q \log a$ coefficient; DRU7 supplies uniform interval control with the displayed, larger allowance.

9.3 The allocations, complements and proper source

The exact receiving maps in FTR10 and ATA34 continue to read

$$S_K + S_R = \mathcal{T}, \quad 4S_K = \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A. \quad (\text{DRU8})$$

These equalities apply to the same labelled row occurrences as well. For a set of $d \leq q$ occurrences, PCB18 therefore gives a common finite budget of size

$$O_{\text{actual}}(ad\{1 + \log(q/d)\} + q \log q) \quad (\text{DRU9})$$

for each nonnegative single-copy allocation and four times that budget for each original four-copy CPQ complement and signed profile bound. Empty sets give zero exactly; a possible row-zero atom is handled by PCB13. The proof is the unchanged extension of the actual restriction and quotient frames: Schur factorization splits the full increasing determinant into nonnegative increments, so each quotient increment is bounded by the full increment. This is the exact map transporting the row control into DRU8. The sum in DRU8 is now evaluated by DRU3 or DRU5. Its two summands still retain their separate original subspaces; no individual asymptotic coefficient follows from their sum.

At the independent proper degree k , retain its own source order $s_0 = 1, k$, metric, centre and quotient frames, and its independent finite guards $k \geq 257$ and $k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q_k$, together with its original quartet domain. They hold eventually for the fixed quartet and are not inferred from the target degree's guards. ATA35–39 proves for the actual displaced-cutoff correction

$$\begin{aligned} -E_Q^- &\leq \delta V_Q \leq E_Q^+, & E_Q^- &= O_{\text{actual}}(k^2 \log k), \\ E_Q^+ &= 4(8k + 24)(16k - 48)f(1) + O_{\text{actual}}(k \log q_k). \end{aligned} \quad (\text{DRU10})$$

The two high intervals each have length $2(8k + 24)$; the factor four counts these rows in a single copy. The proper source's own standard-window return remains in PRD's original degree- k Schur minimum. DRU10 controls its displacement through precisely that map and does not assign a value to the standard-window return.

9.4 The complete arithmetic action

Substituting DRU2 and the exact prefix–tail partition into WCF27 recovers the original TAC identity, without selecting any endpoint extremizers independently:

$$J_a = \mathcal{B}_a^{(s)} + \delta_{a,s}^{\text{ar}} + W_a - P_{a,-} - P_{a,+}. \quad (\text{DRU11})$$

The evaluated original window LRS30 and order-one source LRS22 give

$$J_a - \delta_{a,1}^{\text{ar}} = C_B q^2 + 8q \log a + O_{\text{actual},h}(q). \quad (\text{DRU12})$$

For the other source order the exact identity $\delta_{a,a}^{\text{ar}} - \delta_{a,1}^{\text{ar}} = \mathcal{B}_a^{(1)} - \mathcal{B}_a^{(a)}$ cancels its source change. Here h remains the fixed original quartet polynomial of LRS30. The still unevaluated term is the same signed arithmetic correction in DRU11–12. The exterior lower benchmark is a statement about J_a itself; DRU12 therefore provides a precise receiving equation for the next signed calculation, with no arithmetic sign inserted.

References

- [H23] Stephen Boyd and Lieven Vandenbergh, *Convex Optimization*, Cambridge University Press, 2004. Appendix A.5.5, p. 650 (Schur minimum and determinant factorization), and Appendix C.4.1, p. 673 (block elimination). Author-hosted edition: https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf. Directly inspected PDF pages 664–665 and 687–688 on 17 September 2026. The complex Hermitian form required here is proved explicitly in STF8–12.
- [H25] R. Roy, F. W. J. Olver, R. A. Askey, R. Wong and W. P. Reinhardt, “Algebraic and Analytic Methods,” Chapter 1 of the *NIST Digital Library of Mathematical Functions*, version 1.2.8 (15 September 2026). Cauchy’s formula: equations (1.9.30)–(1.9.31), <https://dlmf.nist.gov/1.9.E30>, <https://dlmf.nist.gov/1.9.E31>. Taylor and binomial series: equations (1.10.1), (1.10.4), <https://dlmf.nist.gov/1.10.E1>, <https://dlmf.nist.gov/1.10.E4>. Laurent coefficients and removable singularities: §1.10(iii), equations (1.10.6)–(1.10.7), <https://dlmf.nist.gov/1.10.iii>. Maximum-modulus principle: §1.10(v), <https://dlmf.nist.gov/1.10.v>. Chapter authors: <https://dlmf.nist.gov/1>. Accessed 18 September 2026.
- [H21] T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, “Orthogonal Polynomials,” Chapter 18 of the *NIST Digital Library of Mathematical Functions*, version 1.2.8 (15 September 2026). Christoffel–Darboux and its confluent form: equations (18.2.12)–(18.2.13), <https://dlmf.nist.gov/18.2.E12>, <https://dlmf.nist.gov/18.2.E13>. Classical norms and leading coefficients: Table 18.3.1, <https://dlmf.nist.gov/18.3.T1>. Rodrigues: equation (18.5.5) and Table 18.5.1, <https://dlmf.nist.gov/18.5.E5>, <https://dlmf.nist.gov/18.5.T1>. Laguerre series and zero value: equations (18.5.12) and (18.6.1), <https://dlmf.nist.gov/18.5.E12>, <https://dlmf.nist.gov/18.6.E1>. Classical recurrence: equation (18.9.1) and Table 18.9.1, <https://dlmf.nist.gov/18.9.E1>, <https://dlmf.nist.gov/18.9.T1>. Meixner–Pollaczek weight, norm and recurrence: equations (18.19.8)–(18.19.9) and (18.22.8), <https://dlmf.nist.gov/18.19>, <https://dlmf.nist.gov/18.22.E8>. Chapter authors: <https://dlmf.nist.gov/18>. Accessed 17 September 2026.
- Generating function: equation (18.23.7), <https://dlmf.nist.gov/18.23.E7>. This locator was directly rechecked on 18 September 2026.
- [H22] R. A. Askey and R. Roy, “Gamma Function,” Chapter 5 of the *NIST Digital Library of Mathematical Functions*, version 1.2.8 (15 September 2026). Defining integral, recurrence and reflection: equations (5.2.1), (5.5.1) and (5.5.3), <https://dlmf.nist.gov/5.2.E1>, <https://dlmf.nist.gov/5.5.E1>, <https://dlmf.nist.gov/5.5.E3>. Binet’s Laplace integral: equation (5.9.10_2), https://dlmf.nist.gov/5.9.E10_2. Stirling expansion and finite positive-real remainder: equation (5.11.1) and §5.11(ii), first paragraph, <https://dlmf.nist.gov/5.11.E1>, <https://dlmf.nist.gov/5.11.ii>. Chapter authors and acknowledgement of P. J. Davis’s contribution: <https://dlmf.nist.gov/5>. Accessed 17 September 2026.
- Euler’s beta integral: equation (5.12.1), <https://dlmf.nist.gov/5.12.E1>. This locator was directly rechecked on 18 September 2026.
- [H19] *A Note on the Globalization Semiring $G(\mathbb{Z})$* , Zenodo, 12 March 2026. Deposited creators: ChatGPT, 5.4 and Gemini, 2.5 DeepThink. Originating programme directed by KokunoYumeto with AI assistance, as expressly identified by the programme author. <https://doi.org/10.5281/zenodo.18976982>. <https://zenodo.org/records/18976982>. This attribution does not claim global novelty.

- [H3] Russell Lyons, “Determinantal probability measures,” *Publications Mathématiques de l’IHÉS* **98** (2003), 167–212. doi:10.1007/s10240-003-0016-0. Inspected author version: arXiv:math/0204325v4, 18 June 2003. Locators here refer to that version: §4, (4.1), p. 11.
- [H20] The Clankers (deposited creator), *A Secondary Note on Split-Zero Globalization: Semimodule Classification, Multiplicative Reconstruction, and All-Orders Zeta Deformation*, Zenodo, record publication date 20 March 2026. Originating programme directed by KokunoYumeto with AI assistance. <https://doi.org/10.5281/zenodo.19120905>. <https://zenodo.org/records/19120905>. The deposited PDF prints 29 April 2026; record and printed dates are retained separately. No global priority claim is made.
- [H24] Gábor Szegő, *Orthogonal Polynomials*, fourth edition, American Mathematical Society Colloquium Publications, vol. 23, American Mathematical Society, Providence, Rhode Island, 1975. Theorem 2.5 and equations (2.5.1)–(2.5.4), pp. 29–30. doi:10.1090/coll/023. The book-level DOI does not specify the edition; the directly inspected fourth-edition copy and its edition-history page are recorded in SLC’s primary-source receipt. The source’s credit to Christoffel is retained.